Shape bias is a readout property: a developmental two-stream model, frozen networks and human cortex

Manuscript draft, compiled 2026-09-22 18:31 from `docs/PAPER_v2.md` at commit `1a791aa`.

Abstract

Humans name objects by shape when shape and texture disagree, yet the ventral stream is well described by texture statistics. We test this with a two-stream network modelling magnocellular and parvocellular input: one ResNet-18 receives a cone-opponent image's luminance under a developmental diet, the other its L−M channel. The magno stream's shape bias starts in its input and survives training only under the diet and beside the second stream; the parvo stream's texture bias is learnt. Fitted to eight Natural Scenes Dataset subjects, the texture-biased stream carries every extrastriate region. Behaviour, reading out from both, is shape-biased; one readout weight moves the rate from 0.40 to 0.90, and the weight best predicting extrastriate cortex (0.33) is not the one best predicting human choices (0.67). In 38 frozen ImageNet networks without a stream split, a reciprocal gain on each network's shape- and texture-predictive subspaces moves the rate by 0.08–0.29 on unseen exemplars within five points of top-1, re-ordering 19% of pairs, twice the resampling noise; it acts mostly on texture evidence, except in a Stylized-ImageNet network with a shape-leaning representation; and agreement with observers follows the rate. Shape-biased behaviour does not require shape-biased representations; the cue-conflict rate is a readout statistic.

1. Introduction

Humans presented with an image whose shape and texture point to different categories name the shape: on the cue-conflict images of Geirhos et al. (2018a), ten observers did so on 96% of decided trials, while ImageNet-trained convolutional networks named the texture. Closing that gap has become a standard target for biologically motivated models (Hermann et al., 2020; Jang & Tong, 2021), and a developmental visual diet — training on input whose acuity, contrast sensitivity and chromatic sensitivity follow the human developmental trajectory — is among the most effective routes to it (Lu et al., 2026).

A second body of evidence sits uneasily beside the first: along the ventral stream, object representations are well described by texture statistics — category-selective human cortex fails to discriminate natural images from texture-matched scrambles yet still discriminates categories (Jagadeesh & Gardner, 2022), macaque V4 responses to natural textures are predicted by their image statistics (Okazawa et al., 2015), and in inferotemporal cortex texture is the more decodable cue (Jagadeesh & Livingstone, 2024; Komatsu & Goda, 2018). The premise is scoped — one human study with one manipulation, the rest macaque, and "texture-like" meaning that a texture-statistics description suffices, not that shape is absent — and the argument needs only that the ventral description is substantially texture-like.

If behaviour is shape-biased and the ventral stream texture-biased, either the two measures disagree about what shape bias is, or behaviour is not a readout of one representation. The second horn is a hypothesis about architecture that single-trunk models of the ventral stream cannot express, and two facts make it concrete: the visual system receives not one image but a luminance-dominated, spatially coarse magnocellular signal and a chromatically opponent, finer parvocellular one, separated before cortex and later recombined (Livingstone & Hubel, 1987; Kaplan &

Shapley, 1982; Derrington et al., 1984; Nassi & Callaway, 2009; Ferrera et al., 1994), and that input is not stationary over development, so the system first learns from a low-passed world. The two-stream hypothesis (Mishkin et al., 1983; Goodale & Milner, 1992) and the emergence of ventral- and dorsal-like streams in a two-pathway network trained on one objective (Bakhtiari et al., 2021) motivate an architecture whose streams are defined by their subcortical-like inputs rather than by their cortical targets.

This paper asks two questions, in the order the originating study posed them (Manocchio, 2026): can two streams fed magnocellular- and parvocellular-like inputs diverge while optimising a single objective and producing shape-biased behaviour, and does such a system explain how the ventral stream can be texture-like while behaviour is shape-biased? We build it — two ResNet-18 trunks receiving the luminance and L−M channels of a cone-opponent image, with a developmental diet on the luminance channel (Fig. 1) — and examine it with linear probes and cue-conflict behaviour on the streams (Fig. 2), voxel-wise encoding models of eight subjects' 7T fMRI responses (Fig. 3) and the behavioural readout that combines the streams (Fig. 4). The streams diverge by different routes; the texture-biased one carries extrastriate cortex; and the integrated behaviour is shape-biased because a behavioural rate is a property of a readout over components, not a report of any one of them — a property that holds, in kind, in networks with no stream split at all (Fig. 6). Because the account predicts that properties bundled with shape bias need not travel with it, we test that too: robustness, agreement with human decisions and representational dimensionality all come apart from the rate (Fig. 5).

Relative to the originating thesis, every condition is trained with three seeds and every claim states its replication unit; the representational-similarity analysis that had found no stream-to-brain mapping is replaced by variance partitioning of encoding models, which removes the global stream preference that had defeated it; and the stream weighting the thesis fixed by hand is swept, so that the dependence of behaviour on the readout is measured rather than assumed.

2. Results

2.1 The model, and what the diet costs

The network splits its input once, before any convolution (Fig. 1a): each image is converted to the DKL cone-opponent space, stream 1 receives the luminance channel and stream 2 the L−M channel, and each is a ResNet-18 trunk. Under the developmental visual diet the luminance channel passes through a Gaussian blur (σ = 4 px) and the scale-free contrast-sensitivity cut of Lu et al. (2026) at a fixed age of 48 months, separating the transform from the trajectory (Methods 4.1). The trunks are initialised bit-identically — a control in which both streams receive the same image leaves all 120 tensor pairs identical after 150 epochs — so every stream difference below is attributable to the inputs, and a linear classifier reads out $w \cdot y_1 + (1-w) \cdot y_2$ of the two streams' penultimate features, with $w = 0.5$ during training.

Four front-end conditions are used throughout (Fig. 1a): the diet; the no-diet control; a channel-swap control applying the identical transform to the L−M channel instead; and a `parvo_highsf` control in which stream 2 additionally receives untransformed luminance, because real parvocellular cells carry both. Networks were trained on MiniEcoset (112 classes; Thorat & Kietzmann, 2024) for 150 epochs, three seeds per condition (Methods 4.2).

The diet costs accuracy (Fig. 1c): the no-diet conditions converge to 0.70 validation top-1, the diet arm reaches 0.556 at its best saved checkpoint against 0.701 and, uniquely, degrades after its peak, ending at 0.445 (range 0.347–0.527), so behavioural and out-of-distribution results use each seed's best saved checkpoint, stated wherever the policy matters (Methods 4.7). Two single-

stream controls trained with the identical diet — a ResNet-18 (11.2M parameters) and a ResNet-34 parameter-matched to the two-stream model (21.3M against 22.4M) — reach 0.599 and 0.623, so the accuracy cost of the split is not a capacity artefact (Fig. 4c).

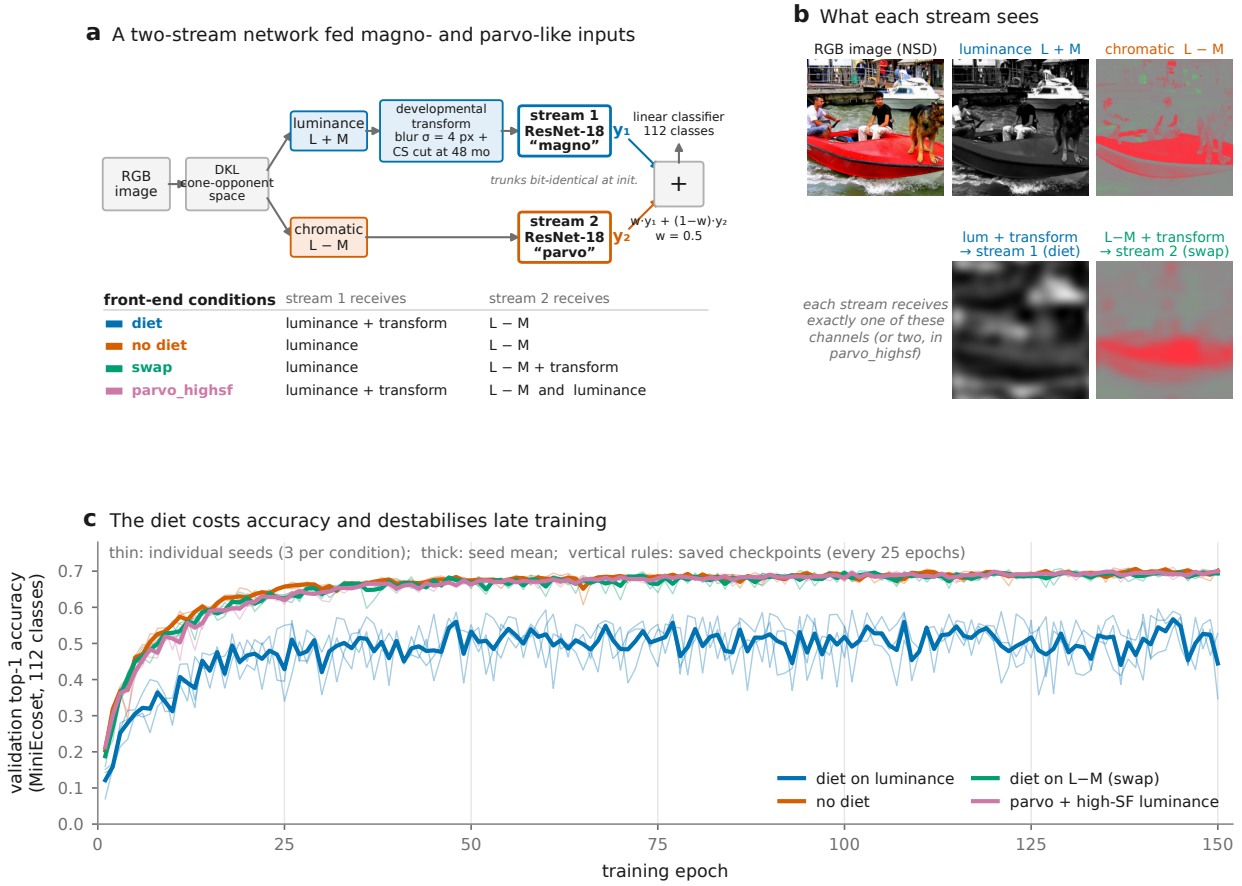


Figure 1 | A two-stream network fed magnocellular- and parvocellular-like inputs under a developmental visual diet.

(**a**) The model. Each RGB image is converted to the DKL cone-opponent space and split once, before any convolution: stream 1 ("magno") receives the luminance channel, stream 2 ("parvo") the L−M chromatic channel. Under the developmental visual diet the luminance channel additionally passes through a Gaussian blur ($\sigma = 4$ px) and the scale-free contrast-sensitivity cut of Lu et al. (2026), evaluated at a fixed age of 48 months. The two ResNet-18 trunks are initialised bit-identically, so any difference between the trained streams is attributable to their inputs. A linear classifier reads out the weighted sum $w \cdot y_1 + (1-w) \cdot y_2$ of the two streams' penultimate features, with w fixed at 0.5 during training. The table lists the four front-end conditions used throughout: the diet, the no-diet control, the channel-swap control (the identical transform moved onto L−M) and the `parvo_highsf` control (stream 2 additionally receives untransformed luminance).

(**b**) What each stream sees, for one NSD image (id 15504, the most chromatic of a fixed candidate grid), rendered with the exact training front end. Top: the RGB image and its luminance and L−M channels; bottom: the two channels after the developmental transform. The L−M channel is contrast-stretched for display only.

(**c**) Validation top-1 accuracy over 150 epochs of training on MiniEcoset (112 classes), three seeds per condition. Thin lines are individual seeds, thick lines the seed mean; vertical rules mark the epochs at which weights were saved. The diet costs accuracy — 0.556 against 0.701 at the best saved checkpoint — and, uniquely among the four conditions, degrades after its per-epoch peak (final-epoch mean 0.445, range 0.347–0.527). Every behavioural and cortical result in this paper states which checkpoint policy it uses.

2.2 The two streams specialise in opposite directions, by different routes

Read out separately with a linear probe on the cue-conflict images, the two streams of the diet-trained network end on opposite sides of the shape–texture boundary: scored on exemplars the

probe was not trained on, shape − texture accuracy is +0.337 for the magno stream and −0.168 for the parvo stream, means of three seeds, every seed on its side (Fig. 2a; +0.819 and −0.518 with exemplars shared; Methods 4.3); the magno stream decodes a new exemplar's texture at chance (0.08, against 0.0625). Behaviour agrees: read out alone at the final checkpoint, the policy of the cortical fits, the magno stream's cue-conflict shape bias is 0.878 and the parvo stream's 0.418, against 0.675 for the integrated readout (Table 1; best-checkpoint values in Extended Data Table 10) — the divergence the thesis reported on a single model per condition, now replicated across seeds.

Table 1. The two streams end on opposite sides of the boundary, by different routes. Final checkpoint of each seed, the policy of the cortical fits (Methods 4.7); cue-conflict values at the best saved checkpoints are in Extended Data Table 10. Probe values are shape − texture accuracy scored on held-out exemplars (the exemplar-balanced probe of Methods 4.3) and, after the slash, with test images sharing exemplars with the training folds, as originally reported.

	magno stream (luminance + transform)	parvo stream (L−M)	integrated
linear probe, shape − texture (0 = sign flip), 3 seeds: exemplars held out / shared	**+0.337** / +0.819	**−0.168** / −0.518	—
cue-conflict shape bias (0.5 = parity), 3 seeds	**0.878**	**0.418**	**0.675**
linear probe, untrained, seed 0	**+0.210** / +0.796	−0.033 / −0.087	—
linear probe, trained, seed 0	**+0.337** / +0.810	**−0.173** / −0.482	—
contribution of learning (trained, 3 seeds, minus untrained, seed 0)	**+0.127** / +0.024	**−0.135** / −0.432	—

The decomposition arises asymmetrically (Fig. 2a, Table 1). Scored on the untrained network with the same front end, most of the magno stream's shape bias is present before any weight is updated — the low-pass and contrast-sensitivity transform does it — at +0.210 of the trained +0.337 on held-out exemplars; the parvo stream begins nearly unbiased (−0.033) and acquires its texture preference over training. The shape-biased component starts in the input, the texture-biased component is built by learning, and what ordinary training does to the input's contribution is the decisive contrast: with the transform removed, the untrained luminance stream starts lower (+0.090) and training drives it across the sign flip to −0.122 — it does not merely fail to become shape-biased, it becomes texture-leaning — while the parvo stream ends texture-leaning whether or not the other stream is dieted (−0.168 against −0.118). Nor does the dieted input keep its bias alone: a single ResNet-18 trained on the magno stream's exact input and recipe ends at a rate of 0.557 and a penultimate-layer probe of −0.06 to −0.10 on held-out exemplars, where the same input as one of two streams ends at 0.878 and +0.26 to +0.27 — the shape-biased component survives training beside a second stream.

Which property of the input supplies the shape bias? Crossing the carrier a stream receives (achromatic luminance or chromatic L−M) with whether the developmental transform is applied to it gives four cells, two of which the swap and `parvo_highsf` conditions provide (Fig. 2b, c): on the probe, scored on held-out exemplars, the chromatic carrier gives −0.118 without the transform and −0.002 with it, the achromatic carrier −0.139 without and +0.337 with, so only the conjunc-

tion crosses zero, with an interaction of +0.36 over the additive prediction of the other three cells (−0.02) and no seed overlap between the conjunction cell (+0.333 to +0.340) and any other (at most +0.008); with exemplars shared the interaction is +0.73, and cue-conflict shape bias on the same four cells replicates it (Extended Data Table 4). The shape bias requires the developmental transform applied specifically to an achromatic channel — a relational fact about what a stream receives, not a property of a stream called "magno".

Two qualifications belong here. First, cue-conflict shape bias is computed only over images a readout decides as the shape or the texture category, a count that varies four-fold across cells (165 to 598 of 1,200; Fig. 2d), so the three non-conjunction behavioural cells are read as "no strong preference"; the probe, which classifies every image, gives the same pattern, and so does a forced two-way estimator over every image, which compresses values toward 0.5 (the four cells read 0.44, 0.51, 0.35 and 0.76) while preserving every ordering this paper draws a conclusion from, the two no-preference cells trading places (Extended Data Table 10). Second, the diet-trained models are less accurate: we pre-registered the fatal outcome — a no-diet model at the diet's accuracy with a shape-biased magno stream — and retrained the no-diet network with per-epoch checkpointing, selecting by rule, in each of three seeds, the epoch nearest the diet's accuracy (0.546–0.554 against 0.556); matching accuracy closes 4.3% of the probe gap on held-out exemplars (1.5–5.9% across seeds; 7.2% with exemplars shared) and 6.1% of the behavioural gap (2.2–9.1%), and the matched magno stream stays on the texture side in every seed (−0.10 to −0.12; Fig. 2e, f).

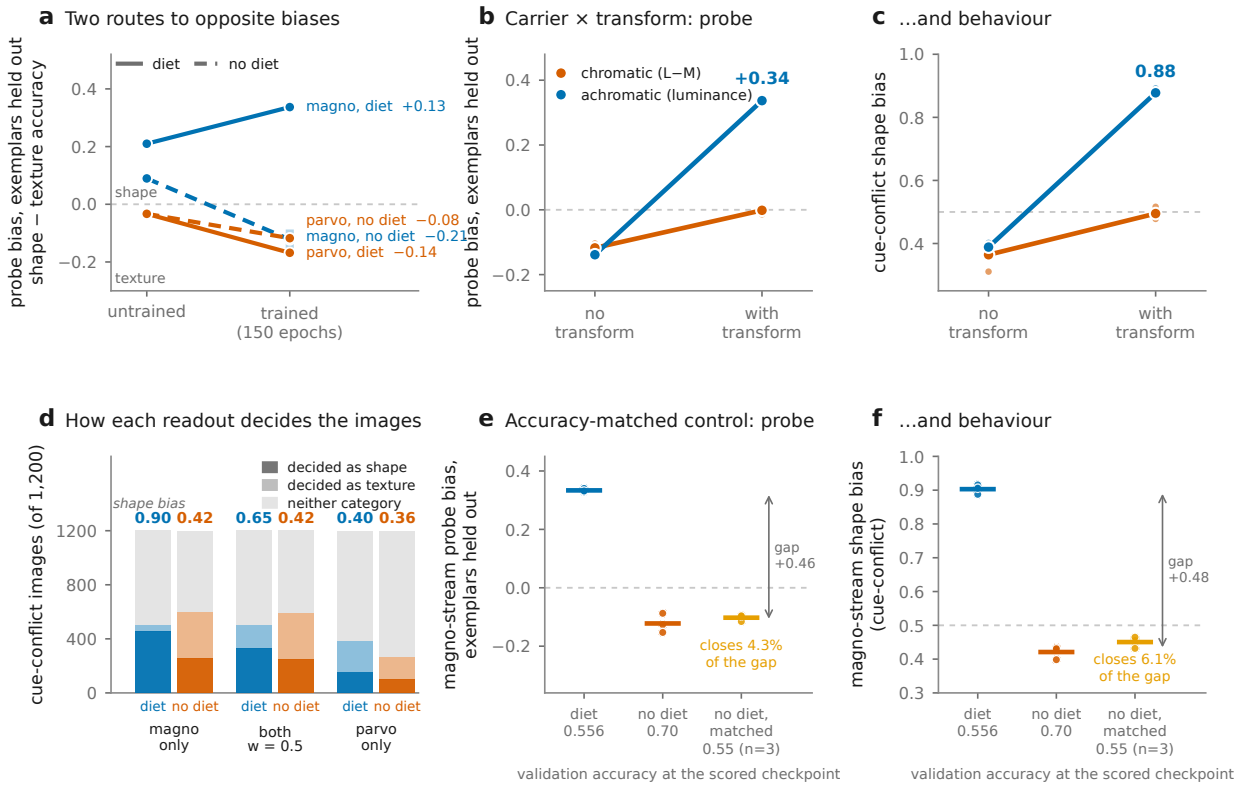


Figure 2 | The two streams end training on opposite sides of the shape–texture boundary, and they get there by different routes.

(**a**) Linear-probe shape−texture bias of each stream's features on exemplars the probe was not trained on — shape-label accuracy minus texture-label accuracy on the Geirhos cue-conflict images, from a probe trained on three exemplars per class for both labels with one exemplar left out per fold and PCA and scaling fitted inside each fold, mean of 20 draws (Methods 4.3); zero is the sign flip — before and after training, with the diet (solid) and without it (dashed). The untrained references are single models with no training seed; trained points are the mean of three seeds with the seed range as a bar. Under the diet the magno stream is already shape-biased before any weight is updated (+0.21) and training adds +0.13; the parvo stream starts near neutral (−0.03) and acquires its texture bias over training (−0.17). Without the diet, training drives the luminance stream from +0.09 across the sign flip to −0.12: ordinary training removes the shape bias the input supplies. The parvo stream ends texture-biased whether or not the other stream is dieted (−0.12 without). Decoding new exemplars is hard — the diet's magno stream reads a new exemplar's shape at 0.41–0.42 and its texture at 0.08, against a chance level of 0.0625 — so the biases are smaller than with the paper's original stratified-fold probe, whose test images share object and texture exemplars with its training folds; that probe gives every direction drawn here (magno +0.80 untrained and +0.82 trained under the diet, parvo −0.09 and −0.52, the undieted luminance stream +0.42 and −0.23; Table 1).

(**b, c**) The carrier × transform 2×2 on the held-out probe (b) and on cue-conflict shape bias (c). Each line is one carrier (chromatic L−M or achromatic luminance); the x-axis is whether the developmental transform is applied to it. Points are the three training seeds; the thick line is their mean. Neither factor alone crosses the boundary; only the conjunction does, on both instruments (probe: chromatic carrier −0.118 without the transform and −0.002 with it, achromatic carrier −0.139 and +0.337; interaction +0.36, and +0.73 with exemplars shared; Extended Data Table 4). Cells are read out on stream 2 in three cases and on the diet's magno stream in the fourth (Methods).

(**d**) How each readout decides the 1,200 cue-conflict images at the best saved checkpoint (diet) and the final checkpoint (no diet): images decided as the shape category, as the texture category, or as neither of the two. Shape bias is shape/(shape + texture); the number of decided images varies about four-fold across readouts, so rates near parity are read as "no strong preference". A forced two-way estimator that keeps every image (shape logit against texture logit) compresses all rates toward 0.5 but preserves every ordering shown here (Extended Data Table 10).

(**e, f**) Accuracy-matched control. The diet-trained models are less accurate, so the conjunction effect could in principle be an accuracy effect. A no-diet network retrained with per-epoch checkpointing and stopped, per seed, at the epoch nearest the diet's accuracy (epochs 10, 10 and 11; 0.546–0.554 against 0.556) closes 4.3% of the magno stream's probe gap on held-out exemplars (e; +0.33 diet at its best checkpoint, −0.12 no diet, −0.10 matched; 1.5–5.9% across three seeds, 4.1–8.0% under the other held-out probe variants, 7.2% and 5.8–8.1% with exemplars shared) and 6.1% of the behavioural gap (f; 2.2–9.1%), and its magno stream stays on the texture side in every seed (−0.10 to −0.12). Three matched seeds, as for the diet and no-diet arms. The six checkpoints of this control were added to the held-out probe after the pre-registration, as a descriptive extension with no prediction (`docs/PLAN_EXEMPLAR_2026-09-21.md`, addendum of 2026-09-22; `results/figures_v2/exemplar_matched.json`).

2.3 In cortex, the texture-biased stream carries the ventral end of the hierarchy

The thesis had asked whether the streams align with different cortical streams by representational similarity analysis (Kriegeskorte et al., 2008) against seven regions of the Natural Scenes Dataset, and found that whichever stream was the more predictive correlated better with every region — a general factor that a correlation cannot separate from regional structure. We replaced the instrument with voxel-wise banded-ridge encoding models for each of the eight NSD subjects, one band per stream, partitioning every voxel's cross-validated explained variance into the part only the magno band accounts for (U_m) and the part only the parvo band does (U_p) (Methods 4.5); the per-voxel index (U_m − U_p)/(U_m + U_p), median over reliable voxels per ROI, across V1v → hV4 is Fig. 3a.

The two trained models sit on opposite sides of the global stream-preference axis — without the diet every retinotopic area is magno-explained, with it every area is parvo-explained — and that global difference, the general factor that defeated the correlational analysis, is not the result. The result is the shape of the profile: only the diet arm descends steeply from V1 to hV4, early visual cortex relatively more magno-explained and hV4 relatively more parvo-explained, the direction a luminance/low-spatial-frequency versus chromatic split predicts, and the descent belongs to the model whose global preference points the other way, which a model simply preferring one stream everywhere could not produce.

Quantified as the contrast hV4 − mean(V1v, V1d) of the per-ROI median (Fig. 3b; Extended Data Table 1), the gradient is −0.745 to −0.834 across the three diet-trained seeds against −0.049 to −0.096 without the diet, with the same separation on the raw statistic U_m − U_p, and all six comparisons reach the exact Wilcoxon floor for eight subjects ($p = .0078$) with complete seed separation. The per-subject Spearman ρ between hierarchical rank and stream preference is −0.964 to −0.973 with the diet and −0.46 to −0.66 without (Fig. 3c; Extended Data Table 2); the rank vector [1, 1, 2, 2, 3, 3, 4] admits exactly 630 labellings of the seven ROIs, and against that enumerable null — which calibrates the extremity of a result that was not pre-specified rather than testing a prediction — the diet arm's ρ is the single most extreme in every seed and on both statistics ($p = 1/630 = .0016$) while the no-diet arm is not ($p = .019$ to $.092$), so we claim nothing there.

The diet does not create the gradient; it decides what training does to the one the input supplies. An untrained network on the diet front end already sits at −0.544 (8/8 subjects, $p = .0078$), and on the no-diet front end at almost the same value (−0.479), so the two front ends alone impose no different gradient; training with the diet steepens it and training without it erodes it to near zero (Fig. 3b). Correcting each arm for the signed bias of the estimator itself — a simulated floor of the wrong sign, +0.04 with the diet and +0.05 without — leaves a paired displacement of −0.871 with the diet against −0.130 without, each non-zero in every subject, and an independently trained checkpoint from the thesis's own pipeline, with its progressive developmental schedule, gives −0.793.

Pushed through models fitted on natural scenes only, the cue-conflict images yield a per-voxel predicted shape-bias index whose magno − parvo difference is +1.53 to +1.57 with the diet against +0.08 to +0.11 without (8/8 subjects, $p = .0078$ in every seed; Fig. 3d; Extended Data Table 3) — a test that needs neither probe nor readout.

The gradient survives its nuisance axes and the instrument's own bias. Receptive-field size, eccentricity coverage and the noise ceiling all change from V1 to hV4, and the diet's magno stream is a low-pass instrument by construction; yet the diet gradient is negative within every one of NSD's five eccentricity bands and every pRF-size tertile while the no-diet arm stays near zero in each, it holds on continuous pRF eccentricity and size within subject, and the estimator's own simulated bias is small, flat across ROIs and of the wrong sign (Extended Data Fig. 3; Extended Data Tables 8 and 9; Supplementary Note S3).

The gradient does not stop at hV4, and it does not turn. Relabelling the stored fits with NSD's full atlases (Extended Data Fig. 4; Extended Data Table 11), the diet index runs from −0.05 in V1v through −0.81 in hV4 and then sits between −0.81 and −0.94 in every extrastriate region — ventral, lateral and dorsal alike, in the Kastner2015 atlas, NSD's stream parcels and the fLoc category regions — with every region-minus-V1 contrast negative in 8/8 subjects ($p = .0078$) and no region-minus-region contrast departing from zero under either arm (all $p > .05$). Beyond V3 the diet's magno band contributes about 0.01 unique variance while the parvo band's rises from 0.10 in V1v to 0.26 in PHC2 (Extended Data Fig. 4b): the texture-biased stream carries extrastriate cortex as a whole, and the thesis's magno→dorsal, parvo→ventral mapping fails for want of any magno variance to map.

Three things this section does not settle. The diet predicts BOLD worse, not better (median r^2 normalised by the noise ceiling 0.405 against 0.464 without the diet, in 8/8 subjects; Fig. 3e), so its case rests on stream differentiation and the gradient, not on prediction accuracy. The two front-end controls are degenerate on this instrument (swap: U_p = 0.006; `parvo_highsf`: U_m = 0.007; degenerate in every one of their three seeds, Methods 4.5), so only the diet on luminance yields two streams that both carry substantial unique cortical variance and a gradient between them (Fig. 3a). And the gradient was not pre-specified: it is the survivor of seven regional hypotheses, six of which failed — among them the thesis's magno→dorsal mapping, an hMT+ specificity and a foveal/peripheral contrast whose sign flipped when the blur was removed — reported because it alone survives removal of that manipulation, and read as sign and ordering rather than as a calibrated effect (Supplementary Notes S1 and S3; Extended Data Table 1 for the subject-01 influence check).

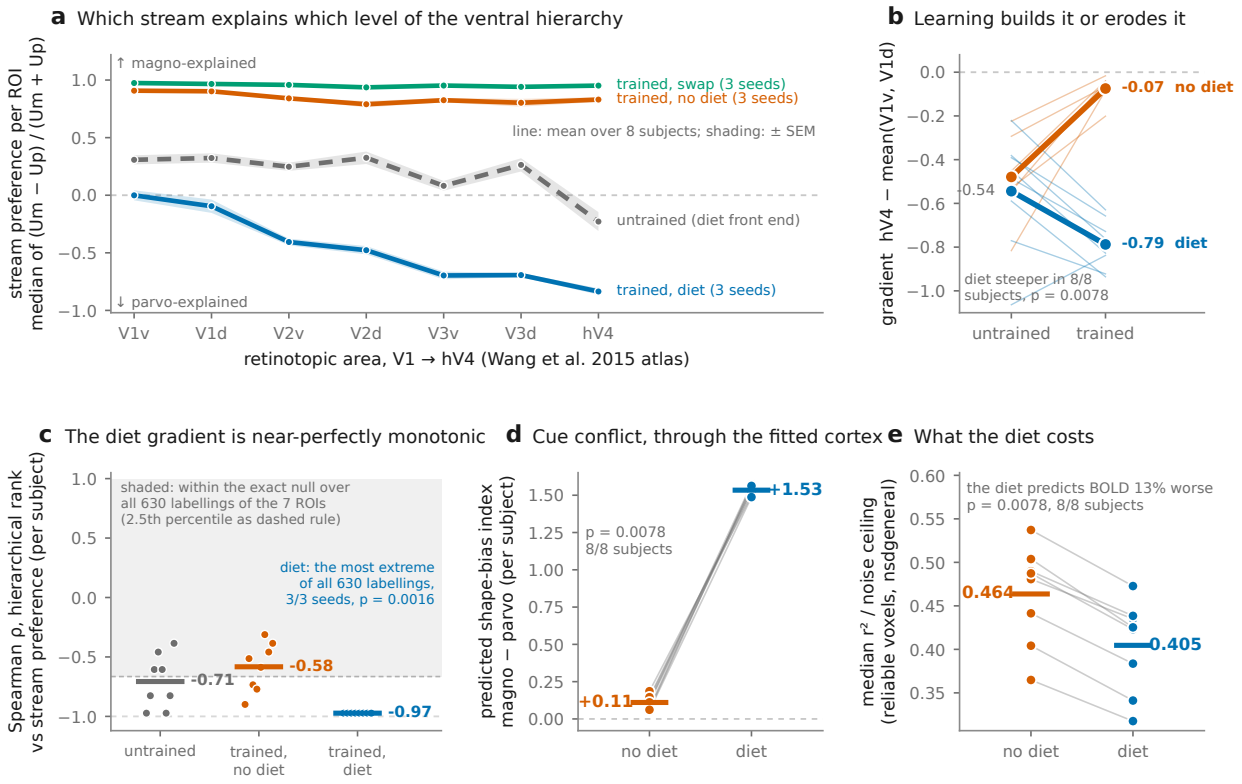


Figure 3 | In human visual cortex the texture-biased stream carries the ventral end of the hierarchy, and the diet decides whether learning builds that gradient or erodes it.

Voxel-wise banded-ridge encoding models were fitted on the Natural Scenes Dataset (8 subjects, `nsdgeneral` mask) with one feature band per stream, and every voxel's cross-validated explained variance was partitioned into the part only the magno band accounts for (U_m) and the part only the parvo band does (U_p). Stream preference per ROI is the median over reliable voxels of the per-voxel index (U_m − U_p)/(U_m + U_p) (noise ceiling > 0.1, U_m + U_p > 10^{-3}, at least 20 voxels); the raw statistic uses the per-voxel difference U_m − U_p.

(**a**) Stream-preference profile across the retinotopic hierarchy V1v → hV4 (Wang et al. 2015 atlas). Lines are means over the 8 subjects, shading ± SEM; the trained diet, no-diet and channel-swap arms average three training seeds per subject, the untrained network is a single model. The two trained arms sit on opposite sides of the global stream-preference axis (no-diet magno-explained everywhere, diet parvo-explained), and only the diet arm descends steeply from V1 to hV4. The swap control leaves the parvo stream contributing almost no unique variance anywhere (median U_p = 0.006 at seed 0, and 0.005–0.010 of the magno band's in every seed), so its flat profile reflects a degenerate decomposition rather than an absent gradient.

(**b**) The gradient hV4 − mean(V1v, V1d) per subject, from each arm's own untrained front end to the trained network (seed-mean per subject). The two untrained front ends start close together (−0.54 with the diet front end, −0.48 without); training with the diet steepens the gradient to −0.79, training without it erodes it to −0.07 (three seeds each: −0.834/−0.785/−0.745 against −0.079/−0.096/−0.049). The diet arm is steeper in 8/8 subjects (exact Wilcoxon p = .0078), and the same ordering holds on the raw statistic median(U_m − U_p).

(**c**) Monotonicity: Spearman ρ between hierarchical rank (V1 < V2 < V3 < hV4) and stream preference, per subject. The shaded region is the interior of the exact permutation null over all 630 distinct labellings of the seven ROIs; the dashed rule is its 2.5th percentile. The diet arm's observed ρ is the single most extreme of all 630 labellings in every seed (p = 1/630 = .0016); the no-diet arm does not reach that floor (p = .019 to .092).

(**d**) In-silico cue conflict. The 1,200 cue-conflict images pushed through the fitted encoding models yield a per-voxel predicted shape-bias index; plotted is the magno − parvo difference, median over reliable voxels, paired within subject. The effect is carried by the magno stream (+0.79 with the diet against −0.64 without);

the parvo streams are indistinguishable (-0.74 against -0.75).

(**e**) What the diet costs in prediction: median r^2 normalised by the noise ceiling over reliable voxels. The no-diet model predicts BOLD better in 8/8 subjects (0.464 against 0.405, a 13% cost). Panels (d) and (e) use seed 0 of each arm.

2.4 Behaviour reads out from both streams, and the rate is a property of the readout

The two streams are oppositely biased and the texture-biased one carries extrastriate cortex; what remains is what the behavioural readout does with them. The integrated readout's cue-conflict shape bias is 0.675 ± 0.025 with the diet against 0.416 ± 0.020 without (mean $\pm$ s.d. over three seeds, final checkpoints, as in Table 1; the thesis's checkpoint reproduces at 0.697 against its reported 0.677). On the identical images and statistic the ten human observers give 0.958 ± 0.019 (range 0.919–0.987) and the Geirhos et al. (2018a) CNNs 0.429, 0.312 and 0.172, values our pipeline reproduces. The diet-trained magno stream, at 0.878, closes most of the distance from the CNN regime toward the observers without matching the lowest of them, and integration costs 0.203 of shape bias at this checkpoint policy ($0.878 \rightarrow 0.675$; 0.248 at the best saved checkpoints; Fig. 4c, Extended Data Tables 5 and 10).

The integrated readout lands between its two components — an outcome two oppositely biased components nearly force, so not a prediction — and what can be measured is how much of the rate the readout controls. Sweeping w while holding y_1 and y_2 exactly fixed (Fig. 4a) runs cue-conflict shape bias from 0.40 to 0.90 over 21 values of w, three seeds and both checkpoint policies; $w = 0.5$ reproduces each trained model exactly, and the two streams alone sit at 0.41 and 0.78 on the forced two-way estimator, so the span is not an artefact of the decided-image denominator (Extended Data Table 10). Because the affine sweep is exact interpolation between two measured endpoints, under a rule fixed before the test we refitted a two-layer nonlinear head, and a linear head as a control for the refit itself, on the frozen features at each w (Fig. 4b): the nonlinear head's median span over the three best-checkpoint seeds is 0.466 against 0.508 for the frozen affine head (92% of the pre-registered baseline, 86% in the weakest seed, threshold 80%), the refitted-linear and nonlinear medians (0.461 and 0.466) agree within 0.01, and restricted to readouts within 5 points of peak validation accuracy the nonlinear span is 0.283 (0.26–0.31 across seeds): the span is a property of two oppositely biased components under any readout, and the extreme rates near $w = 1$ are bought with accuracy.

On the ladder of Fig. 4c the two streams alone bracket the integrated readout (0.903 and 0.400 around 0.655) and the single-stream controls trained with the identical diet lie between (0.567 and 0.625), inside the 0.058 that capacity alone moves the rate, so the ordering among these three models is attributed to capacity and no shape-bias advantage is claimed for the split.

Nothing here fits w to humans, and it could not be: fitted to each of the ten observers individually on a 0.001 grid, no observer's rate is reachable in any run (0 of 90 observer-runs, the closest approach missing the lowest observer by 0.027), and a weight fitted to an observer's trial-by-trial choices does not cross-validate (held-out agreement +0.026 against +0.023 for an arbitrary weight, better in 45 of 90). The rate is a free parameter of the readout, and nothing in this family of readouts selects the human operating point.

Cortex and behaviour prefer different readout weights. The readout's input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$ is a feature vector like any other, so for each w it was the design of a ridge encoding model of every voxel in NSD's general visual mask, and the w that maximises an ROI's median r^2 is the weight that region prefers (Methods 4.5; Extended Data Fig. 5; Extended Data Table 12). Under the diet that weight descends the hierarchy — 0.58 in V1 to 0.33–0.35 in every extrastriate group, ventral, lateral and dorsal alike, below V1 in 8/8 subjects ($\Delta = -0.252$ $[-0.283, -0.219]$, exact p =

.0078) — whereas the weight that best reproduces the ten observers' trial-by-trial choices is 0.67 (range 0.49–0.89), and the two sets of optima do not overlap (extrastriate 0.27–0.40; exact Mann–Whitney p = .000046, complete separation of the 8 subject optima from the 10 observer optima). On the behavioural sweep the two demands sit on opposite sides of the rate, 0.56 at the weight cortex prefers and 0.78 at the weight behaviour prefers, and without the diet the gradient reverses (0.65 in V1 to 0.75–1.00 beyond hV4; $\Delta = +0.188$ [+0.077, +0.285], 7/8). The curves are shallow (every ROI within 3% of its best r^2 at w = 0.5 with the diet and within 6% without) and their heights sit at the ridge's own reproducibility — the 1,024-dimensional concatenation, which every mixture is a linear map of, is itself exceeded by the best mixture in 36 of 112 subject × ROI × arm cells by a median 0.0019 r^2, more than the 0.0016 the sweep buys — so this is an ordering of optima and never an r^2 difference: the dissociation of Fig. 3 in the readout's own currency.

Without a stream split, a gain on a network's own shape- and texture-predictive subspaces still moves the rate. The span of Fig. 4a is given by construction in a network built around two streams, so we tested networks with none, under plans committed before each analysis was run (Methods 4.8). In any frozen network the penultimate features hold a shape-predictive and a texture-predictive subspace — the fifteen discriminant directions of a 16-class shape and of a 16-class texture linear discriminant, made mutually orthogonal — and a gain of 2w on the first and 2(1 − w) on the second, everything else including the network's own head untouched, is the analogue of the mixing weight; every w is gated on top-1 accuracy on held-out natural images (Fig. 6). Because the 1,200 cue-conflict images reuse 160 object and 48 texture exemplars, the subspaces are fitted on images sharing no exemplar with those the rate is scored on (Methods 4.8); the pre-registered split shared exemplars, and its values are given as exemplar-shared. In the six ImageNet networks of Fig. 6 the rate rises monotonically with w ($\rho \geq 0.999$ on the mean curve) by 0.14–0.20 with top-1 within 1.5 points of the trained value over the whole sweep — about a third of the two-stream span — so the pre-stated threshold (≥ 0.20 in at least four of the six), met in all six with exemplars shared (0.25–0.33), is met in one; four of the six can still be moved past the trained rates of at least two others, but the reverse of their trained order, reachable with exemplars shared, no longer is (Fig. 6a, b; Extended Data Tables 13 and 16). Across the field's zoo the result holds and the ranking does not: in 38 torchvision ImageNet classifiers the whole sweep stays within the five-point gate (top-1 changes by at most 4.4 points), the reachable span is 0.08–0.29 (median 0.19; 0.20–0.50 with exemplars shared, the difference coming mostly through the texture exemplars; Methods 4.8) and unrelated to the trained rate ($\rho = +0.10$), the rankings by trained rate and by the rate at the shape end of the gate agree at Kendall $\tau = 0.63$ with 19% of the 703 pairs reversing (16% on the forced estimator; 27% and 21% with exemplars shared), against 8.9% [6.0%, 12.2%] for the trained ranking re-estimated on resampled images, and the ranking by trained rate agrees with that by a linear probe's shape-minus-texture accuracy on held-out exemplars only at $\tau = 0.43$ (Extended Data Fig. 6; Extended Data Table 16). The span needs the labelled directions: a gain on pairs of random 15-dimensional subspaces of the same features moves no network's rate by more than 0.08 (median 0.01 over ten draws each; Extended Data Table 18), below every network's span on unseen exemplars. Error consistency with the ten observers (Results 2.5), added after the pre-registered analysis, is 2–23% of the human–human ceiling at the trained readouts (median 8%); it rises with the trained rate across networks ($\rho = 0.80$) and, within every network, from the texture end to the shape end of its gate (38 of 38; median +14 points of the ceiling), yet no gated readout of an ImageNet-trained network exceeds 37% of the ceiling — a rise that is mostly the measure's own bound moving with the rate (Results 2.5; Extended Data Table 17). The informative exception is a ResNet-50 trained on Stylized-ImageNet (Geirhos et al., 2019): the highest trained rate of the 42 networks tested (0.81), the only one whose probe leans to shape on exemplars it was not trained on (+0.23 [+0.13, +0.34], against −0.52 to +0.01 for the ImageNet-trained networks, 37 of 38 below zero; −0.11 against −0.30 to −0.66 with exemplars shared), and the highest agreement with

the observers (40% of the ceiling at its trained readout, 53% at its shape end) — its bias sits in the representation, which the rate alone could not have shown. Its span is among the smallest (0.12 on unseen exemplars, the 3rd smallest of 39; 0.17 and the smallest with exemplars shared, short of its pre-registered 0.20 threshold, P-Z1–P-Z4, the other three predictions met), but that ranking belongs to the estimator — in sample it is the 9th smallest of 39, and on the forced estimator the 16th — and what sets the network apart is which evidence the gain acts on (below). Our own single trunks move by 0.23–0.34 ungated (0.17–0.28 on unseen exemplars) but lose 7–15 points of MiniEcoset top-1 at the ends of the sweep (seed means; up to 12–17 points in single seeds, Extended Data Table 13), 0.14–0.22 within the gate (0.10–0.15 on unseen exemplars, Fig. 6a), and a head refitted at each w on the lossy sum of the two subspaces, the residual passed unchanged, keeps validation accuracy within 1.2 points of its ceiling in every seed (0.5–0.8 points as family means) while the rate still moves by 0.10–0.15 (two-layer head; 0.08–0.13 linear). The information the gain uses is distributed — the two subspaces hold 4–15% (shape) and 20–33% (texture) of the feature variance — and the pre-specified unit-level rule, a median split of the units by their shape-minus-texture η^2, reaches gated spans of only 0.07–0.16 (Extended Data Table 14); on the two-stream network the data-driven subspaces are not the streams (63–70% of the shape basis's energy on stream 1, against the pre-stated 90%) and the subspace gain moves the rate by 0.32 (0.25 on unseen exemplars) where the streams themselves give 0.46 and the readout weight of Fig. 4a 0.50, so the architectural split remains that network's most effective factorisation.

What the gain acts on. Post hoc, under a fifth plan committed before the analysis ran (`docs/PLAN_SPAN_2026-09-18.md`), we decomposed each network's decisions at its trained readout. Because the gain adds $(2w - 1)\cdot W(P_S - P_T)y$ to the logits, its first-order effect on each image's shape-minus-texture margin is a shape term, the evidence the head receives through the shape subspace, plus a texture term, the texture evidence it receives through the texture subspace, the first added and the second removed as w rises (Methods 4.8). This account reproduces every network's forced-choice curve (median largest error 0.009) and, reduced to two numbers per network, ranks the forced-choice spans at $\rho = 0.95$, but it ranks the reported, held-out spans only at $\rho = 0.59$, and of seven predictions two were met, four failed and one held in half (Extended Data Table 18). What it does show is which evidence the gain moves. In 37 of the 38 ImageNet networks the texture term is the larger (median 89% of the effect; ViT-B/32 is the exception, at 45%): the gain moves an ImageNet network toward shape by removing texture evidence more than by adding shape evidence. The texture share tracks the probe's representational bias about as closely as it tracks the trained rate when the probe is scored on held-out exemplars ($\rho = -0.58$ $[-0.78, -0.29]$ against +0.59), and more closely only with exemplars shared (-0.81 $[-0.91, -0.62]$); the Stylized-ImageNet network is its extreme at 11%, its texture term below every ImageNet network's (0.31 against 0.93–3.95 margin units, as predicted), the share returning to 75% and 91% as ImageNet is added to its training and then fine-tuned on.

Cortex loads on the texture-leaning units, not on the discriminant subspaces. With the two unit halves of the no-diet single trunk as encoding bands, the texture-leaning half carries every extrastriate group relative to V1 in 8/8 subjects ($\Delta = -0.32$, exact p = .0078), and the cortex-optimal weight of the lossy sum falls from 0.80 in V1 to 0.37–0.48 beyond hV4 (7/8, p = .023) while the behavioural rate rises toward w = 1 (Fig. 6c, d; Extended Data Table 15); the diet-trained single ResNet-34 shows the same contrast (7/8, p = .016). In the ImageNet networks the texture-leaning half carries early visual cortex as well — V1 through hV4 and ventral cortex in ResNet-50, whose units are almost all texture-leaning, and every region in ConvNeXt-B (index -0.09 to -0.45) and ViT-B/16 (-0.13 to -0.31), two networks fitted after the pre-registered analysis — so the V1-relative prediction fails there (extrastriate − V1 = +0.24, p = .039; +0.22, 2/8; −0.08, 6/8, p = .25). These unique variances are small (≤ 0.04 for the texture-leaning half against 0.10–0.25 for

the parvo band with the split, 80–96% of the explained variance being shared between the halves against 45–63% between the streams; ViT-B/16 is the least redundant, at up to 0.09 and 64–72%), and under the subspace rule they all but vanish: entered as three bands with the residual, the two 15-dimensional subspaces each add at most 1.3% of noise-ceiling-normalised unique variance in any region against 23–63% carried by the shared and residual variance, their index sits at noise level, and the pre-registered cortical predictions fail — the extrastriate − V1 contrast is negative in 2/8 subjects each in the no-diet trunk, the diet ResNet-34 and ResNet-50, in 7/8 in ConvNeXt-B ($\Delta = -0.44$, $p = .016$) and in 3/8 in ViT-B/16, and the 15-dimensional lossy sum's cortex-optimal weight is lower beyond V1 in no family at the pre-registered 7/8. The directions that move the rate are nearly silent as unique cortical predictors — which component cortex loads on is decided by how the representation is partitioned — so the cortical claim is made under the unit rule only. The architecture is therefore not what makes the rate a readout property, nor, in the networks trained on our images, what makes extrastriate cortex load on the texture-biased component; it is what makes the components separable enough for one weight to move the rate across the whole range models occupy at no cost in accuracy, and for cortex to be carried almost entirely by one of them.

This is the mechanism we propose for the paradox: a system holding a shape-biased and a texture-biased component, whose behaviour reads out from both, will look shape-biased behaviourally and texture-biased wherever the texture-biased component dominates — and in this model of the subcortical split, the component that dominates the ventral end is the texture-biased one. It is an existence proof with a cortically grounded demonstration, not a measurement of the human readout.

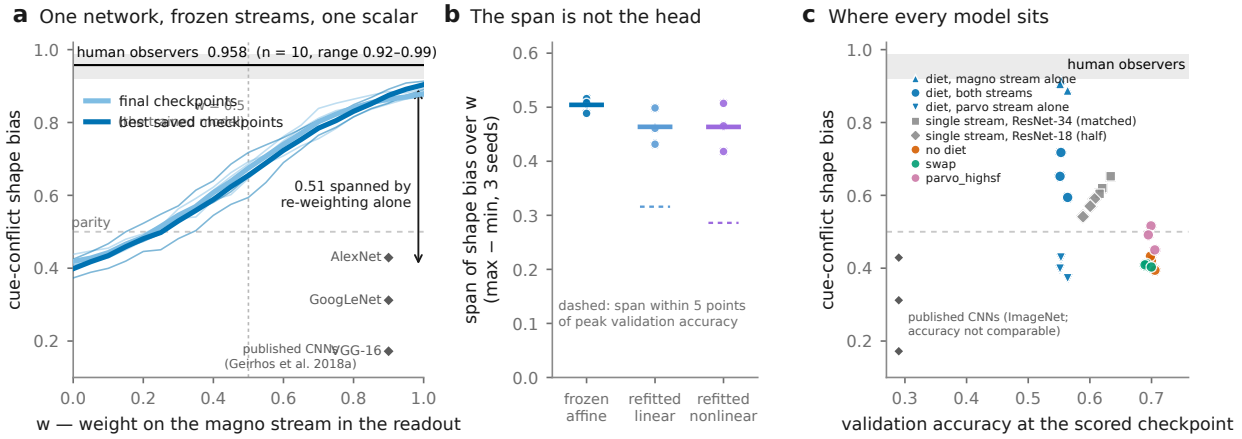


Figure 4 | Behaviour reads out from both streams, and the shape-bias rate is a property of that readout.

(**a**) Cue-conflict shape bias as a function of w, the weight on the magno stream in the readout full = classifier($w \cdot y_1 + (1-w) \cdot y_2$), at 21 values of w, for three diet-trained seeds under both checkpoint policies. Thin lines are seeds, thick lines the seed mean. Nothing is retrained, refitted or reselected along the x-axis: y_1 and y_2 do not depend on w, so the sweep moves the readout while both representations stay bit-identical. Shape bias runs from 0.40 to 0.90 at the best saved checkpoints; w = 0.5 is the trained model and reproduces its published per-seed values exactly. The ten human observers who viewed the identical images (mean 0.958, range 0.919–0.987; grey band) and the three CNNs of the Geirhos et al. (2018a) release are drawn on the same axis. No value of w reaches the lowest individual observer.

(**b**) The span is not a property of the affine head. Span of shape bias over w (max − min) per seed, best saved checkpoints, for the frozen affine head of (a), a linear head refitted on the frozen features at each w, and a two-layer nonlinear head refitted likewise (identical budget, same initialisation seed at every w). Median spans 0.508, 0.461 and 0.466; the refitted-linear and nonlinear heads agree to within 1%. Dashed bars: the span restricted to readouts within 5 points of peak validation accuracy (0.283 for the nonlinear head), the range over which the rate can be moved without buying it with accuracy.

(**c**) Where every model sits: cue-conflict shape bias against validation accuracy at the scored checkpoint, one marker per training seed. The diet-trained network's two streams read out alone bracket its integrated readout (0.903, 0.400 and 0.655); the two single-stream controls trained with the identical diet (one ResNet-18 trunk, 11.2M parameters; one ResNet-34 trunk, 21.3M, parameter-matched to the 22.4M two-stream model) are more accurate than the two-stream model and lie at 0.567 and 0.625. The three no-diet conditions cluster near parity at 0.70 accuracy. The published CNNs were trained on ImageNet and their accuracy is not on this axis; they are drawn at the left margin for their shape bias only.

2.5 Properties that do not follow from shape bias

If shape bias is a property of a readout over components rather than of a representation, quantities the field treats as travelling with it need not. Over identical inputs, three do not (Fig. 5).

Robustness. The standard reading of Geirhos et al. (2018a, 2021) is that shape bias buys out-of-distribution robustness. We pre-registered that prediction and tested it on corruptions applied before the front end, scored in the models' own label space at each seed's best saved checkpoint (Fig. 5a; Extended Data Fig. 1; Extended Data Table 6). The diet-trained model is less robust on four of the five corruption families it was not trained on — uniform noise, high-pass, false colour and rotation (relative-accuracy AUC 0.634 against 0.729, 0.678 against 0.827, 0.854 against 0.938 and 0.574 against 0.615), with complete seed separation on each — equivalent within $\pm$0.033 on phase scrambling, more robust only to low-pass (0.444 against 0.356), declared trained-in before the run because the diet is a low-pass blur, and not to contrast reduction (0.481 against 0.549) although its front end includes a contrast-sensitivity cut. Robustness appears exactly where it was trained in and nowhere else, the strongest available evidence that the nulls are real rather

than a broken measurement. This does not contradict Geirhos et al., whose stylisation route yields shape bias together with robustness — it shows the coupling belongs to that route — and it differs from the source of the diet, whose title reports robust, shape-based vision: we reproduce the shape-based half (their released model 0.822, our diet-trained magno stream 0.878) and not the robust half, and Methods 4.6 lists three implementation differences any of which could carry the divergence.

Agreement with human decisions. Two decision-makers with the same marginal rate can agree on no individual image beyond what those rates force; error consistency — Cohen's κ on trial-by-trial agreement — subtracts that forced agreement (Geirhos et al., 2020), and across the ten observers the human–human ceiling is $\kappa = 0.234$. We pre-registered that the diet-trained model would be more human-consistent than its control; it is not, on either readout, and the sharpest true statement is stronger: our most shape-biased model, the diet-trained magno stream at 0.878, has, averaged over seeds, the lowest error consistency of the seven models compared ($\kappa = 0.0082$, 3.5% of the ceiling, seeds spanning −6.1% to 15.5%, against 7.3% for VGG-16 at shape bias 0.172; Extended Data Table 7); across all 189 readouts of the w sweep, with w chosen post hoc per run to maximise agreement, the best κ never exceeds 20.7% of the ceiling (Fig. 5b), and equivalence intervals fixed in advance exclude a κ change larger than 0.038 across the 0.71 span of shape bias the readouts cover; across the field's zoo the same measure does rise with the rate, to at most 37% of the ceiling (Results 2.4) — and that rise is the measure's own bound. Cohen's κ cannot exceed a value fixed by the two marginals, which with the observers at 0.958 runs from 0.006 to 0.219 across the 798 gated readouts of the 38 networks; one quadratic in the rate accounts for the agreement ($R^2 = 0.90$) and, normalised by the bound, its rank dependence on the rate disappears (0.001 against 0.891). Agreement is thus readout-controlled as the rate is: over the 33 networks that reach a common rate of 0.40, ranking by agreement at the trained readouts against at that rate gives $\tau = 0.33$ with 34% of pairs reversing, the 1.9-fold residual spread is unrelated to the probe ($\rho = +0.01$), and the Stylized-ImageNet network is not separable from ViT-B/32 (135% against 120% of the human-pair value). Four of six predictions of a fourth plan failed in this way and a fifth was untestable as specified (`docs/PLAN_KAPPA_2026-09-17.md`, post hoc here; Extended Data Table 17). The measure does detect a correspondence when one is there — the same machinery recovers 24.1% of the split-half ceiling on whether an image resolves to either cue at all, against 8.7% on which cue is picked (Fig. 5c) — a positive control that invites a diet reading we do not make, because the crossed front-end 2×2 shows a train/test-matching pattern rather than a weights effect (Fig. 5d).

Representational dimensionality. The participation ratio of each stream's features on 1,000 NSD images per subject gives a magno/parvo ratio whose ordering across the four front ends is reproduced by a learning-free Gabor bank, so it tracks the bandwidth of the carrier a stream is handed; what the diet decides is which side of parity training ends on — every diet-trained seed ends magno-dominant (24/24 subject-seeds in each of three diet arms), including the `parvo_highsf` front end, which starts inverted in 8/8 subjects and which we had pre-registered would stay inverted, while every no-diet seed collapses to parity (Extended Data Fig. 2). No level is claimed, because the within-arm seed range exceeds the gaps between arms; and `parvo_highsf`'s second stream becomes the more dimensional stream while remaining texture-leaning (probe −0.139 on held-out exemplars, −0.271 shared), which separates dimensionality from shape bias.

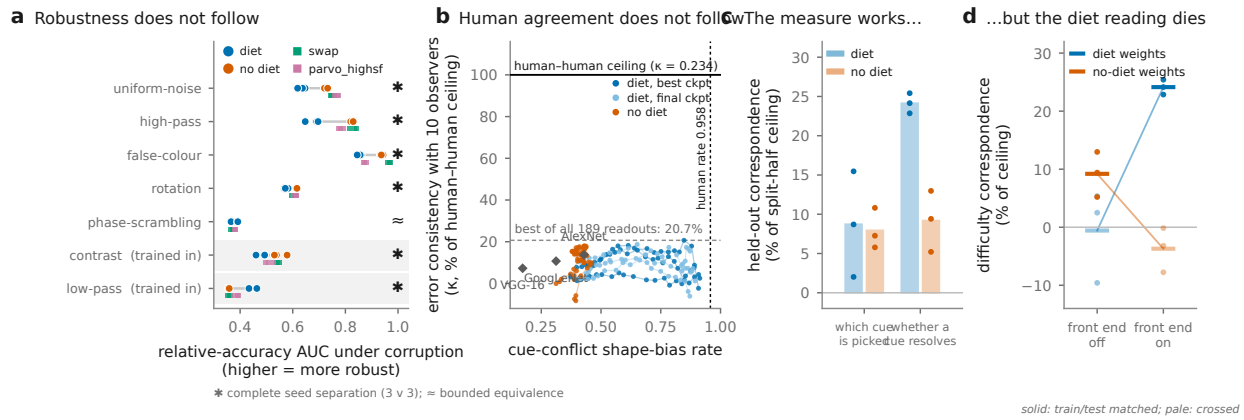


Figure 5 | Three properties the field treats as travelling with shape bias do not follow from it.

(**a**) Out-of-distribution robustness. Corruptions are applied to the image before the front end and scored in the models' own 112-class label space at each seed's best saved checkpoint; plotted is the area under the curve of accuracy relative to each seed's own clean accuracy, one marker per seed, for the diet (blue) and no-diet (vermillion) arms, with the swap and `parvo_highsf` controls as small squares. $*$ marks complete seed separation (the exact 3 v 3 Mann–Whitney floor, $p = 0.100$); $\approx$ marks the one family where the arms are equivalent to within ± 0.033. The diet-trained model is less robust on four of the five corruption families it was not trained on and more robust on exactly one family, low-pass, which the diet installs. Shaded rows are the two families the diet has effectively seen; it is not more robust to contrast reduction even though its front end includes a contrast-sensitivity cut. Full degradation curves: Extended Data Fig. 1.

(**b**) Trial-level agreement with the same ten human observers (error consistency, Cohen's κ, as a percentage of the human–human ceiling of $\kappa = 0.234$ over 45 observer pairs) against cue-conflict shape-bias rate, for all 189 readouts of the w sweep (9 runs × 21 weights) and the three published CNNs (Geirhos et al. 2020). The rate spans five-fold (0.17–0.90, the published CNNs included); agreement never exceeds 20.7% of the ceiling even with w chosen post hoc, per run, to maximise it. No regression line is drawn, because the x-position of every point was set by a free parameter.

(**c**) Positive control. On the same 1,200 images and the same 10 observers, with shape- and texture-category effects removed and w chosen on half the images and scored on the held-out half, the same machinery recovers 24.1% of the split-half ceiling for whether an image resolves to either cue at all against 8.7% for which cue is picked (diet arm; three seeds as dots, bar = mean). Each target is expressed against its own split-half ceiling.

(**d**) The crossed front-end 2×2 that withdraws the diet reading of (c). Difficulty correspondence for diet weights and no-diet weights evaluated with the developmental front end on and off. Both train/test-matched cells are positive (24.1%, 9.2%) and both crossed cells are not (−0.6%, −3.7%), so the diet arm's advantage in (c) is a train/test-matching pattern, not an effect of the learned weights.

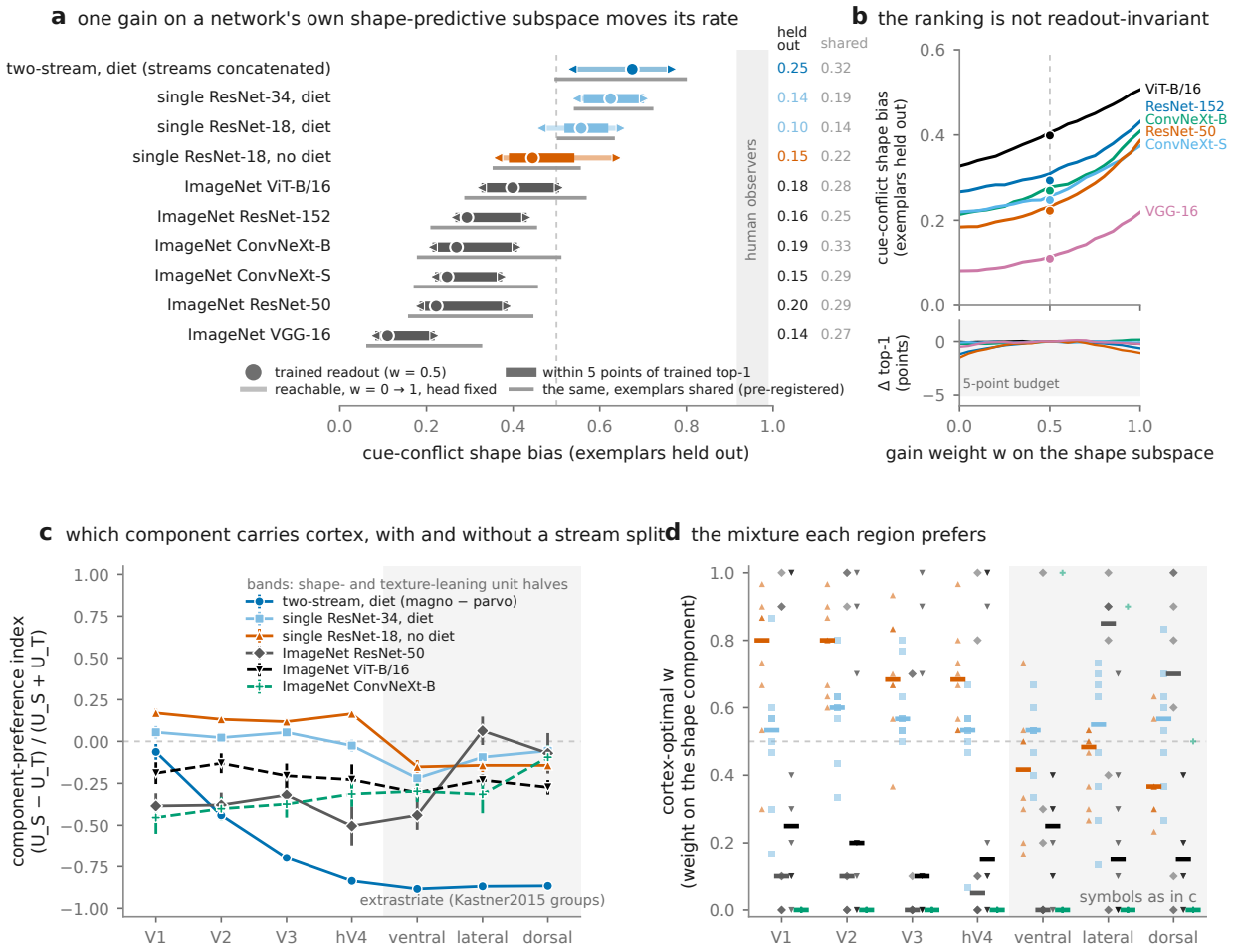


Figure 6 | The readout account beyond the two-stream architecture.

Every network here ends in logits = W·y + b for some penultimate feature vector y, and nothing is retrained. y is reduced by PCA and a 16-class shape and a 16-class texture linear discriminant give fifteen shape-predictive and fifteen texture-predictive directions, made mutually orthogonal by symmetric (Löwdin) orthogonalisation: bases B_S and B_T, with B_R the residual (Methods 4.8). Because the 1,200 cue-conflict images reuse 160 object and 48 texture exemplars, the curves drawn fit the subspaces on images whose object and texture exemplars both lie outside one of three exemplar folds and score the rate on the images whose exemplars both lie inside it (ten repetitions, mean curve; seeds averaged for our networks). The pre-registered estimator, fitted on one stratified half of the images and scored on the other, shares 98% of object and all texture exemplars between the halves; it is drawn as the grey brackets and gives spans about 1.6 times larger (Extended Data Tables 13 and 16).

(**a**) A reciprocal gain on a network's own shape- and texture-predictive subspaces moves its rate on exemplars the subspaces were not fitted on. logits(w) = W·(P_R y + 2w·P_S y + 2(1 − w)·P_T y) + b, the trained head untouched; w = 0.5 is the trained network (dot, its rate on all 1,200 images). Thin bars: the rate reachable as w runs from 0 to 1. Thick segments: the part of the sweep within 5 points of the trained top-1 on held-out natural images (MiniEcoset validation for our networks, a stratified 10-per-class ILSVRC-2012 validation subset for the ImageNet networks), with top-1 measured on bases fitted on all 1,200 images and the held-out fits inheriting that w-range. Grey brackets beneath: the same gated range under the pre-registered estimator, whose halves share exemplars. The numbers at right are the gated spans of the seed-mean curves with exemplars held out and shared (the latter, and the ungated held-out spans, in Extended Data Table 13). Our networks by the paper's primary estimator, the ImageNet networks by the Geirhos 16-class rule (Methods 4.3). In the six ImageNet networks the whole sweep lies inside the gate (top-1 changes by at most 1.5 points) and the rate moves by 0.14–0.20 with exemplars held out (0.25–0.33 shared), monotonically ($\rho \geq 0.99$); the pre-registered 0.20 is reached in one of the six. In our three single trunks the 16-class heads lose 7–15 points of top-1 at the ends of the sweep and the gated spans are 0.10–0.15 held out (0.14–0.22

shared). The pre-specified unit-level rule — a median split of the units by their shape-minus-texture η^2, the special case in which the two subspaces are disjoint sets of units, estimated with exemplars shared — reaches 0.07–0.16 within the same gates (Extended Data Table 14; no longer drawn). In the two-stream network the discriminant subspaces are not the streams (63–70% of the shape basis's energy lies on stream 1) and the subspace gain moves the rate by 0.25 with exemplars held out (mean over repetitions and seeds; the seed-mean curve runs $0.54 \rightarrow 0.77$) and by 0.32 with exemplars shared ($0.50 \rightarrow 0.80$), where the streams themselves, which the unit-level rule recovers, give 0.46 and the readout weight of Fig. 4a 0.50. Grey band, the ten human observers (0.92–0.99).

(**b**) The held-out curves of the six ImageNet networks (top) and the change of top-1 along each sweep (bottom; grey band, the five-point budget). Dots are the trained rates; the field's ranking is not invariant to the readout. With exemplars held out, four of the six networks can be moved past the trained rates of at least two of the other five within the budget, but the six can no longer be placed in the reverse of their trained order; with exemplars shared, five of the six pass the trained rates of four others (ViT-B/16, the highest, one) and the reverse order is reachable (Results 2.4).

(**c**) Which component carries cortex. For the single trunks the two unit halves of the pre-specified rule are the two bands of a banded ridge model of every NSD voxel (eight subjects, Methods 4.5); the index (U_S − U_T)/(U_S + U_T) is the median over reliable voxels per region, mean ± s.e.m. over subjects (seeds averaged per subject). The two-stream diet model's magno − parvo profile (Fig. 3, Extended Data Fig. 4) is drawn on the same axis. In the no-diet single trunk the texture-leaning half carries every extrastriate group relative to V1 (8/8 subjects, $\Delta = -0.32$, exact p = .0078) with no descent within V1–hV4; the diet-trained single ResNet-34 shows the V1-relative contrast (7/8, p = .016), a descent already within V1–hV4 (7/8, p = .016) and ventral cortex as its most texture-carried region (dorsal − ventral +0.16, 8/8); in ImageNet ResNet-50, whose units are almost all texture-leaning, that half carries V1 through hV4 and ventral cortex alike (−0.32 to −0.50) while lateral and dorsal cortex are balanced, so the V1-relative contrast reverses (+0.24, p = .039). In ConvNeXt-B and ViT-B/16, fitted after the pre-registered analysis, the texture-leaning half carries every region (index −0.09 to −0.45 and −0.13 to −0.31) with no V1-relative contrast (+0.22, 2/8 subjects negative; −0.08, 6/8, p = .25). Unique variances of the two halves in the single trunks and ConvNeXt-B are 0.006–0.04 (ViT-B/16 up to 0.09), against 0.10–0.25 for the parvo band with the split; most of a single trunk's explained variance is shared between its halves (Extended Data Table 15).

(**d**) The mixture each region prefers. The lossy sum $z(w) = w \cdot y_S + (1 - w) \cdot y_T$ (halves paired by rank) is the design of a ridge model at 11 values of w (0, 0.1, ..., 1), following Extended Data Fig. 5; points are subjects, bars medians, symbols as in (c). In the no-diet trunk the optimum falls from 0.80 in V1 to 0.37–0.48 beyond hV4 (7/8, p = .023); in the diet ResNet-34 the optimum sits near 0.55 in every region (the curves stay within 5% of their maximum over w in all three networks, as in Extended Data Fig. 5); in ResNet-50, V1–hV4 and ventral cortex prefer the texture-leaning half (0.0–0.1) while lateral and dorsal cortex prefer the shape-leaning one (0.7–0.85); ConvNeXt-B prefers the texture-leaning half in every region (median 0.0) and ViT-B/16 nearly so (0.10–0.25). Prediction P5 therefore holds in one of the three pre-specified networks and in neither of the two added after them.

Pre-specification and predictions: `docs/PLAN_GENERALITY_2026-09-14.md` (P1–P5, the unit-level rule) and `docs/PLAN_SUBSPACE_2026-09-15.md` (P-A, P-A′, P-B, P-C, P-D, the subspace rule). Panels c–d show the unit-level fits because the subspace rule's cortical predictions P-C1 and P-C3 failed and P-C2 held only in ConvNeXt-B: entered as three bands, the two 15-dimensional subspaces add $\leq$ 1.3% unique cortical variance each and their index is at noise level (Extended Data Table 15, upper block); per-network values: Extended Data Tables 13–15, of which Table 14's last two columns hold a post hoc sensitivity analysis with the shape-leaning set defined as the units with $d_j > 0$ or as the top decile (Methods 4.8).

3. Discussion

The two questions. The thesis asked whether magnocellular- and parvocellular-driven streams inside one network could diverge while the network behaves shape-biased, and whether such a system bears on why the ventral stream can be texture-like while behaviour is not. To the first the answer is yes, replicated across seeds and with the mechanism located: divergence requires the developmental transform on an achromatic carrier, the shape-biased component starts in the

input and is kept, and extended, by training only under the diet and beside a second stream, and the texture-biased component is built by learning. To the second the answer is an existence proof the thesis could not reach with its instruments: fitted to human cortex, the texture-biased stream's unique variance carries the extrastriate end of the hierarchy — ventral, lateral and dorsal alike — while behaviour, reading out from both, is shape-biased. Read as competing claims about a single representation the two cannot both be right; read as claims about a readout over components, they are not in competition.

What the readout account licenses. A model matching the human shape-bias rate is routinely described as having human-like representations; that inference fails twice over here. The same network holds components at $+0.34$ and -0.17 on held-out exemplars ($+0.82$ and -0.52 with exemplars shared) and behaves at 0.675, and one scalar over frozen representations moves the rate across most of the range the field's models occupy, so the rate does not identify a representation; and across readouts spanning a five-fold range of shape bias, trial-by-trial agreement with ten observers never exceeds 21% of the human–human ceiling in our networks and is lowest, 3.5%, for our most shape-biased stream; across the zoo the two do move together ($\rho = 0.80$ over 38 networks, every network's agreement rising from the texture to the shape end of its gate), yet no gated readout exceeds 37% of the ceiling and that rise is the bound on κ moving with the rate (Extended Data Table 17), so the rate is at best a weak proxy for the decision process — as is agreement, read as the field reads it. Prior work established that a representation can hold more shape than its behaviour reveals (Hermann et al., 2020; Islam et al., 2021), that the measured rate depends on the test design (Tartaglini et al., 2022), that a prompt — a readout-level manipulation — steers it in vision–language models (Gavrikov et al., 2024), and that it tracks a network's spatial-frequency channel (Subramanian et al., 2023), the quantity the diet constrains; new here is the form of the dissociation — both representations held fixed, the rate moved by the readout alone, a refitted nonlinear head preserving the span — and its cortical grounding: the same mixing weight prefers the parvo-heavy end when fitted to every region beyond V1 and the magno-heavy end when fitted to human choices (Extended Data Fig. 5). Nor does the account need the split: in 38 frozen ImageNet networks with none, a reciprocal gain on the network's own shape- and texture-predictive subspaces moves the rate by 0.08–0.29 on exemplars those subspaces never saw, within five points of top-1, re-ordering 19% of network pairs, twice the resampling noise, and in our single trunks the texture-leaning units again carry extrastriate cortex relative to V1 (Fig. 6); the split amplifies the dissociation (0.50 against 0.32 on the same network, 0.25 on unseen exemplars), it does not create it.

How to report shape bias. A shape-bias claim should carry three numbers, each answering a different question: the rate at the trained readout (an operating point); the span the same frozen features reach under a reciprocal gain on their own shape- and texture-predictive subspaces within an accuracy budget (how much of the rate the readout controls); and a probe's shape-minus-texture accuracy (what the representation holds) — the last two scored on exemplars the subspaces and probe were not fitted on, since the standard cue-conflict set reuses 48 texture images across 1,200 stimuli and exemplar-shared estimates inflate both. Across 38 ImageNet networks the three rank the zoo differently ($\tau = 0.63$ and 0.43), and the Stylized-ImageNet network shows what their agreement looks like when a bias really is representational — seen from the readout, the difference is not how far the gain moves it but which evidence the gain moves (Results 2.4); error consistency does not make a fourth: read at each network's own readout it is largely a re-expression of the rate, because κ is bounded by the mismatch with the observers' 0.958, and at a matched rate it separates neither the zoo nor the Stylized-ImageNet network from it, so where human trial data exist it should be read at a matched rate (Extended Data Table 17). None of these numbers alone licenses a claim about representations. One command in the repository prints all three, with the

exemplar-shared values beside them, for any torchvision network (Methods 4.8).

What a developmental constraint does. The intuitive account of low-pass developmental input is informational — coarse input shields the learner from texture, so it learns shape — and our results do not support it: the transform removes bandwidth rather than adding structure, most of the shape bias it produces is present before training, and what the diet decides is whether learning builds on that bias or destroys it (on held-out exemplars, training drives the luminance stream from +0.09 to −0.12 without it and from +0.21 to +0.34 with it, and a second instrument replicates the side of parity rather than a level). This is a constraint on the solution learning converges to, not a better input representation, consistent with the thesis's observation that the transformed magno stream becomes less predictive of the category as the parvo stream becomes more so.

What comes apart, and what the diet costs. Shape bias, robustness, dimensionality and human-likeness are discussed as a package; over identical inputs this system separates them three times, and a readout account says why it should: none of the three is a property of a readout. The diet is also expensive — less accurate (0.556 against 0.701), less stable late in training, less robust on four of five untrained-on corruptions, a worse predictor of BOLD — so nothing here recommends it as a training method; its interest is as a model of development, and the thesis's central mechanistic reading, that divergence and shape bias are bought by degrading the magno stream's evidence, survives in every instrument.

Limitations. The load-bearing cortical result rests on eight subjects from one dataset; its magnitude is read as sign and ordering only, and it is the survivor of seven regional hypotheses, disclosed above. Beyond hV4 the index saturates near its bound in every atlas, so the extension establishes that the effect is hierarchy-wide but cannot order extrastriate regions against one another. The mapping between our chromatic stream and texture-selective cortex is by explanatory role, not content: nothing here asserts that the ventral stream is chromatic, the prediction that the chromatic stream should win in colour-biased cortex fails (pooled VO +0.005, bounded within $\pm$0.017), and substituting the koniocellular axis makes it worse (Supplementary Note S2). The `parvo_highsf` condition varies architecture alongside the front end. The diet is a fixed 48-month front end rather than a trajectory, though a schedule-trained checkpoint behaves like it on every instrument we tried. Whether a two-stream architecture specialises on its own is untestable here, because the trunks are bit-identical at initialisation. Human comparisons are against ten published observers and of aggregate rate; the readout weight is a hyperparameter that nothing selects. The single-trunk subspaces are linear discriminants of one cue-conflict set, and their spans are lower bounds on what a readout can reach, not measurements of the best one. That set reuses its exemplars, so spans and probes are reported with exemplars held out (about 60% and two thirds of their exemplar-shared values in the zoo); the unit rule's η^2 split, fitted on all images, has not been recalibrated this way. The generality test was pre-registered three times and all three ledgers are complete (Methods 4.8; outcomes in the plan files): of the unit-level plan's five predictions, the behavioural thresholds were met by direction but not by magnitude, the stream-recovery check was met, the V1-relative cortical contrast held in our two trunks and reversed in ImageNet ResNet-50, and the mixing-weight ordering held in one network of three; of the subspace plan's, the magnitude threshold was met in all six ImageNet networks but in one of our three trunks, the stream-recovery check failed (63–70% of the shape basis on stream 1, against 90%), the subspace span exceeded the unit-level span in every network (Extended Data Tables 13–16), and of its three cortical predictions only the ConvNeXt-B contrast held, the two discriminant subspaces carrying almost no unique cortical variance; of the zoo plan's four predictions three were met and the Stylized-ImageNet span fell short of its threshold, read, as pre-stated, as a bias held in the representation rather than in the readout — a reading of the span that a fifth plan does not support (Results 2.4). A fourth plan, committed before its own analysis ran, predicted that agreement with

the observers would separate representations at a matched rate; four of its six predictions failed, for the reason given in Results 2.5. A fifth, on what sets the span, met two of its seven predictions; which evidence the gain moves is a post hoc reading of it (Results 2.4). A sixth, re-estimating spans and probes with exemplars held out, met 15 of 17 predictions; the two failures concern the probe, whose ImageNet networks stay texture-leaning where we predicted they would not (`docs/PLAN_EXEMPLAR_2026-09-21.md`). Of the main study's four pre-registered predictions, two failed and are reported as failures; claims deliberately not made are listed in Supplementary Note S1.

4. Methods

4.1 Front end

RGB images are converted to the DKL cone-opponent space through RGB→XYZ (D65)→LMS (Hunt–Pointer–Estévez)→DKL at 256 × 256 px, each channel normalised to $[0, 1]$ with the training set's global range and quantised to 8 bits exactly as the training data were stored (Derrington et al., 1984). Stream 1 receives the luminance channel; under the diet it passes through an isotropic Gaussian blur ($\sigma = 4$ px, kernel 33, reflect padding) and the scale-free contrast-sensitivity cut of Lu et al. (2026) evaluated at a fixed age of 48 months (`cs_logspan_start` $= 5 \times 10^{-3}$, decomposition mode "keep"). Stream 2 receives the L−M channel untransformed. The four conditions are: `dkl_bcs` (diet: luminance + blur + CS | L−M); `dkl_base` (no diet: luminance | L−M); `dkl_bcs_swap` (luminance | L−M + blur + CS); `dkl_bcs_parvo_highsf` (luminance + blur + CS | L−M stacked with untransformed luminance, two channels). The last differs architecturally in its first-layer fan-in, and that difference is measured rather than argued away (Extended Data Fig. 2; Results 2.5). Evaluation stimuli, including NSD images and cue-conflict images, pass through the identical front end.

4.2 Architecture and training

Two ResNet-18 trunks (He et al., 2016), one per stream, initialised bit-identically, with a joint linear classification head over the weighted sum of their penultimate features (w = 0.5; cross-stream mixing off). MiniEcoset at 256 px, 112 classes (Thorat & Kietzmann, 2024). 150 epochs, batch 256, Adam (lr 1×10^{-4}, $\beta_1 = 0.9$; Kingma & Ba, 2017), cross-entropy, with the augmentation set of Lu et al. (2026) applied on GPU (horizontal flip, rotation $\pm 15°$, colour jitter, random equalise, perspective, sharpness; random grayscale for RGB inputs only). Three seeds per condition; weights saved every 25 epochs. Each run used one NVIDIA RTX A5000 (24 GB); the 21 training runs behind this paper took 578 GPU-hours in total (19.5–51.4 h each, from the runs' own timing logs). All encoding, statistical and plotting work ran on CPU. Over the whole project the scheduler allocated 29,000 core-hours and 605 GPU-hours to 1,222 jobs between 2026-08-25 and 2026-09-17: 4,700 core-hours and 584 GPU-hours to the 21 training runs, 20 GPU-hours to the accuracy-matched retraining, 14,800 core-hours to the two-band encoding fits and their controls, 4,600 to the subspace-rule cortex fits, 2,000 to the single-trunk generality analysis, 980 to the cortex-optimal readout weight, 55 to the model zoo and 1,800 to smoke tests, caches, figures, the robustness runs and the verifier (`tmp_analysis/compute_accounting.py` reads the scheduler's own accounting and writes `results/figures_v2/compute_accounting.json`). Estimated with the Green Algorithms method (Lannelongue et al., 2021) at 12 W per allocated core, 0.3725 W per allocated gigabyte, the A5000's 230 W, a power usage effectiveness of 1.67 and Germany's grid intensity of 0.35 kg CO_2e per kWh, that is 930 kWh and 325 kg CO_2e. It is an upper bound: allocated cores and memory are counted as fully used for the whole of each job's wall-clock time. Not included are the pretraining of the public networks analysed in Results 2.4 and the acquisition of the two human datasets. Single-stream controls use one ResNet-18 or one ResNet-34 trunk fed the diet's luminance input

with the identical schedule. The accuracy-matched control retrained `dkl_base` for 20 epochs with per-epoch checkpointing at three seeds and selected by rule, per seed, the epoch whose validation top-1 is nearest the diet arm's best-checkpoint accuracy of 0.556 (epochs 10, 10 and 11: 0.548, 0.546 and 0.554). No hyperparameter search was run for any arm: every condition uses the training recipe of Lu et al. (2026) unchanged, and the only selection is the checkpoint policy stated above.

4.3 Behaviour and probe

Cue-conflict shape bias is computed on the Geirhos et al. (2018a) style-transfer set restricted to the 16 categories present in MiniEcoset and to images whose shape and texture categories differ (1,200 images). Logits are restricted to the 16 columns; the argmax is scored as a shape or texture response; images decided as neither are excluded from the denominator and their count is reported. Bootstrap 95% confidence intervals are over images. A second, forced two-way estimator keeps every image and scores whether the logit of the image's shape category exceeds that of its texture category; it is reported beside the primary estimator for every readout (Extended Data Table 10) and is computed from the same stored logits. The readout is full = classifier($w \cdot y_1 + (1-w) \cdot y_2$), magno_only = classifier($w \cdot y_1$), parvo_only = classifier($(1-w) \cdot y_2$), all through the same classifier and bias term; the w sweep evaluates full at 21 values of w on stored features. The refitted heads (Linear(512, 112) and Linear(512, 512)→ReLU→Linear(512, 112)) were trained on $w \cdot F_1 + (1-w) \cdot F_2$ of MiniEcoset training features at each w with an identical budget (Adam, lr 10^{-3}, batch 1024, 20 epochs) and the same initialisation seed at every w. Human observers are the ten of the same release (1,280 trials each), scored by the identical rule; our recomputation reproduces the published 0.958 and the published AlexNet, GoogLeNet and VGG-16 values. Error consistency is Cohen's κ (Cohen, 1960) on the binary shape/texture choice over images both parties decided, with expected agreement $p_m \cdot p_h + (1-p_m)(1-p_h)$, reported as a percentage of the human–human κ over all 45 observer pairs (Geirhos et al., 2020). For the frozen networks of Methods 4.8 the same κ was computed at every gain w (`src/analysis/generality/kappa_sweep.py`), together with the largest κ attainable at the two marginals, $\kappa_{max} = (1 - |p_m - p_h| - p_e)/(1 - p_e)$, against which agreement is also reported as κ/κ_{max}, as a percentage of the same ratio for human pairs (0.353). Per-image correspondence uses the observers' per-image shape rate and cue-resolution rate (split-half reliabilities r = .658 and .794 after category removal), the models' shape-minus-texture logit margin, w chosen on half the images and scored on the held-out half, as a percentage of each target's own ceiling. The linear probe reduces each stream's features to 256 PCA components and fits multinomial logistic regression (C = 1.0) under 5-fold stratified cross-validation to shape and texture labels separately; bias is shape accuracy minus texture accuracy on the same folds. Because the 1,200 images are built from 160 object and 48 texture exemplars, stratified folds leave every test image's exemplars in the training folds, so the primary probe is exemplar-balanced: three exemplars per category for both labels (three of each shape category's ten, drawn at random; the texture set has three), texture images subsampled to the same per-category count, scored leave one exemplar out with PCA ($\min(256, n_{train} - 1)$ components) and scaling fitted inside each fold, mean of 20 draws with the 5–95% interval over draws (`src/analysis/generality/exemplar.py`; `docs/PLAN_EXEMPLAR_2026-09-21.md`). The stratified-fold probe is reported beside it as exemplar-shared. With about 250 training images per fold the balanced probe is sensitive to preprocessing: PCA fitted on all images and then standardised compresses every bias toward zero (median +0.03 over the 38 ImageNet networks), whereas fewer components or no standardisation agree in sign with the primary estimator (medians −0.14 and −0.39; $\rho = 0.94$ with the latter over the 42 zoo networks).

4.4 Effective dimensionality

Participation ratio $(\Sigma\lambda)^2/\Sigma\lambda^2$ of the stream feature matrix on 1,000 NSD images per subject (Gao et al., 2017), computed Gram-side, reported as the magno/parvo ratio with parity at 1.0. Subjects are the replication unit within a seed; seeds are reported separately and never pooled. A learning-free reference is the same ratio computed on a Gabor oriented-energy bank applied to each stream's own transformed input.

4.5 Encoding

NSD (Allen et al., 2022) `betas_fithrf_GLMdenoise_RR` single-trial betas, converted to percent signal change and z-scored per session per voxel, `nsdgeneral` mask, eight subjects. Splits are by 73k image id with the shared-515 images held out. Stream features for each subject's stimuli are reduced by PCA to 1,000 components per stream (fit on training images only) and entered as two bands into a banded ridge regression (himalaya; Dupré la Tour et al., 2022) with 3-fold GroupKFold by image id, per-voxel alphas and per-voxel band weights. Unique variance U_m (U_p) is the drop in cross-validated r^2 when the magno (parvo) band is removed. Noise ceilings are NSD's `ncsnr`; voxels below 0.1 are excluded and any ROI cell with fewer than 20 surviving voxels is reported as missing. ROIs are the `prf-visualrois` labels of the Wang et al. (2015) atlas as distributed with NSD (V1v, V1d, V2v, V2d, V3v, V3d, hV4), with hierarchical rank [1, 1, 2, 2, 3, 3, 4]. The canonical-correlation spectrum between the two bands runs from ≈0 to 0.93, so each band retains directions the other cannot predict and unique variance is identifiable; a recovery simulation with the fitted feature geometry and a known across-ROI split returns the true ordering exactly ($\rho = +1.000$ in 8/8 subjects, both arms), which is why the gradient is reported as sign and ordering rather than as calibrated magnitude. The swap and `parvo_highsf` arms are fitted for all three seeds, as the diet and no-diet arms are, and their behavioural and probe measures use the same three seeds (Extended Data Table 4). Both are degenerate by construction and in every seed — the swapped network leaves the parvo band a median unique variance 0.005–0.010 of the magno band's, the parvo + high-SF network leaves the magno band 0.003–0.005 of the parvo band's — so they enter no cortical claim beyond the sign and size of their unique variances (Results 2.3; Extended Data Tables 1 and 11). The fits of this section were originally submitted at about 160 GB of memory per subject, which is not the size of the data: the peak is the group-ridge random search, which forms an (n_alphas × n_test × n_train) array per gamma sample and fold, about 30 GB in float64 with all 21 alphas at once. Batching that search seven alphas at a time changes no estimate — a refit of the diet seed-0 subject-01 fit reproduces the stored one at voxelwise $r = 1.00000$ on both unique variances, with the largest per-ROI index difference below 10^{-4} — and the same fits then peak at 32.6 GB (`slurm/encode_equiv_probe.sh`, `slurm/encode_cc_controls.sh`). Eccentricity bands are NSD's `prf-eccrois` labels (0–0.5°, 0.5–1°, 1–2°, 2–4°, > 4°); continuous pRF eccentricity, size and R^2 are NSD's population-receptive-field estimates in the same functional space, and the within-subject regressions of Extended Data Table 9 use z-scored log(eccentricity + 0.5) and log(size) over voxels with pRF $R^2 > 5\%$. The per-voxel predicted shape-bias index is obtained by pushing the cue-conflict set through each fitted encoding model. For Extended Data Fig. 4 and Table 11 the same voxelwise fits were relabelled, without refitting, with the full Kastner2015 atlas as distributed with NSD (adding VO1/2, PHC1/2, LO1/2, TO1/2, V3A/B and IPS0–2; IPS3–5, SPL1 and FEF have no voxels inside `nsdgeneral`), with NSD's `streams` parcellation and with the fLoc face, place, body and word ROIs; regional means are unweighted means of the ROI medians, and region contrasts are tested per subject with the exact Wilcoxon test. For the cortex-optimal readout weight (Extended Data Fig. 5; Extended Data Table 12) the classifier's own input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$, where y_1 and y_2 are the 512-dimensional stream outputs the classifier mixes, was the design of a single-band ridge model of every `nsdgeneral` voxel for $w \in \{0, 0.05, \ldots, 1\}$,

with the same split, grouped folds and alpha grid as the banded model and columns z-scored on the training trials; because $z(w)$ sums the streams dimension by dimension the ridge cannot undo the mixture, so w is a constraint on the design rather than a re-parameterisation of the two-band model. w*_cortex is the w that maximises the median test r^2 over an ROI's reliable voxels (seeds averaged per subject); the 1,024-dimensional concatenation $[y_1, y_2]$ is fitted as a reference and not as a bound, because although every mixture is a linear map of it the two models are regularised separately on a shared alpha grid, and the best mixture exceeds the fitted concatenation in 36 of the 112 subject × ROI × arm cells by a median 0.0019 r^2 — more than the 0.0016 the sweep itself buys from w = ½ to w, *so w*_cortex is read as the location of an optimum and never as an r^2 difference (Extended Data Table 12); and the behaviour-optimal weight is the trial-fit w of Fig. 4 (Methods 4.4) for each of the ten observers. The koniocellular control substitutes S−(L+M) for L−M in stream 2 of the untrained network at each of three seeds, each against the L−M network of its own seed, contrast-matched to L−M per image, with the magno band bit-identical across arms (gate $\max|\Delta r^2_magno| < 10^{-4}$, passed at 0.000).

4.6 Out-of-distribution robustness

Corruptions are applied to RGB before the front end and scored in the native 112-class label space. Five families were declared untrained-on before the run (uniform noise, high-pass, phase scrambling, rotation, false colour) and two as trained-in positive controls (low-pass, contrast). Reported as the area under the relative-accuracy curve across severities, at each seed's best saved checkpoint. Three implementation differences from Lu et al. (2026) are declared: our front end is evaluated at a fixed 48 months rather than scheduled (the one we can rule out, since a schedule-trained arm behaves like the static diet); our diet is a transform on one stream's input of a two-stream network rather than an augmentation on a single trunk; and we train on MiniEcoset and score corruptions in that label space, applied before the front end.

4.7 Statistics

Within-subject comparisons use the paired exact Wilcoxon signed-rank test, n = 8 (exact two-sided floor p = .0078); seed-level comparisons use the exact Mann–Whitney U, 3 v 3 (floor p = 0.100, reached only on complete separation), and seeds are never pooled with subjects. Every cortical contrast is computed both on the raw per-voxel difference U_m − U_p and on the normalised index (U_m − U_p)/(U_m + U_p), each summarised by its median over reliable voxels per ROI, and a claim is made only where the two agree; disagreements are reported as unresolved. Equivalence uses two one-sided tests with bounds fixed before any interval was computed (Schuirmann, 1987; Lakens, 2017). No multiple-comparison correction is claimed: with 28 p-values and an exact floor of .0078, Holm–Bonferroni over any family larger than six cannot yield a rejection, so claims rest on complete separation, agreement of both statistics, replication across seeds and effect size (dz = −6.60 for the diet gradient against −1.09 without the diet). Behavioural and robustness results use each seed's best saved checkpoint unless stated; cortical fits use the final checkpoint. Four predictions — the front-end 2×2, the inverted-front-end prediction, the corruption families and the error-consistency hypothesis — were written into a version-controlled plan and committed before the corresponding analysis was run (internal pre-specification, not registry-based); two failed and are reported as failures.

4.8 Readout re-weighting in networks without a stream split

Pre-specified three times — in `docs/PLAN_GENERALITY_2026-09-14.md` (the unit-level rule; five directional predictions P1–P5), `docs/PLAN_SUBSPACE_2026-09-15.md` (the subspace rule; predic-

tions P-A, P-A′, P-B, P-C and P-D) and `docs/PLAN_ZOO_2026-09-15.md` (the model zoo; P-Z1–P-Z4) — each committed before its analysis was run, with every outcome reported (Results 2.4; Extended Data Tables 13–16). Both rules act on a frozen network with logits = W·y + b, y its D-dimensional penultimate features, and neither retrains anything. Subspace rule: on a fitting set of cue-conflict images y is centred and reduced by PCA to min(256, D) components; a 16-class linear discriminant analysis on the shape labels (scikit-learn, eigen solver, Ledoit–Wolf shrinkage) gives 15 shape-discriminant directions and one on the texture labels 15 texture-discriminant directions; mapped back to unit space and orthonormalised, the two bases are made mutually orthogonal by symmetric (Löwdin) orthogonalisation, which treats shape and texture identically (sequential Gram–Schmidt in either order is the reported sensitivity; Extended Data Table 13), giving B_S, B_T and an orthonormal basis B_R of the residual, with projectors P_S, P_T and P_R. Fixed-head gain sweep: logits(w) = W·(P_R y + 2w·P_S y + 2(1 − w)·P_T y) + b, w ∈ {0, 0.05, …, 1}, so that w = 0.5 is the trained network exactly; the bases are fitted on one half of the 1,200 images, stratified by shape × texture pair, and the rate scored on the other, over 10 random splits in both directions. These halves share exemplars (98% of scored images share their object exemplar with the fitting half and all of them their texture exemplar), so every span is also computed on exemplar-disjoint folds (`docs/PLAN_EXEMPLAR_2026-09-21.md`, predictions P-X1–P-X16, committed before its code existed): in each of ten repetitions every texture category's three exemplars and every shape category's ten object exemplars are assigned at random to three folds, the bases are fitted on the images whose object and texture exemplars both lie outside fold k and the rate is scored on the images whose exemplars both lie inside it, and the three folds' scored images (about 400) form one curve; holding out only the texture or only the object exemplars gives the two components (median spans 0.195 and 0.293 over the 38 ImageNet networks, against 0.19 for both and 0.30 for neither). As for the exemplar-shared estimator, the reported span is max − min of the mean curve over the gated w; the pre-registered predictions were scored on the mean of the repetitions' own spans (median 0.189 against 0.187). The ranking's noise floor is the fraction of pairs reversing between two independent bootstrap resamples of the 1,200 images at the trained readouts (2,000 pairs of resamples, shared across networks). Unit-level rule, the special case in which the two subspaces are spanned by disjoint sets of units: each unit j receives d_j = η^2_shape(j) − η^2_texture(j), the one-way fraction of its variance over the images lying between the 16 shape or the 16 texture categories; the median split of d gives the shape-leaning half S and the texture-leaning half T (equal-sized, as the two streams are), and logits(w) = W·(g(w) ⊙ y) + b with g_j = 2w for j ∈ S and 2(1 − w) for j ∈ T, on the same image splits. At a referee's request, and not pre-specified, this rule was also run with S the absolute shape-leaning set (units with d_j > 0) or the top decile of units by d_j, T the remainder (Extended Data Table 14, last two columns; `slurm/generality_rules.sh`): both leave S small — 2–89 units under the absolute rule (a single unit, hence undefined, for ConvNeXt-S) and a tenth of the units (51–409) under the decile rule — so that w → 1 silences at least nine tenths of the representation; the ungated spans are 0.11–0.40 with positive mean ρ (0.29–0.99) but not every split is monotone in our trunks, and within 5 points of the trained top-1 — a gate that closes at w = 0.72–0.95 in every case but ConvNeXt-B under the decile rule, where the whole sweep passes — the reachable span is 0.02–0.21. Estimators: our networks by Methods 4.3; torchvision ImageNet networks (ResNet-50, ResNet-152, VGG-16, ViT-B/16, ConvNeXt-S/B, IMAGENET1K_V1 weights, model-vs-human preprocessing) by the Geirhos 16-class rule, mean softmax probability over each category's ImageNet classes and argmax over the 16, which assigns every image to one of the 16 categories; as in Methods 4.3, the rate is computed over the images assigned to their shape or their texture category (814 of 1,200 for ResNet-50; per-network counts in Extended Data Table 16). Accuracy gate: top-1 of the gained features, with the bases or the split defined on all 1,200 images, on held-out natural images — the MiniEcoset validation split for our networks and a stratified 10-per-class subset of the ILSVRC-2012 validation set for the ImageNet networks; the gated span is max −

min of the seed-mean held-out curve over the w at which top-1 lies within 5 points of the trained value. Model zoo (Extended Data Fig. 6, Extended Data Table 16): the same pipeline on 42 networks — 38 torchvision `IMAGENET1K_V1` classifiers, ResNet-50 under torchvision's `IMAGENET1K_V2` recipe, and the three ResNet-50s of Geirhos et al. (2019) trained on Stylized-ImageNet, on Stylized-ImageNet plus ImageNet, and on both then fine-tuned on ImageNet — with y the input of each network's last linear layer, captured on the network's own forward pass and checked against its logits; Resize 256 / CenterCrop 224 except Inception-v3 (342/299), EfficientNet-B3 (320/300) and EfficientNet-V2-S (384/384), which were not trained at 224. The probe bias is the linear probe of Methods 4.3 on the same 1,200 features, predicting the shape and the texture class in turn, under both of its estimators. Ranking statistics over the 38 ImageNet-trained networks are Kendall τ between rankings and the fraction of the 703 pairs whose order reverses, with no p-values, the networks being a fixed set rather than a sample. `python -m src.analysis.generality.report --model tv:<arch>` prints the three numbers for any torchvision classifier. Refit: because a refitted linear head can undo a per-direction gain, the refit test mixes the two components lossily, as the two-stream model does, and passes the rest of the representation unchanged — $z(w) = [w{\cdot}s + (1 - w){\cdot}t, B_R^{T}y]$ with $s = B_S^{T}y$ and $t = B_T^{T}y$ paired by discriminant rank (under the unit-level rule $z(w) = w{\cdot}y_S + (1 - w){\cdot}y_T$, the halves paired by rank of d) — and refits the linear and two-layer heads of Methods 4.3 at each w, with a head refitted on the unmixed y as the accuracy ceiling. Cortex: under the unit-level rule S and T (split on all 1,200 images) are the two bands of the banded ridge model of Methods 4.5 on the penultimate units of the single-trunk networks, giving U_S, U_T and the index $(U_S - U_T)/(U_S + U_T)$ per voxel; the cortex-optimal w of the lossy sum follows the protocol of Extended Data Fig. 5 on an 11-point grid (0, 0.1, ..., 1). Under the subspace rule the bands are $s = B_S^{T}y$, $t = B_T^{T}y$ and the residual $r = B_R^{T}y$ (15, 15 and $D - 30$ dimensions; bases fitted on all 1,200 images), with $U_S = r^2_\{STR\} - r^2_\{TR\}$ and $U_T = r^2_\{STR\} - r^2_\{SR\}$ from the three-band model and the two two-band models (random search with 30 and 20 samples), and the w sweep on the 15-dimensional lossy sum $w{\cdot}s + (1 - w){\cdot}t$ alone; the five families are `rgb_tsn` and `dkl_bcs_single34` (three seeds each, averaged per subject), ResNet-50, and ViT-B/16 and ConvNeXt-B, whose NSD features were extracted for this test (predictions P-C1–P-C3; Extended Data Table 15, upper block). The `rgb_tsn` network — two trunks that receive the identical three-channel image and remain bit-identical — is one ResNet-18 on the untransformed image and is used as the no-diet single trunk. All of our networks enter this analysis at their final checkpoints, the policy of the cortical fits (Methods 4.7).

Evidence decomposition (post hoc; `docs/PLAN_SPAN_2026-09-18.md`, predictions S-1 to S-7 written before the analysis; Extended Data Table 18). For image i with shape category s and texture category t the Geirhos rule's margin is $\Delta_i = \log p_s - \log p_t$, p_c the mean softmax probability over c's ImageNet classes, and its gradient with respect to the logits is $u_i = q_s - q_t$, q_c the softmax restricted to c's classes. Because $\text{logits}(w) = \text{logits}(½) + (2w - 1){\cdot}W(P_S - P_T)y$ exactly, $e_S,i = u_i{\cdot}W\ P_S\ y_i$ and $e_T,i = u_i{\cdot}W\ P_T\ y_i$ give the first-order margin $\Delta_i + (2w - 1)(e_S,i - e_T,i)$ (model M0) and, with $z = \text{mean}\ \Delta\ /\ \text{sd}\ \Delta$ and $\lambda = \text{mean}(e_S - e_T)\ /\ \text{sd}\ \Delta$ over the 1,200 images, the two-number forced rate $\Phi(z + (2w - 1)\lambda)$ (M1); the shape term is mean e_S, the texture term $-$mean e_T and the texture share the second over their sum. All use the bases fitted on all 1,200 images and are compared with the forced-choice span in sample. Null: ten pairs of mutually orthogonal random 15-dimensional subspaces (seeds 0–9) inside each network's 256-dimensional PCA space of the cue-conflict features, under the same gain, estimator and gate (`src/analysis/generality/span_model.py`, `slurm/zoo_span.sh`).

4.9 Use of large language models

No large language model is part of the scientific pipeline: no result, number, figure or statistical decision in this paper is produced by one. A large language model (Anthropic Claude, through the Claude Code command-line interface) was used as a coding and drafting assistant under the authors' direction — writing and refactoring analysis, figure and job scripts, drafting and revising manuscript text, and producing internal referee simulations (`docs/NMI_REVIEW_*.md`) used only to prioritise revisions. Its use is recorded commit by commit in the repository's `Co-Authored-By` trailers. Every pre-registration in `docs/PLAN_*.md` was written and committed before the corresponding analysis ran, every reported number is regenerated from committed result files by the scripts named in each caption and checked by `tmp_analysis/verify_new_claims.py`, and the authors take full responsibility for all content, including any text or code an assistant drafted.

Data availability

The Natural Scenes Dataset (Allen et al., 2022) is available at naturalscenesdataset.org under its own data-use agreement, which requires registration; we used the `betas_fithrf_GLMdenoise_RR` (b3) preparation, the `nsdgeneral` mask, the `prf-visualrois` and `prf-eccrois` labels and the pRF estimates as distributed with NSD. The Geirhos cue-conflict stimuli and the ten observers' trial-level responses are distributed at github.com/rgeirhos/texture-vs-shape (directory `raw-data/style-transfer-512-nomask-experiment/`) under CC BY 4.0; the same experiment is mirrored at github.com/bethgelab/model-vs-human, and the two mirrors were parsed by two independently written scripts that agree to four decimals on the human reference. MiniEcoset is at osf.io/msna2 (Thorat & Kietzmann, 2024). Derived result tables and fitted encoding-model summaries sufficient to reproduce every figure and Extended Data table are in the code repository, and will be deposited with a DOI on acceptance.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article; a draft is `docs/REPORTING_SUMMARY.md` in the repository.

Code availability

Training, front-end, encoding, behavioural and analysis code is at github.com/LuZeking/Magno-Parvo-Konio-Net (branch `mpk-rewrite`, commit as stamped on this manuscript's title page), under the BSD-3-Clause licence, with a pinned environment manifest (`requirements.txt`, Python 3.11) and the committed result files behind every figure; the figures of this paper regenerate with the scripts in `tmp_analysis/figs_v2/` and the manuscript's numbers, including those of Fig. 6 and Extended Data Tables 13–16, are locked by `tmp_analysis/verify_new_claims.py`, which reads the committed result files (for the generality analysis, `results/figures_v2/generality_behaviour.json` and `generality_cortex.json`). The per-subject encoding outputs that Fig. 3 reads are large and are deposited alongside the derived tables. The 21 trained networks (final and best saved checkpoints, $\approx$ 17 GB) that Figs. 4 and 6 and Extended Data Tables 10–15 read are not in the repository; they will be deposited in the same DOI archive at acceptance and are available to referees through the private link. A private link is available to referees on request; a DOI-archived release accompanies acceptance.

Ethics and human participants

This study collected no new human data. Both human datasets are existing, de-identified public releases analysed under their own terms. NSD was collected under the approval of the University of Minnesota institutional review board with written informed consent from all eight participants (Allen et al., 2022); the behavioural data were collected under the approval described in Geirhos et al. (2018a). Secondary analysis of two de-identified public datasets required no additional approval under our institution's policy, and no participant-identifying information was accessed.

Sex and demographics. Participant sex was not under our control and is not a design factor; no analysis is disaggregated by sex, and at eight participants and ten observers none could be, because the exact tests used already sit at their resolution floor on the full samples. We do not restate sex or age: the Geirhos per-observer files carry only trial-level fields, so no per-observer demographic exists in the data we analysed, and for NSD we have not verified values against the released subject-information file and therefore point to the NSD release rather than transcribe from prose.

Constraints on generality. The neuroimaging sample is eight adults with normal or corrected-to-normal vision in the United States; the behavioural reference is ten adults tested in a German university laboratory. Both are small, Western, convenience-recruited adult samples. Claims about "human observers" should be read as claims about these samples; nothing here establishes cross-cultural, developmental or clinical generality, and the developmental motivation of the model's front end is a motivation for the model, not a claim about any developmental population we measured.

Author contributions

[To be completed at submission per Nature-portfolio policy: conceptualisation, model and training code, encoding analyses, behavioural analyses, figures, writing, supervision. The originating thesis (Manocchio, 2026) supplied the architecture, the front-end conditions and the first behavioural and RSA analyses; the seed replications, encoding models, controls and manuscript were developed subsequently.]

Competing interests

The authors declare no competing interests.

Extended Data tables

Extended Data Table 1 | The cortical gradient hV4 − mean(V1v, V1d), per seed and statistic. "normalised" is the per-voxel index (U_m − U_p)/(U_m + U_p), "raw" the per-voxel difference U_m − U_p, each summarised by its median over reliable voxels per ROI. Mean over 8 subjects of the per-subject contrast; "sep." is the number of subjects in which the diet arm is more negative than the no-diet arm, and for the rows with no paired comparison the number in which the contrast is negative; p is the exact Wilcoxon, paired for the diet against no-diet rows and one-sample otherwise (floor .0078). The n = 7 column drops subject 01 (Results 2.3). The swap and parvo_highsf arms are degenerate by construction, in every seed: the swapped network leaves the parvo band a median unique variance 0.005–0.010 of the magno band's, and the parvo + high-SF network leaves the magno band 0.003–0.005 of the parvo band's, so their gradients describe one surviving band and are not comparable with the two-band arms. Sources: `results/encoding_`

`cc/` (diet, no diet, thesis) and `results/encoding/` (untrained, swap, parvo_highsf).

statistic	seed	diet	no diet (n = 8)	no diet (n = 7)	sep.	p
normalised	0	−0.833	−0.078	−0.054	8/8	0.0078
	1	−0.785	−0.096	−0.084	8/8	0.0078
	2	−0.745	−0.049	−0.031	8/8	0.0078
raw	0	−0.166	−0.027	−0.021	8/8	0.0078
	1	−0.145	−0.031	−0.030	8/8	0.0078
	2	−0.153	−0.023	−0.021	8/8	0.0078
untrained, diet front end	—	−0.544			8/8 < 0	0.0078
untrained, no-diet front end	—		−0.479		8/8 < 0	0.0078
swap (degenerate: U_p ≈ 0)	0	−0.006			5/8 < 0	0.4609
	1	−0.034			7/8 < 0	0.0234
	2	−0.014			6/8 < 0	0.1953
parvo_highsf (degenerate: U_m ≈ 0)	0	−0.143			8/8 < 0	0.0078
	1	−0.082			8/8 < 0	0.0078
	2	−0.108			8/8 < 0	0.0078
thesis checkpoint (progressive schedule)	—	−0.793				

Extended Data Table 2 | Monotonicity of the profile and the exact permutation null. Spearman ρ between hierarchical rank [1,1,2,2,3,3,4] and stream preference, mean over subjects; the null enumerates all 630 distinct labellings of the seven ROIs (one-sided in the pre-stated direction, floor $1/630 = .0016$). "per-voxel index" is the manuscript's statistic (Fig. 3c; Results 2.3): the median over reliable voxels of $(U_m - U_p)/(U_m + U_p)$ per ROI, as in Extended Data Table 1; "ratio of medians" is $\text{median}(U_m - U_p)/\text{median}(U_m + U_p)$ per ROI, the statistic of an earlier draft, kept for comparison. Source: `results/hierarchy_permutation_null.json` (which labels the two `raw` and `norm`).

arm	seed	statistic	observed mean ρ	exact p	most extreme of 630	null 2.5th pct
diet	0	per-voxel index	-0.964	0.0016	yes	-0.743
diet	0	ratio of medians	-0.964	0.0016	yes	-0.754
diet	1	per-voxel index	-0.973	0.0016	yes	-0.743
diet	1	ratio of medians	-0.973	0.0016	yes	-0.750
diet	2	per-voxel index	-0.966	0.0016	yes	-0.756

arm	seed	statistic	observed mean ρ	exact p	most extreme of 630	null 2.5th pct
diet	2	ratio of medians	-0.966	0.0016	yes	-0.748
no diet	0	per-voxel index	-0.571	0.0206	no	-0.559
no diet	0	ratio of medians	-0.502	0.0286	no	-0.502
no diet	1	per-voxel index	-0.661	0.0190	no	-0.594
no diet	1	ratio of medians	-0.606	0.0190	no	-0.546
no diet	2	per-voxel index	-0.463	0.0921	no	-0.594
no diet	2	ratio of medians	-0.580	0.0365	no	-0.616

Extended Data Table 3 | In-silico cue conflict through the fitted encoding models, per seed. Median over reliable voxels of the per-voxel predicted shape-bias index for each stream, mean over 8 subjects; the contrast is magno − parvo, paired within subject against the no-diet arm of the same seed. Source: `results/encoding_cc/`.

seed	arm	magno stream	parvo stream	magno − parvo	diet − no diet	sep.	p
0	diet	+0.791	−0.743	+1.534	+1.424	8/8	0.0078
0	no diet	−0.638	−0.748	+0.110			
1	diet	+0.796	−0.770	+1.566	+1.468	8/8	0.0078
1	no diet	−0.601	−0.699	+0.098			
2	diet	+0.799	−0.759	+1.558	+1.474	8/8	0.0078
2	no diet	−0.613	−0.696	+0.083			

Extended Data Table 4 | The carrier × transform 2×2 on the probe and on behaviour. Mean over training seeds with the seed range in brackets; "decided" is the mean number of the 1,200 cue-conflict images the readout classifies as the shape or the texture category (the shape-bias denominator). Cells are read out on stream 2 in three cases and on the diet's magno stream in the fourth. Source: `results/figures/fig1_data.npz` (from `results/shape_texture/`, `results/cue_conflict/`). "probe bias, exemplars held out" is the exemplar-balanced probe of Methods 4.3, scored on object and texture exemplars the probe was not trained on (three exemplars per category for both labels, leave one exemplar out, PCA inside the folds, mean of 20 draws; `docs/PLAN_EXEMPLAR_2026-09-21.md`, `results/figures_v2/exemplar_streams.json`); "probe bias" is the original stratified-fold probe, whose test images share exemplars with its training folds.

carrier	transform	condition, stream	probe bias	probe bias, exemplars held out	cue-conflict shape bias	decided / 1,200	seeds
chromatic (L−M)	none	no diet, parvo	−0.504 [0.035]	−0.118 [0.021]	0.364 [0.080]	270	3
chromatic (L−M)	yes	swap, parvo	−0.145 [0.043]	−0.002 [0.022]	0.495 [0.038]	165	3

carrier	transform	condition, stream	probe bias	probe bias, exemplars held out	cue-conflict shape bias	decided / 1,200	seeds
achromatic (luminance)	none	parvo_highsf, stream 2	−0.271 [0.023]	−0.139 [0.012]	0.389 [0.028]	598	3
achromatic (luminance)	yes	diet, magno	+0.819 [0.014]	+0.337 [0.007]	0.878 [0.022]	488	3

Extended Data Table 5 | Single-stream controls trained with the identical diet, and the readout ladder. Best saved checkpoint per seed, except the no-diet row, which is scored at the final checkpoint because its accuracy does not degrade (Fig. 2d; Methods 4.7); mean [range] over three seeds. Accuracy is validation top-1 at the scored checkpoint. Sources: `results/cue_conflict_bestep/`, `results/cue_conflict_single34_bestep/`, `results/cue_conflict_single_bestep/`, run histories, `results/param_matched/`.

model / readout	parameters	shape bias	decided / 1,200	accuracy
diet, magno stream alone	22,397,936	0.903 [0.888, 0.916]	504	0.556 [0.552, 0.564]
diet, both streams (w = 0.5)	22,397,936	0.655 [0.594, 0.718]	502	0.556 [0.552, 0.564]
diet, parvo stream alone	22,397,936	0.400 [0.372, 0.429]	388	0.556 [0.552, 0.564]
single stream, ResNet-34 (matched)	21,338,992	0.625 [0.604, 0.652]	537	0.623 [0.616, 0.634]
single stream, ResNet-18 (half)	11,230,832	0.567 [0.541, 0.590]	501	0.599 [0.589, 0.608]
no diet, both streams	22,397,936	0.416 [0.394, 0.433]	593	0.701 [0.698, 0.706]

Extended Data Table 6 | Out-of-distribution robustness: relative-accuracy AUC per corruption family and condition. Mean over three seeds [range]; the thesis's schedule-trained checkpoint is a single run. "sep." marks complete seed separation between the diet and no-diet arms (the exact 3 v 3 floor, p = 0.100). Families marked ‡ were declared trained-in before the run. Source: `results/ood_bestep/`.

family	diet	no diet	swap	parvo_highsf	thesis schedule	diet − no diet	sep.
uniform-noise	0.634 [0.619, 0.647]	0.729 [0.720, 0.735]	0.749 [0.744, 0.758]	0.767 [0.758, 0.773]	0.640	−0.095	yes
high-pass	0.678 [0.647, 0.696]	0.827 [0.820, 0.831]	0.831 [0.817, 0.841]	0.783 [0.775, 0.791]	0.680	−0.149	yes
false-colour	0.854 [0.845, 0.860]	0.938 [0.933, 0.943]	0.966 [0.960, 0.971]	0.874 [0.868, 0.880]	0.839	−0.084	yes

family	diet	no diet	swap	parvo_highsf	thesis schedule	diet − no diet	sep.
rotation	0.574 [0.567, 0.582]	0.615 [0.613, 0.616]	0.605 [0.597, 0.612]	0.611 [0.602, 0.616]	0.572	−0.041	yes
phase-scrambling	0.382 [0.365, 0.391]	0.373 [0.369, 0.377]	0.371 [0.365, 0.375]	0.382 [0.380, 0.383]	0.362	+0.009	no
contrast ‡	0.481 [0.460, 0.493]	0.549 [0.530, 0.578]	0.543 [0.539, 0.550]	0.511 [0.498, 0.526]	0.489	−0.068	yes
low-pass ‡	0.444 [0.434, 0.464]	0.356 [0.353, 0.359]	0.359 [0.353, 0.362]	0.385 [0.375, 0.392]	0.436	+0.089	yes

Extended Data Table 7 | Error consistency (Cohen's κ) with the ten human observers. The trained readout (w = 0.5), the magno stream alone (w = 1; the $\kappa = 0.0082$, 3.5% of Results 2.5 is the mean of the three final-checkpoint runs) and the best readout over the 21 values of w, per run; κ as a percentage of the human–human ceiling ($\kappa = 0.234$, 45 pairs). Published CNN values are from Geirhos et al. (2020). Source: `results/sweep_error_consistency/`.

run	shape bias at w = 0.5	κ at w = 0.5	% ceiling	κ at w = 1 (magno alone)	% ceiling	rate at w = 1	best κ over w	% ceiling	w at best	rate at best
dkl_base_seed0_final	0.421	+0.0309	13.2	+0.0404	17.3	0.438	+0.0404	17.3	1.00	0.438
dkl_base_seed1_final	0.433	+0.0333	14.3	+0.0390	16.7	0.431	+0.0421	18.0	0.75	0.429
dkl_base_seed2_final	0.392	+0.0372	15.9	+0.0349	15.0	0.400	+0.0412	17.6	0.35	0.410
dkl_bcs_seed0_best	0.594	+0.0096	4.1	+0.0178	7.6	0.890	+0.0178	7.6	1.00	0.890
dkl_bcs_seed0_final	0.692	+0.0234	10.0	+0.0361	15.5	0.872	+0.0405	17.3	0.90	0.857
dkl_bcs_seed1_best	0.718	+0.0269	11.5	+0.0058	2.5	0.913	+0.0425	18.2	0.35	0.614
dkl_bcs_seed1_final	0.686	+0.0319	13.7	+0.0028	1.2	0.894	+0.0437	18.7	0.45	0.651
dkl_bcs_seed2_best	0.652	+0.0372	15.9	-0.0055	-2.3	0.908	+0.0483	20.7	0.85	0.847
dkl_bcs_seed2_final	0.626	+0.0280	12.0	-0.0143	-6.1	0.871	+0.0405	17.3	0.75	0.784
AlexNet (published)	0.429	+0.0320	13.7							
GoogLeNet (published)	0.312	+0.0250	10.7							
VGG-16 (published)	0.172	+0.0170	7.3							

Extended Data Table 8 | The estimator's own bias, and the residual-misfit variants. Synthetic voxels were generated from each arm's fitted 1,000-component feature spaces with the measured per-ROI noise ceilings imposed and the true magno − parvo difference set to zero everywhere; the recovered contrast is the instrument's signed floor. "Displacement" is the real contrast minus the floor, paired per subject. The six variants differ in how the floor is paired (group or per-subject) and whether the residual misfit of the recovery simulation is also removed (before or after rescaling by the recovered slope). Sources: `results/vp_recovery_realnc/diet_floor_paired_2026-`

`09-07.txt`, `floor_pairing_2026-09-07.txt`, `prediction_scored_2026-09-07.txt`.

quantity	diet	no diet
floor, normalised contrast (true = 0)	+0.0375 (8/8 > 0, p = .0078)	+0.0511 (8/8 > 0, p = .0078)
floor, raw contrast	−0.0021 (p = .15)	+0.0009 (p = .84)
real contrast, normalised (seed 0)	−0.8335	−0.0785
displacement real − floor	−0.871 (8/8, p = .0078)	−0.130 (8/8, p = .0078); n = 7: −0.103 (7/7, p = .0156)
recovered slope of estimate vs truth	1.210	1.281
variant: group floor, minimal	−0.720 (8/8, p = .0078)	−0.101 (8/8, p = .0078)
variant: per-subject floor, minimal	−0.720 (8/8, p = .0078)	−0.101 (8/8, p = .0078)
variant: group floor + residual before scaling	−0.626 (8/8, p = .0078)	−0.034 (6/8, p = .055)
variant: per-subject floor + residual before scaling	−0.626 (8/8, p = .0078)	−0.034 (5/8, p = .31)
variant: group floor + residual after scaling	−0.607 (8/8, p = .0078)	−0.015 (4/8, p = 1.00)
variant: per-subject floor + residual after scaling	−0.607 (8/8, p = .0078)	−0.015 (5/8, p = .84)
per-ROI normalised floor, V1v ... hV4 (no-diet geometry)		+0.025, +0.025, +0.026, +0.025, +0.027, +0.026, +0.026 (ρ vs rank +0.54, p = .22)

Extended Data Table 9 | Stream preference regressed on pRF eccentricity and pRF size within subject. Per-voxel stream-preference index regressed on log pRF eccentricity and log pRF size (NSD pRF estimates, voxels with pRF $R^2 > 5\%$, noise ceiling > 0.1); "ecc | size" is the partial slope for eccentricity controlling for size and vice versa; mean over 8 subjects, with the number of subjects on the stated side and the exact Wilcoxon p. Collinearity between the two predictors is modest (mean VIF 1.66). The arm interaction is the paired difference of partial slopes, seed-averaged within arm. Sources: `results/prf_continuous/dissociation_2026-09-08.txt`, `interaction_2026-09-08.txt`, `input_null_2026-09-08.txt`. The paired trained − untrained rows compare each subject's partial slope in the trained seed-0 network with the untrained network on the diet front end; training with the diet changes neither statistic significantly and the two disagree in sign, while training without the diet removes the dissociation on the raw statistic.

model	statistic	ecc alone	size alone	ecc | size	size | ecc
diet, seed 0	normalised	+0.066 (0/8 neg, .0078)	−0.013 (5/8 neg, .31)	+0.119 (0/8 neg, .0078)	−0.086 (8/8 neg, .0078)
diet, seed 1	normalised	+0.065 (0/8, .0078)	−0.029 (7/8, .039)	+0.132 (0/8, .0078)	−0.110 (8/8, .0078)
diet, seed 2	normalised	+0.064 (0/8, .0078)	−0.012 (5/8, .25)	+0.115 (0/8, .0078)	−0.083 (8/8, .0078)
no diet, seed 0	normalised	+0.018 (4/8, .64)	+0.033 (1/8, .039)	−0.004 (4/8, 1.00)	+0.035 (0/8, .0078)
no diet, seed 1	normalised	+0.004 (4/8, .95)	+0.015 (4/8, .38)	−0.009 (6/8, .46)	+0.021 (2/8, .11)
no diet, seed 2	normalised	−0.007 (5/8, .38)	+0.012 (3/8, .31)	−0.022 (6/8, .25)	+0.024 (1/8, .016)

model	statistic	ecc alone	size alone	ecc | size	size | ecc
diet, seed 0	raw	+0.103 (1/8, .039)	−0.158 (8/8, .0078)	+0.325 (0/8, .0078)	−0.356 (8/8, .0078)
no diet, seed 0	raw	+0.130 (1/8, .023)	+0.181 (0/8, .0078)	+0.028 (3/8, .84)	+0.163 (0/8, .0078)
Gabor bank, diet front end (input null)	normalised			+0.035 (6/8 pos, .15)	+0.002 (4/8 neg, .95)
untrained network, diet front end	normalised			+0.083 (7/8 pos, .016)	−0.038 (6/8 neg, .039)
untrained network, undieted front end	normalised			+0.089 (7/8 pos, .023)	−0.038 (8/8 neg, .0078)
trained diet − untrained (diet front end), paired	normalised			+0.036 (6/8 pos, .11)	
trained diet − untrained (diet front end), paired	raw			−0.077 (2/8 pos, .11)	
trained no diet − untrained (diet front end), paired	normalised			−0.087 (2/8 pos, .15)	
trained no diet − untrained (diet front end), paired	raw			−0.374 (0/8 pos, .0078)	
arm interaction, diet − no diet (seed-averaged)	normalised			+0.134 (8/8, .0078)	−0.119 (0/8 pos, .0078)
arm interaction, diet − no diet (seed-averaged)	raw			+0.305 (8/8, .0078)	−0.502 (0/8 pos, .0078)

Extended Data Table 10 | Cue-conflict shape bias under the forced two-way estimator. The primary estimator (Methods 4.3) restricts the logits to the 16 categories, takes the argmax, and scores shape against texture over the images decided as one of the two, so images decided as a third category drop out of the denominator ("decided"). The forced estimator keeps all 1,200 images and scores, per image, whether the logit of the shape category exceeds the logit of the texture category. Mean over seeds [range]. Source: the stored per-image logits in `results/cue_conflict*/`.

model / readout	restricted (primary)	decided / 1,200	forced two-way	seeds
diet, magno stream alone (best ckpt)	0.903 [0.888, 0.916]	504	0.775 [0.767, 0.788]	3
diet, both streams w = 0.5 (best ckpt)	0.655 [0.594, 0.718]	502	0.588 [0.574, 0.609]	3
diet, parvo stream alone (best ckpt)	0.400 [0.372, 0.429]	388	0.414 [0.394, 0.428]	3
diet, magno stream alone (final ckpt)	0.878 [0.870, 0.892]	488	0.761 [0.757, 0.764]	3
diet, both streams w = 0.5 (final ckpt)	0.675 [0.646, 0.692]	475	0.608 [0.596, 0.618]	3
diet, parvo stream alone (final ckpt)	0.418 [0.406, 0.438]	353	0.449 [0.411, 0.472]	3
no diet, both streams	0.416 [0.394, 0.433]	593	0.375 [0.357, 0.388]	3
no diet, magno stream alone	0.421 [0.398, 0.434]	602	0.378 [0.352, 0.398]	3
no diet, parvo stream alone	0.364 [0.311, 0.391]	270	0.442 [0.410, 0.473]	3
swap, both streams	0.408 [0.403, 0.412]	586	0.365 [0.362, 0.367]	3
swap, parvo stream alone (L−M + transform)	0.495 [0.478, 0.516]	165	0.511 [0.491, 0.531]	3
parvo_highsf, both streams	0.486 [0.450, 0.516]	618	0.450 [0.438, 0.457]	3
parvo_highsf, stream 2 alone (L−M + luminance)	0.389 [0.374, 0.402]	598	0.346 [0.324, 0.358]	3
single stream, ResNet-34 (matched)	0.625 [0.604, 0.652]	537	0.561 [0.552, 0.573]	3
single stream, ResNet-18 (half)	0.567 [0.541, 0.590]	501	0.523 [0.506, 0.538]	3
no diet, accuracy-matched (epochs 10, 10, 11), magno alone	0.450 [0.432, 0.465]	533	0.364 [0.357, 0.370]	3
thesis checkpoint (progressive schedule), both streams	0.697	620	0.636	1
untrained, diet front end, magno alone	0.513	154	0.510	1
untrained, no-diet front end, magno alone	0.568	169	0.508	1

Extended Data Table 11 | Stream preference beyond hV4, on every NSD atlas. Median over reliable voxels of the per-voxel index (Um − Up)/(Um + Up) (noise ceiling > 0.1, Um + Up > 10^{-3}, ≥ 20 voxels per ROI per subject), mean over the eight subjects (n = subjects with enough voxels); the no-diet, diet, swap and parvo + high-SF arms are each the mean over three training seeds per subject, and the thesis column is the thesis's one checkpoint (Methods 4.5). Positive = magno band, negative = parvo band. Atlases: Wang et al. (2015) Kastner2015 ROIs beyond hV4 (the V1–hV4 rows are Extended Data Table 1 and Fig. 3), NSD's `streams` atlas, and the fLoc category-selective ROIs. Regional contrasts and the exact tests are in Extended Data Fig. 4c and `results/figures_v2/ext_gradient.md`. Source: the stored voxelwise fits in `results/encoding_cc/` and `results/encoding/`, relabelled; no refitting.

atlas	ROI	untrained (diet front end)	no diet (3 seeds)	diet (3 seeds)	swap (3 seeds)	parvo + high-SF (3 seeds)	thesis ckpt (n = 1)	voxels (median, diet)
Kastner2015	VO1	+0.00 (n=8)	+0.92 (n=8)	−0.87 (n=8)	+0.98 (n=8)	−0.99 (n=8)	−0.94 (n=8)	196
Kastner2015	VO2	+0.12 (n=8)	+0.97 (n=8)	−0.90 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.94 (n=8)	220
Kastner2015	PHC1	+0.41 (n=8)	+0.98 (n=8)	−0.89 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.94 (n=8)	148
Kastner2015	PHC2	+0.46 (n=8)	+0.99 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.93 (n=8)	150
Kastner2015	LO1	+0.13 (n=8)	+0.92 (n=8)	−0.83 (n=8)	+0.98 (n=8)	−0.98 (n=8)	−0.90 (n=8)	230
Kastner2015	LO2	−0.14 (n=8)	+0.92 (n=8)	−0.90 (n=8)	+0.98 (n=8)	−1.00 (n=8)	−0.94 (n=8)	166
Kastner2015	TO1	+0.25 (n=8)	+0.96 (n=8)	−0.86 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.91 (n=8)	280
Kastner2015	TO2	+0.20 (n=8)	+0.96 (n=8)	−0.89 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.90 (n=8)	36
Kastner2015	V3A	+0.39 (n=8)	+0.91 (n=8)	−0.81 (n=8)	+0.98 (n=8)	−0.98 (n=8)	−0.84 (n=8)	278
Kastner2015	V3B	+0.24 (n=8)	+0.96 (n=8)	−0.87 (n=8)	+0.99 (n=8)	−0.99 (n=8)	−0.94 (n=8)	254
Kastner2015	IPS0	+0.15 (n=8)	+0.98 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.92 (n=8)	317
Kastner2015	IPS1	−0.06 (n=8)	+0.98 (n=8)	−0.86 (n=8)	+1.00 (n=8)	−0.99 (n=8)	−0.92 (n=8)	239
Kastner2015	IPS2	+0.02 (n=8)	+1.00 (n=8)	−0.92 (n=8)	+1.01 (n=8)	−1.00 (n=8)	−0.94 (n=8)	47
streams	early	+0.27 (n=8)	+0.86 (n=8)	−0.42 (n=8)	+0.96 (n=8)	−0.93 (n=8)	−0.45 (n=8)	3133
streams	midventral	−0.15 (n=8)	+0.87 (n=8)	−0.84 (n=8)	+0.97 (n=8)	−0.99 (n=8)	−0.86 (n=8)	708
streams	midlateral	−0.04 (n=8)	+0.91 (n=8)	−0.86 (n=8)	+0.97 (n=8)	−0.99 (n=8)	−0.90 (n=8)	571
streams	midparietal	+0.32 (n=8)	+0.94 (n=8)	−0.84 (n=8)	+0.99 (n=8)	−0.99 (n=8)	−0.91 (n=8)	512
streams	ventral	+0.08 (n=8)	+0.96 (n=8)	−0.91 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.95 (n=8)	2724
streams	lateral	+0.25 (n=8)	+0.97 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.91 (n=8)	1842

atlas	ROI	untrained (diet front end)	no diet (3 seeds)	diet (3 seeds)	swap (3 seeds)	parvo + high-SF (3 seeds)	thesis ckpt (n = 1)	voxels (median, diet)
streams	parietal	+0.12 (n=8)	+0.99 (n=8)	−0.87 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.93 (n=8)	1082
floc-faces	OFA	−0.07 (n=8)	+0.90 (n=8)	−0.89 (n=8)	+0.98 (n=8)	−0.99 (n=8)	−0.92 (n=8)	367
floc-faces	FFA-1	+0.01 (n=8)	+0.95 (n=8)	−0.91 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.94 (n=8)	399
floc-faces	FFA-2	+0.22 (n=7)	+0.98 (n=7)	−0.92 (n=7)	+1.00 (n=7)	−1.00 (n=7)	−0.95 (n=7)	260
floc-places	OPA	+0.34 (n=8)	+0.98 (n=8)	−0.86 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.93 (n=8)	1070
floc-places	PPA	+0.44 (n=8)	+0.99 (n=8)	−0.88 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.94 (n=8)	620
floc-bodies	EBA	+0.17 (n=8)	+0.97 (n=8)	−0.88 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.92 (n=8)	2174
floc-bodies	FBA-1	−0.07 (n=6)	+0.96 (n=6)	−0.91 (n=6)	+1.00 (n=6)	−1.00 (n=6)	−0.95 (n=6)	314
floc-bodies	FBA-2	+0.13 (n=8)	+0.98 (n=8)	−0.92 (n=8)	+1.00 (n=8)	−1.00 (n=8)	−0.94 (n=8)	252
floc-words	OWFA	−0.29 (n=8)	+0.88 (n=8)	−0.85 (n=8)	+0.96 (n=8)	−0.99 (n=8)	−0.88 (n=8)	392
floc-words	VWFA-1	−0.16 (n=8)	+0.96 (n=8)	−0.91 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.96 (n=8)	454
floc-words	VWFA-2	−0.02 (n=5)	+0.95 (n=6)	−0.94 (n=6)	+0.99 (n=6)	−1.00 (n=6)	−0.96 (n=6)	150
floc-words	mfs-words	−0.08 (n=8)	+0.97 (n=8)	−0.93 (n=8)	+0.99 (n=8)	−1.00 (n=8)	−0.96 (n=8)	121

Extended Data Table 12 | The readout weight cortex prefers against the one behaviour prefers. For each mixing weight w the classifier's own input $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$ (y_1 magno stream output, y_2 parvo stream output, 512-d each) is the design of a ridge encoding model of every nsdgeneral voxel (same split, folds and alpha grid as the banded model; Methods 4.5). w*_cortex is the w that maximises the median test R^2 over reliable voxels of the ROI (mean over seeds per subject; median [range] over 8 subjects). "R^2 at w = 0.5 / max" is the median R^2 at the paper's readout weight relative to the best w. "concat" is the median R^2 of the 1,024-d concatenation $[y_1, y_2]$, fitted as a reference and not as a bound: every mixture is a linear map of the concatenation, but the two models are regularised separately on a shared alpha grid, and the best mixture exceeds the fitted concatenation in 36 of the 112 subject × ROI × arm cells, by a median 0.0019 R^2 (1.5% of the concatenation; largest 0.0054). That excess is the reproducibility scale of these curves and exceeds the median R^2 the sweep itself buys, 0.0016 from w = ½ to w, *so w*_cortex is read as the location of an optimum and never as an R^2 difference. What the table supports is the ordinal contrast: extrastriate below V1 in 8/8 subjects ($\Delta = -0.252$ [−0.283, −0.219], 10,000-sample percentile bootstrap over subjects, exact Wilcoxon p = .0078), reversed without the diet (+0.188 [+0.077, +0.285], 7/8 positive, p = .023). "Voxels magno / parvo" is the fraction of the ROI's voxels whose own best w is ≥ 0.75 / ≤ 0.25. The behaviour-optimal weight (trial fit to each of 10 human observers' cue-conflict choices, diet final checkpoints, 3 seeds) is given in the last row for comparison, with the cue-conflict shape-bias rate the diet network produces at each weight.

arm	ROI	w*_cortex median [range]	R^2 at w = 0.5 / max	max R^2 (median)	concat R^2	voxels best w ≥ 0.75	voxels best w ≤ 0.25	rate at w*_cortex	n
diet	V1	0.58 [0.50, 0.70]	0.994	0.1313	0.1402	0.19	0.11	0.72	8
diet	V2	0.50 [0.45, 0.55]	0.999	0.1024	0.1066	0.12	0.20	0.67	8
diet	V3	0.40 [0.30, 0.50]	0.987	0.0847	0.0864	0.08	0.29	0.61	8
diet	hV4	0.35 [0.20, 0.50]	0.976	0.0945	0.0948	0.05	0.41	0.57	8
diet	ventral (VO, PHC)	0.33 [0.25, 0.35]	0.971	0.1339	0.1325	0.02	0.46	0.56	8
diet	lateral (LO, TO)	0.33 [0.25, 0.45]	0.974	0.1067	0.1060	0.03	0.40	0.56	8
diet	dorsal (V3A/B, IPS0–2)	0.35 [0.25, 0.45]	0.982	0.0683	0.0673	0.05	0.39	0.57	8
no diet	V1	0.65 [0.55, 0.80]	0.987	0.1221	0.1259	0.43	0.09	nan	8
no diet	V2	0.58 [0.40, 0.95]	0.995	0.0968	0.1022	0.36	0.13	nan	8
no diet	V3	0.58 [0.50, 1.00]	0.991	0.0845	0.0890	0.40	0.13	nan	8
no diet	hV4	0.62 [0.40, 0.70]	0.988	0.0965	0.1014	0.40	0.09	nan	8
no diet	ventral (VO, PHC)	0.75 [0.65, 1.00]	0.975	0.1557	0.1535	0.59	0.02	nan	8
no diet	lateral (LO, TO)	0.88 [0.60, 1.00]	0.949	0.1190	0.1214	0.59	0.03	nan	8
no diet	dorsal (V3A/B, IPS0–2)	1.00 [0.75, 1.00]	0.944	0.0805	0.0792	0.61	0.04	nan	8
behaviour (diet)	10 human observers, trial fit	0.67 [0.49, 0.89]	—	—	—	—	—	0.78	10

Extended Data Table 13 | The readout span without a stream split: one gain on a network's own shape- and texture-predictive subspaces, head fixed. For each frozen network with logits = W·y + b, the 15 discriminant directions of a 16-class shape LDA and of a 16-class texture LDA on

the penultimate features y (PCA to 256 dimensions, Ledoit–Wolf shrinkage; fitted on one stratified half of the 1,200 cue-conflict images) are made mutually orthogonal by symmetric (Löwdin) orthogonalisation, giving orthonormal bases B_S and B_T (15 directions each) and the residual B_R. logits(w) = W·(P_R y + 2w·P_S y + 2(1 − w)·P_T y) + b with the network's own head untouched; w = 0.5 is the trained network. The rate is scored on the other half of the images (10 random splits, both directions; mean, with the range over splits and seeds in brackets), by the paper's primary estimator (112-class, decided images) for our networks and the Geirhos 16-class probability aggregation for the ImageNet networks; "forced-estimator span" is the same sweep on the forced two-way estimator. "gated span" is max − min of the seed-mean held-out curve over the w at which top-1 of the gained features (bases fitted on all 1,200 images) on held-out natural images (MiniEcoset validation for our networks; a stratified 10-per-class ILSVRC-2012 validation subset for the ImageNet networks) stays within 5 points of the trained value, followed by that w-range and the largest change of top-1 anywhere in the sweep. The two stratified halves share object and texture exemplars (98% and 100% of scored images; the 1,200 images reuse 160 object and 48 texture exemplars), so the span is also given with exemplars held out: fitted on images whose object and texture exemplars both lie outside one of three exemplar folds and scored on the images whose exemplars both lie inside it, ten repetitions, ungated (`docs/PLAN_EXEMPLAR_2026-09-21.md`; `results/figures_v2/exemplar_zoo.json`, `exemplar_streams.json`). "refit span" is the block-form lossy sum $z(w) = [w{\cdot}s + (1-w){\cdot}t, B_R^{T}y]$, $s = B_S^{T}y$ and $t = B_T^{T}y$ paired by discriminant rank, with a linear and a two-layer head refitted at each w (Methods 4.3 budget), our networks only; in parentheses the largest shortfall of validation accuracy, over w, from a head refitted on the unmixed features. "variance in S / T" is the fraction of the feature variance on the fitting half lying in each subspace; "order sensitivity" replaces the symmetric orthogonalisation by sequential Gram–Schmidt with the shape basis or the texture basis first. "median-split gated span" is the gated span of the pre-specified unit-level rule of Extended Data Table 14 on the same networks and images. For the two-stream concatenation, which has no held-out natural-image features, the fraction of the shape basis's energy lying on stream-1 units (all-image bases; held-out bases in parentheses; range over seeds) replaces the gate. Pre-specification: `docs/PLAN_SUBSPACE_2026-09-15.md` (predictions P-A, P-A′, P-B, P-D). Source: `results/figures_v2/subspace_behaviour.json`, aggregated by `tmp_analysis/figs_v2/subspace_behaviour.py` from the outputs of `src/analysis/generality/subspace_factor.py`, `gain_accuracy.py` and `refit_sum.py` (`slurm/subspace_{cc,refit}.sh`); the per-network files are outside the repository.

network	D	trained rate	w = 0	w = 1	span, exemplars shared [range]	span, exemplars held out [seed range]	ρ(w, rate)	gated span (w-range; largest change of top-1)	forced-estimator span	refit span linear / two-layer (largest shortfall of top-1)	variance in S / T	order sensitivity, S first / T first	median-split gated span	seeds
two-stream, diet, streams concatenated (validation)	1024	0.675	0.495	0.801	0.320 [0.18, 0.45]	0.249 [0.24, 0.27]	0.98	— (shape basis on stream 1: 0.63–0.70)	0.216	—	0.11 / 0.29	0.35 / 0.25	—	3
single ResNet-34, diet	512	0.625	0.516	0.735	0.226 [0.12, 0.37]	0.173 [0.14, 0.20]	0.98	0.19 (0.10–0.90; 11.9 pts)	0.163	0.08 / 0.10 (0.8 pts)	0.14 / 0.20	0.24 / 0.20	0.07	3
single ResNet-18, diet	512	0.557	0.433	0.674	0.249 [0.13, 0.41]	0.200 [0.18, 0.23]	0.98	0.14 (0.25–0.73; 16.7 pts)	0.189	0.08 / 0.10 (0.6 pts)	0.15 / 0.23	0.25 / 0.20	0.09	3
single ResNet-18, no diet (`rgb_tsn`)	512	0.445	0.311	0.651	0.344 [0.22, 0.47]	0.279 [0.23, 0.32]	0.99	0.22 (0.20–0.78; 13.4 pts)	0.242	0.13 / 0.15 (0.5 pts)	0.13 / 0.26	0.29 / 0.34	0.08	3
ImageNet ViT-B/16	768	0.399	0.287	0.570	0.282 [0.22, 0.34]	0.180	1.00	0.28 (0.00–1.00; 0.2 pts)	0.318	—	0.11 / 0.23	0.22 / 0.31	0.08	1
ImageNet ResNet-152	2048	0.293	0.209	0.455	0.246 [0.20, 0.33]	0.166	1.00	0.25 (0.00–1.00; 1.2 pts)	0.271	—	0.07 / 0.24	0.21 / 0.27	0.16	1
ImageNet ConvNeXt-B	1024	0.269	0.178	0.512	0.334 [0.29, 0.39]	0.196	1.00	0.33 (0.00–1.00; 0.2 pts)	0.371	—	0.08 / 0.33	0.31 / 0.35	0.11	1

network	trained rate	w = 0	w = 1	span, exemplars shared [range]	span, exemplars held out [seed range]	ρ(w, rate)	gated span (w-range; largest change of top-1)	forced-estimator span	refit span linear / two-layer (largest shortfall of top-1)	variance in S / T	order sensitivity, S first / T first	median-split gated span	seeds
ImageNet-768 ConvNeXt-S	0.248	0.170	0.458	0.287 [0.24, 0.34]	0.156	1.00	0.29 (0.00–1.00; 0.1 pts)	0.353	—	0.08 / 0.33	0.27 / 0.30	0.12	1
ImageNet-2048 ResNet-50	0.222	0.157	0.447	0.290 [0.21, 0.34]	0.208	1.00	0.29 (0.00–1.00; 1.5 pts)	0.310	—	0.07 / 0.28	0.28 / 0.29	0.13	1
ImageNet-4096 VGG-16	0.110	0.061	0.329	0.268 [0.23, 0.31]	0.138	1.00	0.27 (0.00–1.00; 0.5 pts)	0.322	—	0.04 / 0.28	0.26 / 0.27	0.09	1

Extended Data Table 14 | The readout span without a stream split under the pre-specified unit-level rule: fixed-head gain sweep on the shape- and texture-leaning halves of the units. The special case of Extended Data Table 13 in which the two subspaces are spanned by disjoint sets of units. Each unit of a network's penultimate layer is scored by the fraction of its variance between shape categories minus that between texture categories on the 1,200 cue-conflict images (one-way η^2); the median split gives a shape-leaning half S and a texture-leaning half T. logits(w) = head(g(w) $\odot$ y) with g = 2w on S and 2(1 − w) on T, the network's own head untouched; w = 0.5 is the trained network. The split is defined on one stratified half of the images and the rate scored on the other (10 random splits, both directions, the same splits as Extended Data Table 13; mean, with the range over splits and seeds in brackets). Rates by the paper's primary estimator (112-class, decided images) for our networks and the Geirhos 16-class probability aggregation for the ImageNet networks; "forced-estimator span" is the same sweep on the forced two-way estimator. "shape-leaning units" is the fraction of units with η^2_shape > η^2_texture on all images; for the two-stream concatenation the fraction of S that belongs to stream 1 is given in parentheses (P3). "w within 5 pts" is the w-range over which top-1 on held-out natural images (MiniEcoset validation; a stratified 10-per-class ILSVRC-2012 validation subset for the ImageNet networks) stays within 5 points of the trained value. "refit span" is the lossy-sum protocol z(w) = w·y_S + (1 − w)·y_T with a linear and a two-layer head refitted at every w (Methods 4.3 budget), our networks only. The last two columns are a sensitivity analysis added after the pre-specified analysis, at a referee's request: the same held-out sweep with S defined as the absolute shape-leaning set (units with d_j > 0) or as the top decile of units by d_j, and T as the remainder; given are the held-out span with its range over splits and seeds, the minimum Spearman ρ, the mean size of S and the w-range within 5 points of the trained top-1 (no gate exists for the two-stream concatena-

tion, which has no held-out natural-image features). Pre-specification: `docs/PLAN_GENERALITY_2026-09-14.md`. Source: `results/figures_v2/generality_behaviour.json`, aggregated by `tmp_analysis/figs_v2/generality_behaviour.py` from the per-network outputs of `src/analysis/generality/` (`slurm/generality_{cc,mini,inval,rules}.sh`); the per-network files are outside the repository.

network	units	trained rate	w = 0	w = 1	span (held-out) [range]	ρ(w, rate)	forced-estimate span	shape-learning units	w within 5 pts of trained top-1	refit span linear / two-layer	span, S = {d > 0}	span, top decile	seeds
two-stream, diet, streams concatenated (validation)	1024	0.675	0.420	0.879	0.461 [0.39, 0.57]	1.00	0.307	0.49 (S from stream 1: 0.99)	—	—	0.46 [0.39, 0.59] ($\rho \geq$ 0.98;	S	= 504)
single ResNet-34, diet	512	0.625	0.565	0.688	0.136 [0.03, 0.24]	0.71	0.102	0.17	0.20–0.78 (top-1 0.626)	0.06 / 0.05	0.12 [0.03, 0.23] ($\rho \geq$ -0.78;	S	= 89; gate 0.00–0.73)
single ResNet-18, diet	512	0.557	0.467	0.652	0.194 [0.11, 0.31]	0.97	0.131	0.09	0.28–0.72 (top-1 0.547)	0.09 / 0.06	0.11 [0.04, 0.18] ($\rho \geq$ -0.63;	S	= 44; gate 0.00–0.73)
single ResNet-18, no diet (`rgb_tsn`)	512	0.445	0.374	0.566	0.195 [0.11, 0.39]	0.96	0.164	0.05	0.27–0.75 (top-1 0.678)	0.10 / 0.09	0.15 [0.03, 0.31] ($\rho \geq$ -0.57;	S	= 24; gate 0.00–0.78)
ImageNet ViT-B/16	768	0.399	0.371	0.455	0.086 [0.07, 0.11]	0.98	0.138	0.01	0.00–1.00 (top-1 0.815)	—	0.21 [0.10, 0.34] ($\rho \geq$ 0.53;	S	= 6; gate 0.00–0.95)
ImageNet ResNet-152	2048	0.293	0.227	0.440	0.214 [0.18, 0.25]	1.00	0.196	0.02	0.15–0.90 (top-1 0.785)	—	0.27 [0.22, 0.33] ($\rho \geq$ 0.86;	S	= 45; gate 0.00–0.85)
ImageNet ConvNeXt-B	1024	0.269	0.229	0.343	0.115 [0.09, 0.14]	0.99	0.173	0.00	0.00–1.00 (top-1 0.843)	—	0.28 [0.20, 0.35] ($\rho \geq$ 0.14;	S	= 2; gate 0.00–0.95)

network	units	trained rate	w = 0	w = 1	span (held-out) [range]	ρ(w, rate)	forced-estimate span	shape-leaning units	w within 5 pts of trained top-1	refit span linear / two-layer	span, S = {d > 0}	span, top decile	seeds
ImageNet ConvNeXt-S	768	0.248	0.211	0.335	0.124 [0.11, 0.15]	0.99	0.179	0.00	0.00–1.00 (top-1 0.840)	—	undefined (empty half)	0.17 [0.12, 0.20] ($\rho \geq$ 0.72;	S
ImageNet ResNet-50	2048	0.222	0.167	0.346	0.180 [0.14, 0.23]	0.99	0.202	0.02	0.15–0.90 (top-1 0.760)	—	0.27 [0.20, 0.32] ($\rho \geq$ 0.75;	S	= 34; gate 0.00–0.85)
ImageNet VGG-16	4096	0.110	0.096	0.189	0.095 [0.07, 0.11]	0.96	0.106	0.01	0.00–1.00 (top-1 0.711)	—	0.40 [0.34, 0.46] ($\rho \geq$ -0.38;	S	= 56; gate 0.00–0.95)

Extended Data Table 15 | Cortex without a stream split: the shape- and texture-predictive components as encoding bands. Upper block (subspace bands): the three bands are the shape-predictive subspace S = Y·B_S (15 dimensions), the texture-predictive subspace T = Y·B_T (15) and the residual R = Y·B_R of Extended Data Table 13 (bases fitted on all 1,200 images), with U_S = r²_{STR} − r²_{TR} and U_T = r²_{STR} − r²_{SR} from the three-band model and the two two-band models (random search, 30 and 20 samples); "shared" is r²_{STR} − U_S − U_T and "r²" the three-band model's noise-ceiling-normalised r². w*_cortex there is the weight of the 15-dimensional lossy sum z(w) = w·s + (1 − w)·t alone. Lower block (unit halves): for each single-trunk network the penultimate units were split as in Extended Data Table 14 (all 1,200 images) and the two halves entered as two bands of a banded ridge model of every `nsdgeneral` voxel (same split, folds and alpha grid as Methods 4.5); U_shape and U_texture are the noise-ceiling-normalised unique variances and the index is (U_shape − U_texture)/(U_shape + U_texture), median over reliable voxels (noise ceiling > 0.1, ≥ 20 voxels), mean over eight subjects (seeds averaged per subject). Regions: V1–hV4 from `prf-visualrois`; ventral (VO1/2, PHC1/2), lateral (LO1/2, TO1/2) and dorsal (V3A/B, IPS0–2) from Kastner2015. The two-stream diet row applies the same computation to the magno (shape-biased) and parvo (texture-biased) bands of Fig. 3. "shared" is the cross-validated variance the bands account for jointly (the full model's r² minus the unique variances) and "r²" the full model's noise-ceiling-normalised r², medians over the same voxels. w*_cortex is the mixing weight of the lossy sum z(w) = w·y_S + (1 − w)·y_T that maximises an ROI's median test R² (Extended Data Fig. 5 protocol). Exact paired Wilcoxon over subjects. The unit-rule rows for ImageNet ViT-B/16 and ConvNeXt-B were run after the pre-registered analysis, at the identical protocol, to ask whether the unit-rule cortical result depends on architecture; they are post hoc and are marked as such in `docs/PLAN_GENERALITY_2026-09-14.md`. Source: `results/figures_v2/generality_cortex.json` and `results/figures_v2/subspace_cortex.json`, aggregated by `tmp_analysis/figs_v2/generality_cortex.py` from the per-subject fits of `src/analysis/generality/cortex_bands.py` (`slurm/generality_{nsd,cortex}.sh`, `slurm/subspace_{nsd,cortex}.sh`), which are deposited alongside the encoding outputs of Fig. 3 (Code availability).

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
subspace bands S, T, R											
single ResNet-18, no diet	index	+0.12	-0.01	+0.01	+0.03	+0.35	+0.13	+0.32	-0.088 (5/8, 0.4609)	+0.151 (2/8, 0.0781)	-0.25
	U_shape	0.011	0.009	0.010	0.010	0.013	0.009	0.011			
	U_texture	0.008	0.009	0.009	0.009	0.006	0.007	0.006			
	shared	0.329	0.311	0.305	0.327	0.521	0.409	0.375			
	r^2 (full model, nc-normalised)	0.351	0.332	0.327	0.351	0.542	0.429	0.396			
	w*_cortex (median)	0.63	0.68	0.68	0.63	0.33	0.33	0.33		Δw* extrastriate − V1 -0.23 (6/8 neg, 0.0391)	
single ResNet-34, diet	index	-0.03	-0.13	-0.08	+0.13	+0.19	-0.02	+0.14	+0.153 (2/8, 0.1094)	+0.130 (2/8, 0.3828)	+0.25
	U_shape	0.007	0.007	0.007	0.010	0.011	0.007	0.010			
	U_texture	0.007	0.008	0.008	0.007	0.008	0.007	0.008			
	shared	0.325	0.298	0.293	0.316	0.503	0.379	0.358			
	r^2 (full model, nc-normalised)	0.341	0.316	0.313	0.337	0.524	0.397	0.378			
	w*_cortex (median)	0.62	0.57	0.55	0.45	0.55	0.68	0.48		Δw* extrastriate − V1 -0.01 (4/8 neg, 0.9453)	
ImageNet ResNet-50	index	-0.33	-0.19	-0.23	+0.07	-0.16	-0.19	-0.24	+0.395 (1/8, 0.0156)	+0.131 (2/8, 0.1484)	+0.60
	U_shape	0.004	0.005	0.005	0.008	0.005	0.004	0.004			

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
	U_texture	0.007	0.006	0.007	0.006	0.006	0.005	0.006			
	shared	0.333	0.311	0.307	0.365	0.635	0.496	0.462			
	r^2 (full model, nc-normalised)	0.348	0.325	0.324	0.384	0.649	0.508	0.479			
	w*_cortex (median)	0.45	0.40	0.45	0.40	0.40	0.45	0.40		Δw* extrastriate − V1 -0.02 (5/8 neg, 0.6406)	
ImageNet ViT-B/16	index	-0.45	-0.54	-0.45	-0.25	-0.50	-0.43	-0.70	+0.193 (1/8, 0.0781)	-0.097 (3/8, 0.8438)	+0.37
	U_shape	0.004	0.003	0.004	0.007	0.004	0.003	0.002			
	U_texture	0.006	0.006	0.007	0.009	0.010	0.005	0.008			
	shared	0.254	0.242	0.254	0.292	0.555	0.404	0.382			
	r^2 (full model, nc-normalised)	0.267	0.256	0.271	0.310	0.573	0.416	0.398			
	w*_cortex (median)	0.50	0.40	0.45	0.15	0.50	0.40	0.55		Δw* extrastriate − V1 +0.08 (3/8 neg, 0.3828)	
ImageNet ConvNeXt-B	index	+0.17	+0.01	-0.17	+0.38	+0.00	-0.30	-0.51	+0.212 (3/8, 0.1953)	-0.440 (7/8, 0.0156)	+0.08
	U_shape	0.006	0.005	0.005	0.008	0.006	0.004	0.003			
	U_texture	0.004	0.003	0.004	0.003	0.004	0.006	0.005			
	shared	0.256	0.240	0.233	0.285	0.540	0.408	0.369			
	r^2 (full model, nc-normalised)	0.267	0.253	0.246	0.301	0.554	0.422	0.385			

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
	w*_cortex (median)	0.45	0.55	0.55	0.00	0.75	0.70	0.75		Δw* extrastriate − V1 +0.20 (1/8 neg, 0.0391)	
unit halves S, T											
two-stream, diet (magno − parvo)	index	-0.06	-0.44	-0.70	-0.84	-0.88	-0.87	-0.87	—	—	—
	U_shape	0.087	0.045	0.026	0.016	0.014	0.014	0.013			
	U_texture	0.099	0.119	0.145	0.180	0.246	0.190	0.181			
	shared	0.289	0.286	0.261	0.219	0.230	0.184	0.168			
	r^2 (full model, nc-normalised)	0.485	0.457	0.439	0.428	0.496	0.390	0.373			
single ResNet-34, diet	index	+0.05	+0.02	+0.05	-0.03	-0.22	-0.09	-0.06	-0.080 (7/8, 0.0156)	-0.178 (7/8, 0.0156)	-0.25
	U_shape	0.032	0.030	0.028	0.025	0.017	0.016	0.020			
	U_texture	0.028	0.028	0.025	0.027	0.027	0.020	0.022			
	shared	0.263	0.245	0.245	0.274	0.458	0.353	0.320			
	r^2 (full model, nc-normalised)	0.327	0.304	0.300	0.329	0.506	0.392	0.366			
	w*_cortex (median)	0.53	0.60	0.57	0.53	0.53	0.55	0.57		Δw* extrastriate − V1 +0.01 (4/8 neg, 0.9453)	

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
single ResNet-18, no diet	index	+0.17	+0.13	+0.12	+0.17	-0.15	-0.14	-0.14	-0.005 (5/8, 0.9453)	-0.316 (8/8, 0.0078)	-0.15
	U_shape	0.038	0.035	0.030	0.031	0.019	0.016	0.018			
	U_texture	0.026	0.026	0.024	0.022	0.025	0.022	0.024			
	shared	0.272	0.258	0.262	0.288	0.483	0.371	0.335			
	r^2 (full model, nc-normalised)	0.340	0.321	0.319	0.347	0.530	0.415	0.381			
	w*_cortex (median)	0.80	0.80	0.68	0.68	0.42	0.48	0.37		Δw* extrastriate − V1 -0.32 (7/8 neg, 0.0234)	
ImageNet ResNet-50	index	-0.38	-0.38	-0.32	-0.50	-0.44	+0.06	-0.07	-0.120 (5/8, 0.3828)	+0.235 (1/8, 0.0391)	-0.13
	U_shape	0.008	0.008	0.008	0.006	0.006	0.010	0.009			
	U_texture	0.019	0.018	0.017	0.020	0.015	0.009	0.011			
	shared	0.312	0.292	0.297	0.365	0.606	0.489	0.446			
	r^2 (full model, nc-normalised)	0.343	0.322	0.327	0.397	0.633	0.512	0.471			
	w*_cortex (median)	0.10	0.10	0.00	0.05	0.00	0.85	0.70		Δw* extrastriate − V1 +0.07 (3/8 neg, 0.8125)	
ImageNet ViT-B/16	index	-0.19	-0.13	-0.20	-0.23	-0.31	-0.23	-0.27	-0.039 (5/8, 0.6406)	-0.081 (6/8, 0.2500)	-0.35
	U_shape	0.036	0.036	0.032	0.037	0.047	0.042	0.038			
	U_texture	0.053	0.047	0.050	0.062	0.092	0.070	0.069			
	shared	0.162	0.154	0.161	0.189	0.381	0.292	0.258			

network	statistic	V1	V2	V3	hV4	ventral	lateral	dorsal	hV4 − V1 (n neg, p)	extrastriate − V1 (n neg, p)	ρ(rank, index) V1–hV4
	r^2 (full model, nc-normalised)	0.252	0.239	0.245	0.290	0.527	0.409	0.366			
	w*_cortex (median)	0.25	0.20	0.10	0.15	0.25	0.15	0.15		Δw* extrastriate − V1 -0.18 (5/8 neg, 0.5469)	
ImageNet ConvNeXt-B	index	-0.45	-0.40	-0.37	-0.31	-0.30	-0.32	-0.09	+0.140 (2/8, 0.1094)	+0.218 (2/8, 0.0547)	+0.42
	U_shape	0.012	0.013	0.013	0.017	0.022	0.020	0.025			
	U_texture	0.030	0.029	0.029	0.035	0.039	0.035	0.029			
	shared	0.214	0.200	0.203	0.248	0.476	0.357	0.319			
	r^2 (full model, nc-normalised)	0.259	0.246	0.249	0.304	0.540	0.420	0.374			
	w*_cortex (median)	0.00	0.00	0.00	0.00	0.00	0.00	0.00		Δw* extrastriate − V1 +0.10 (0/8 neg, 0.5000)	

Extended Data Table 16 | The readout span across the field's model zoo. Frozen ImageNet networks under the subspace rule of Extended Data Table 13 (PCA to 256 dimensions, 16-class shape and texture LDA, 15 directions each, symmetric orthogonalisation; bases fitted on one stratified half of the 1,200 cue-conflict images and the rate scored on the other, 10 splits both directions), each network's own head untouched, the Geirhos 16-class probability aggregation throughout; "images decided" is the number of the 1,200 assigned to their shape or their texture category at the trained readout, the denominator of the rate. "gate" is the w-range over which top-1 on a stratified 10-per-class ILSVRC-2012 validation subset (10,000 images; bases fitted on all cue-conflict images) stays within 5 points of the trained value, with the largest change of top-1 inside it; "texture end / shape end" are the held-out rates at the smallest and the largest gated w, and "gated span" is their difference, under the subspace rule and under the pre-specified unit-level median split of Extended Data Table 14. "probe" is the linear probe of Methods 4.3 (PCA to 256 dimensions, multinomial logistic regression, 5-fold stratified cross-validation) predicting the shape class and,

separately, the texture class from the same features; bias is shape minus texture accuracy (chance 0.0625). The stratified halves and the probe's folds share object and texture exemplars (the 1,200 images reuse 160 object and 48 texture exemplars), so both are also given with exemplars held out (`docs/PLAN_EXEMPLAR_2026-09-21.md`; `results/figures_v2/exemplar_zoo.json`): "gated span, exemplars held out" fits the subspaces on images whose object and texture exemplars both lie outside one of three exemplar folds and scores the rate on the images whose exemplars both lie inside it (ten repetitions); "probe, exemplars held out" is the exemplar-balanced probe of Methods 4.3 (three exemplars per category for both labels, leave one exemplar out, PCA inside the folds, mean of 20 draws). With exemplars held out, over the 38 ImageNet-trained networks: gated span 0.08–0.29 (median 0.19), ≥ 0.20 in 11; 18.6% of the 703 pairs reverse between the trained ranking and the shape-end ranking (forced two-way 16.2%), against 8.9% [6.0%, 12.2%] between two bootstrap resamples of the images at the trained readout; probe bias -0.52 to +0.01 (median -0.36), the 5–95% interval over draws below zero in 37. Preprocessing is Resize 256 / CenterCrop 224 except for Inception-v3 (342/299), EfficientNet-B3 (320/300) and EfficientNet-V2-S (384/384), which were not trained at 224. "κ with observers" is error consistency (Cohen's κ on trial-by-trial agreement, Results 2.5; Extended Data Table 7) with each of the ten human observers, mean over observers, as a percentage of the human–human ceiling $\kappa = 0.234$, at the trained readout and, in parentheses, at the texture end and the shape end of the gate under the same gain; computed under this pipeline's preprocessing, so the VGG-16 value differs from the published response file's 7.3% (Extended Data Table 7) as its trained rate does. Over the 38 ImageNet-trained networks: Kendall τ between the ranking by trained rate and by the shape-end rate 0.47 (by the texture-end rate 0.78); 188 of 703 pairs (27%) reverse order; gated span 0.20–0.50 (median 0.30), ≥ 0.20 in 97% and the whole sweep inside the gate in 100%; Spearman ρ between gated span and trained rate +0.04, and trained top-1 -0.22; error consistency at the trained readout 2.4–23.0% of the ceiling (median 7.9%), at most 36.7% anywhere inside a gate, changing by a median of +13.5 points from the texture end to the shape end; $\rho(\kappa$, trained rate) +0.80; Kendall τ between trained rate and probe bias 0.44. The three Stylized-ImageNet networks are the published checkpoints of Geirhos et al. (2019), all ResNet-50: trained on Stylized-ImageNet (SIN), on SIN + IN, and on SIN + IN then fine-tuned on IN. Pre-specification: `docs/PLAN_ZOO_2026-09-15.md` (predictions P-Z1–P-Z4). Source: `results/figures_v2/zoo_behaviour.json`, aggregated by `tmp_analysis/figs_v2/zoo_behaviour.py` from the outputs of `slurm/zoo_cc.sh`; the per-network files are outside the repository.

network	D	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
ImageNet-trained (torchvision IMAGENET1K_V1)													
ViT-B/32	768	0.758	0.617 (767)	0.00–1.00 (0.2 pts)	0.499 / 0.738	0.239	0.161	1.00	0.069	0.67 / 0.97 (-0.30)	0.46 / 0.46 (+0.01)	23.0 (12.9 → 34.0)	yes
ViT-B/16	768	0.815	0.399 (818)	0.00–1.00 (0.2 pts)	0.287 / 0.570	0.282	0.180	1.00	0.085	0.56 / 0.99 (-0.43)	0.38 / 0.57 (-0.19)	11.6 (7.2 → 19.2)	yes
ViT-L/16	1024	0.801	0.319 (885)	0.00–1.00 (0.2 pts)	0.207 / 0.545	0.337	0.193	1.00	0.078	0.57 / 0.99 (-0.43)	0.35 / 0.58 (-0.23)	11.5 (5.9 → 21.9)	yes
Inception v3	2048	0.759	0.305 (797)	0.00–1.00 (0.6 pts)	0.241 / 0.487	0.246	0.134	0.99	0.099	0.40 / 0.96 (-0.56)	0.29 / 0.63 (-0.35)	11.0 (5.5 → 22.5)	yes
MaxViT-T	512	0.842	0.295 (899)	0.00–1.00 (0.2 pts)	0.230 / 0.469	0.239	0.142	1.00	0.114	0.40 / 0.97 (-0.57)	0.22 / 0.60 (-0.38)	9.1 (4.5 → 20.7)	yes
ResNeXt-101 32x8d	2048	0.796	0.294 (732)	0.00–1.00 (0.6 pts)	0.228 / 0.511	0.283	0.184	0.99	0.220	0.47 / 1.00 (-0.53)	0.30 / 0.61 (-0.32)	7.9 (3.9 → 16.3)	yes
ConvNeXt-L	1536	0.848	0.293 (924)	0.00–1.00 (0.1 pts)	0.196 / 0.528	0.332	0.189	1.00	0.079	0.51 / 1.00 (-0.49)	0.28 / 0.66 (-0.38)	9.2 (4.4 → 18.1)	yes

network	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
ResNet-152	0.785	0.293 (781)	0.00–1.00 (1.2 pts)	0.209 / 0.455	0.246	0.165	1.00	0.159	0.51 / 1.00 (-0.49)	0.31 / 0.64 (-0.33)	9.1 (2.1 → 16.0)	yes
EfficientNet-B3	0.818	0.291 (905)	0.00–1.00 (0.2 pts)	0.204 / 0.514	0.310	0.191	1.00	0.141	0.49 / 0.99 (-0.50)	0.34 / 0.74 (-0.39)	12.5 (6.8 → 26.0)	yes
ResNet-101	0.777	0.284 (770)	0.00–1.00 (1.4 pts)	0.216 / 0.493	0.276	0.206	1.00	0.143	0.48 / 1.00 (-0.52)	0.30 / 0.60 (-0.30)	8.7 (5.1 → 14.8)	yes
Wide-ResNet-101-2	0.790	0.279 (746)	0.00–1.00 (0.7 pts)	0.201 / 0.431	0.231	0.155	1.00	0.119	0.39 / 0.99 (-0.60)	0.26 / 0.57 (-0.31)	7.9 (3.4 → 18.5)	yes
DenseNet-201	0.764	0.273 (792)	0.00–1.00 (1.5 pts)	0.206 / 0.540	0.334	0.192	1.00	0.085	0.51 / 1.00 (-0.49)	0.31 / 0.59 (-0.28)	8.5 (3.0 → 18.2)	yes
EfficientNet-B0	0.777	0.271 (835)	0.00–1.00 (0.4 pts)	0.175 / 0.502	0.327	0.193	1.00	0.110	0.45 / 0.99 (-0.54)	0.31 / 0.67 (-0.36)	7.1 (3.6 → 19.0)	yes
ConvNeXt-B	0.843	0.269 (921)	0.00–1.00 (0.2 pts)	0.178 / 0.512	0.334	0.195	1.00	0.114	0.53 / 1.00 (-0.47)	0.25 / 0.70 (-0.45)	8.0 (2.1 → 21.6)	yes
MobileNet-V3-L	0.741	0.265 (796)	0.00–1.00 (1.2 pts)	0.156 / 0.591	0.435	0.263	1.00	0.102	0.53 / 1.00 (-0.47)	0.33 / 0.63 (-0.29)	10.8 (4.0 → 29.0)	yes

network	D	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
AlexNet	4096	0.552	0.261 (735)	0.00–1.00 (1.4 pts)	0.120 / 0.624	0.504	0.293	1.00	0.142	0.55 / 1.00 (-0.45)	0.34 / 0.58 (-0.24)	7.1 (3.0 → 14.3)	yes
RegNet-Y-16GF	3024	0.808	0.261 (797)	0.00–1.00 (0.6 pts)	0.180 / 0.480	0.300	0.174	1.00	0.208	0.50 / 1.00 (-0.50)	0.33 / 0.63 (-0.30)	7.7 (4.6 → 15.3)	yes
ResNet-34	512	0.732	0.259 (791)	0.00–1.00 (2.7 pts)	0.175 / 0.528	0.353	0.227	1.00	0.084	0.47 / 1.00 (-0.53)	0.24 / 0.56 (-0.31)	6.1 (3.5 → 15.9)	yes
GoogLeNet	1024	0.691	0.251 (876)	0.00–1.00 (1.4 pts)	0.187 / 0.483	0.296	0.189	1.00	0.056	0.47 / 0.99 (-0.52)	0.29 / 0.61 (-0.32)	9.5 (5.4 → 22.1)	yes
ConvNeXt-S	768	0.840	0.248 (901)	0.00–1.00 (0.1 pts)	0.170 / 0.458	0.287	0.155	1.00	0.124	0.47 / 1.00 (-0.53)	0.23 / 0.67 (-0.44)	8.1 (3.6 → 16.9)	yes
RegNet-Y-8GF	2016	0.802	0.244 (788)	0.00–1.00 (0.8 pts)	0.151 / 0.516	0.365	0.241	1.00	0.140	0.50 / 1.00 (-0.50)	0.31 / 0.67 (-0.36)	6.7 (1.8 → 15.9)	yes
ConvNeXt-T	768	0.830	0.241 (924)	0.00–1.00 (0.4 pts)	0.154 / 0.507	0.353	0.200	1.00	0.125	0.50 / 1.00 (-0.50)	0.29 / 0.71 (-0.42)	7.0 (2.4 → 22.3)	yes
Swin-S	768	0.836	0.236 (880)	0.00–1.00 (0.2 pts)	0.154 / 0.440	0.287	0.169	1.00	0.077	0.44 / 0.99 (-0.55)	0.25 / 0.68 (-0.42)	8.8 (5.3 → 16.6)	yes

network	D	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
ShuffleNet V2 x1.0	1024	0.686	0.234 (781)	0.00–1.00 (2.5 pts)	0.146 / 0.527	0.381	0.245	1.00	0.034	0.57 / 1.00 (-0.43)	0.34 / 0.60 (-0.26)	9.2 (4.0 → 25.7)	yes
Swin-B	1024	0.839	0.233 (891)	0.00–1.00 (0.1 pts)	0.177 / 0.406	0.229	0.128	1.00	0.071	0.42 / 1.00 (-0.57)	0.23 / 0.63 (-0.40)	7.7 (4.0 → 17.0)	yes
ResNet-18	512	0.695	0.233 (849)	0.00–1.00 (3.8 pts)	0.163 / 0.479	0.316	0.195	1.00	0.050	0.47 / 0.99 (-0.53)	0.25 / 0.59 (-0.34)	6.3 (3.2 → 16.2)	yes
Wide-ResNet-50-2	2048	0.785	0.230 (792)	0.00–1.00 (0.6 pts)	0.166 / 0.427	0.261	0.177	0.99	0.105	0.38 / 0.99 (-0.62)	0.23 / 0.60 (-0.37)	5.0 (2.4 → 11.6)	yes
MNASNet 1.0	1280	0.734	0.226 (766)	0.00–1.00 (1.3 pts)	0.134 / 0.530	0.395	0.251	1.00	0.042	0.47 / 1.00 (-0.53)	0.29 / 0.66 (-0.37)	8.7 (3.7 → 19.7)	yes
Swin-T	768	0.820	0.225 (902)	0.00–1.00 (0.2 pts)	0.150 / 0.460	0.309	0.165	1.00	0.084	0.48 / 1.00 (-0.52)	0.29 / 0.68 (-0.39)	8.4 (4.2 → 18.4)	yes
DenseNet 121	1024	0.740	0.224 (839)	0.00–1.00 (1.9 pts)	0.147 / 0.494	0.347	0.211	1.00	0.071	0.45 / 1.00 (-0.55)	0.23 / 0.56 (-0.33)	5.7 (3.7 → 14.9)	yes
ResNet-50	2048	0.760	0.222 (814)	0.00–1.00 (1.5 pts)	0.157 / 0.447	0.290	0.203	1.00	0.130	0.43 / 1.00 (-0.56)	0.29 / 0.63 (-0.34)	5.1 (2.7 → 12.2)	yes

network	D	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
ResNeXt-50 32x4d	2048	0.774	0.219 (776)	0.00–1.00 (0.9 pts)	0.145 / 0.461	0.316	0.201	1.00	0.193	0.41 / 0.99 (-0.58)	0.25 / 0.65 (-0.39)	5.6 (2.1 → 14.1)	yes
EfficientNet-V2-S	1280	0.832	0.184 (879)	0.00–1.00 (0.2 pts)	0.134 / 0.332	0.198	0.084	1.00	0.074	0.35 / 0.99 (-0.64)	0.22 / 0.74 (-0.52)	5.6 (3.6 → 8.1)	yes
MobileNet-V2	1280	0.715	0.184 (798)	0.00–1.00 (4.4 pts)	0.128 / 0.474	0.346	0.186	1.00	0.051	0.48 / 1.00 (-0.52)	0.27 / 0.65 (-0.38)	5.6 (1.9 → 18.0)	yes
VGG-19-BN	4096	0.743	0.150 (867)	0.00–1.00 (0.8 pts)	0.094 / 0.335	0.241	0.149	1.00	0.093	0.37 / 0.99 (-0.62)	0.27 / 0.70 (-0.43)	4.0 (2.2 → 6.8)	yes
VGG-16-BN	4096	0.733	0.115 (915)	0.00–1.00 (0.7 pts)	0.068 / 0.294	0.226	0.130	1.00	0.104	0.36 / 1.00 (-0.64)	0.24 / 0.70 (-0.45)	2.4 (2.9 → 5.2)	yes
VGG-16	4096	0.711	0.110 (918)	0.00–1.00 (0.5 pts)	0.061 / 0.329	0.268	0.137	1.00	0.093	0.33 / 0.99 (-0.66)	0.23 / 0.72 (-0.50)	3.4 (2.3 → 6.4)	yes
VGG-11-BN	4096	0.703	0.105 (927)	0.00–1.00 (1.0 pts)	0.059 / 0.294	0.235	0.110	1.00	0.092	0.42 / 1.00 (-0.57)	0.27 / 0.69 (-0.42)	4.2 (2.3 → 4.8)	yes

network	top-1 (subset)	trained rate (images decided)	gate: w-range (largest Δtop-1)	rate at texture end / shape end	gated span, subspace rule	gated span, exemplars held out	ρ(w, rate)	gated span, unit rule	probe: shape / texture accuracy (bias)	probe, exemplars held out: shape / texture (bias)	κ with observers, % of ceiling: trained (texture end → shape end)	whole sweep in gate
same architecture, different training recipe												
ResNet-2048 50 (V2 recipe)	0.814	0.185 (874)	0.00–1.00 (0.2 pts)	0.131 / 0.465	0.334	0.195	1.00	0.056	0.38 / 1.00 (-0.62)	0.26 / 0.72 (-0.47)	0.5 (1.9 → 7.8)	yes
Stylized-ImageNet controls (Geirhos et al., 2019)												
ResNet-2048 50 SIN	0.582	0.810 (758)	0.00–1.00 (3.9 pts)	0.701 / 0.867	0.166	0.124	0.99	0.140	0.79 / 0.90 (-0.11)	0.54 / 0.30 (+0.23)	39.5 (28.7 → 49.7)	yes
ResNet-2048 50 SIN+IN	0.743	0.363 (802)	0.00–1.00 (2.4 pts)	0.256 / 0.606	0.350	0.238	1.00	0.170	0.59 / 0.99 (-0.40)	0.39 / 0.53 (-0.14)	11.7 (7.1 → 18.5)	yes
ResNet-2048 50 SIN+IN, ft IN	0.770	0.227 (797)	0.00–1.00 (1.9 pts)	0.170 / 0.447	0.277	0.175	1.00	0.154	0.41 / 0.99 (-0.58)	0.26 / 0.60 (-0.33)	5.9 (4.6 → 14.7)	yes

Extended Data Table 17 | Error consistency with the ten observers at a matched cue-conflict rate. Every frozen network of Extended Data Table 16 under the same gain (Methods 4.8), swept

over w ∈ {0, 0.05, ..., 1}; for each network the row is the **gated** readout whose rate is closest to 0.40 (tolerance ± 0.05; networks whose gate does not reach it are absent, 5 of 38: EfficientNet-V2-S, VGG-11-BN, VGG-16-BN, VGG-16, VGG-19-BN). κ is Cohen's κ on trial-by-trial agreement over the images both parties decided, mean over the ten observers, as a percentage of the human–human ceiling $\kappa = 0.234$ (Methods 4.3; Extended Data Table 7). "κ at the trained readout" is the same quantity at w = 0.5, the number Extended Data Table 16 reports and the form in which the field reports error consistency. "κ / κ_max" normalises by the largest κ attainable at the two marginals, κ_max = (1 − |p_model − p_human| − p_e)/(1 − p_e), expressed as a percentage of the same ratio for human pairs (35% of their own bound); values above 100% mean a network attains a larger share of its much smaller attainable agreement than two observers do of theirs, not that it is more human-like than a human. Over the 798 gated readouts of the 38 ImageNet-trained networks, raw κ is a function of the rate alone (one quadratic in the rate, $R^2 = 0.90$, RMSE 1.7 points of ceiling; per-network offsets −2.2 to 2.8 points, unrelated to the probe bias, $\rho = 0.04$), the bound itself running from 0.006 to 0.219. Normalised by the bound, the rank dependence on the rate disappears (pooled rank R^2 0.001 against 0.891; within-network median Spearman ρ 0.20 against 0.99). Ranking the 33 networks by κ at their trained readouts against by κ here gives Kendall $\tau = 0.33$ with 178 of 528 pairs (34%) reversing. The Stylized-ImageNet ResNet-50 cannot be gained below 0.65 inside its gate; at a common rate of 0.65 it is ResNet-50 SIN 135% [104, 165]; ViT-B/32 120% [64, 174]; AlexNet 79% [61, 96] on the normalised scale (95% percentile bootstrap over the ten observers). Pre-specification: `docs/PLAN_KAPPA_2026-09-17.md` (predictions K-1 to K-6: four failed and a fifth could not be tested as specified; the normalisation by κ_max is post hoc). Source: `results/figures_v2/kappa_rate.json`, aggregated by `tmp_analysis/figs_v2/kappa_rate.py` from the per-network sweeps of `src/analysis/generality/kappa_sweep.py` (`slurm/zoo_kappa.sh`); the per-network files are outside the repository. * not ImageNet-trained.

network	w at the matched rate	rate	κ, % of human–human ceiling	κ at the trained readout	κ / κ_max, % of the human-pair value	probe bias	trained rate
GoogLeNet	0.90	0.397	18.7	9.5	185.9	−0.52	0.251
ShuffleNet-V2 x1.0	0.80	0.401	18.1	9.2	158.7	−0.43	0.234
Inception-v3	0.80	0.406	17.8	11.0	160.2	−0.56	0.305
Swin-S	0.95	0.413	17.5	8.9	172.7	−0.55	0.236
MobileNet-V3-L	0.80	0.407	17.0	10.8	173.6	−0.47	0.265
Swin-B	1.00	0.418	17.0	7.7	169.1	−0.57	0.233
MaxViT-T	0.85	0.401	16.0	9.1	171.2	−0.57	0.295
Swin-T	0.90	0.407	15.4	8.4	163.6	−0.52	0.225
ViT-L/16	0.75	0.408	15.4	11.5	168.1	−0.43	0.319
EfficientNet-B3	0.75	0.396	15.1	12.5	138.4	−0.50	0.291
DenseNet-201	0.85	0.403	14.9	8.5	137.8	−0.49	0.273
MobileNet-V2	0.95	0.389	14.4	5.6	139.4	−0.52	0.184
Wide-ResNet-101-2	0.90	0.402	13.9	7.9	137.8	−0.60	0.279
EfficientNet-B0	0.85	0.407	13.9	7.1	130.2	−0.54	0.271

network	w at the matched rate	rate	κ, % of human–human ceiling	κ at the trained readout	κ / κ_max, % of the human-pair value	probe bias	trained rate
ResNet-101	0.95	0.405	13.8	8.7	150.6	−0.52	0.284
ResNet-152	0.90	0.402	13.7	9.1	126.5	−0.49	0.293
ResNet-50 SIN+IN, ft IN *	0.95	0.397	13.5	5.9	127.8	−0.58	0.227
ResNet-50 SIN+IN *	0.60	0.399	13.3	11.7	140.6	−0.40	0.363
RegNet-Y-16GF	0.90	0.409	13.0	7.7	131.8	−0.50	0.261
ConvNeXt-B	0.80	0.398	13.0	8.0	132.1	−0.47	0.269
ViT-B/32	0.00	0.418	12.9	23.0	130.4	−0.30	0.617
ResNeXt-50 32x4d	0.90	0.383	12.9	5.6	130.6	−0.58	0.219
ResNet-18	0.90	0.407	12.7	6.3	107.3	−0.53	0.233
ResNet-34	0.85	0.403	12.5	6.1	131.0	−0.53	0.259
DenseNet-121	0.90	0.402	12.5	5.7	114.0	−0.55	0.224
MNASNet-1.0	0.85	0.406	12.5	8.7	113.7	−0.53	0.226
ConvNeXt-S	0.85	0.399	12.4	8.1	122.7	−0.53	0.248
ConvNeXt-T	0.85	0.395	12.3	7.0	115.3	−0.50	0.241
ResNet-50	1.00	0.394	12.2	5.1	116.9	−0.56	0.222
AlexNet	0.75	0.393	12.0	7.1	124.8	−0.45	0.261
ConvNeXt-L	0.80	0.401	11.8	9.2	122.4	−0.49	0.293
ViT-B/16	0.50	0.399	11.6	11.6	126.2	−0.43	0.399
RegNet-Y-8GF	0.90	0.407	11.5	6.7	103.1	−0.50	0.244
Wide-ResNet-50-2	0.95	0.405	10.5	5.0	100.5	−0.62	0.230
ResNeXt-101 32x8d	0.80	0.400	10.0	7.9	107.5	−0.53	0.294
ResNet-50 (V2 recipe) *	1.00	0.356	7.8	0.5	70.8	−0.62	0.185

Extended Data Table 18 | What the gain acts on: the span decomposed. Every frozen network of Extended Data Table 16, at its trained readout. For each cue-conflict image the Geirhos rule gives a margin $\Delta = \log$ p_shape $-\log$ p_texture (p_c the mean softmax probability over category c's ImageNet classes; the forced two-way estimator calls the image "shape" iff $\Delta > 0$). The gain of Methods 4.8 adds $(2w-1)\cdot W(P_S - P_T)y$ to the logits exactly, so to first order it changes Δ by $(2w-1)(e_S - e_T)$, with e_S and e_T the evidence the head receives through the shape and the texture subspace (u·W P_S y and u·W P_T y, u the gradient of Δ with respect to the logits). "Shape term" is the mean of e_S over the 1,200 images (evidence added at the shape end), "texture

term" the mean of −e_T (texture evidence removed there), both in units of the margin; texture share = texture term / (shape term + texture term). M0 is the first-order forced-choice span from the per-image margins, M1 the two-number span $\Phi(z + \lambda) - \Phi(z - \lambda)$ with z = mean Δ / sd Δ and λ = mean(e_S − e_T) / sd Δ. "Reported" is the held-out span of Extended Data Table 16 (primary estimator, on halves that share exemplars); "in sample" is the same estimator and "forced" the forced two-way estimator, both with the bases fitted on all 1,200 images. Random-direction span: ten draws (seeds 0–9) of two mutually orthogonal random 15-dimensional subspaces inside each network's 256-dimensional PCA space of the cue-conflict features, same gain, estimator (in sample) and gate; every random sweep stays inside its gate. Pre-specification: `docs/PLAN_SPAN_2026-09-18.md`, predictions S-1 to S-7, written after Extended Data Table 16 and before this analysis; the whole analysis is post hoc with respect to the manuscript's first three plans. Met: S-1 (M0 reproduces the forced-choice curve, median largest error 0.009; ρ(M0, forced span) = 0.99), S-5b (the Stylized-ImageNet texture term 0.31 is below every ImageNet network's, 0.93–3.95) and S-7 (median absolute mean signed random span 0.002; the reported span exceeds the largest random span in 38 of 38 ImageNet networks and 42 of 42). Failed: S-2 (M1 ranks the forced spans at $\rho = 0.95$ but the reported spans at 0.59, against 0.7), S-3 ($\rho(\lambda$, reported span) = 0.56, against 0.7), S-4 ($\rho(\lambda$, z) = −0.32, against $|\rho| \leq 0.3$), S-5a (the Stylized-ImageNet λ 0.53 lies inside the ImageNet range 0.43–0.85) and S-6 (a decision-free head-weighted subspace share ranks the reported spans at $\rho = 0.35$, against 0.5, no better than the unweighted variance share, 0.40). Post hoc: the reported and the forced spans rank the 38 ImageNet networks at $\rho = 0.58$ (in-sample primary 0.81 with the reported and 0.89 with the forced); the forced span rises with the trained forced rate ($\rho = 0.50$ [0.20, 0.73]) where the reported one does not (0.04); among the 39 networks (the 38 and itself) the Stylized-ImageNet span is the smallest reported, the 9th smallest in sample and the 16th smallest forced, and the 3rd smallest with exemplars held out. The texture term exceeds the shape term in 37 of 38 ImageNet networks (texture share median 0.89, range 0.45–0.95; the exception is ViT-B/32), and the texture share tracks the probe bias of Extended Data Table 16 (ρ = −0.81 [−0.91, −0.62]; −0.86 over all 42) more closely than the trained rate does (0.59 [0.29, 0.80]; difference in $|\rho|$ 0.22 [0.04, 0.44]); against the exemplar-balanced probe scored on held-out exemplars (Extended Data Table 16) it is −0.58 [−0.78, −0.29], no closer than the trained rate. Intervals are 95% percentile bootstraps over networks (10,000 resamples), descriptive of this fixed zoo. On the forced estimator the ranking by trained rate and by the shape-end rate agree at Kendall τ = 0.58 with 146 of 703 pairs (21%) reversing (in-sample primary $\tau = 0.50$, 25%; held-out primary 0.47, 27% with exemplars shared and 0.63, 18.6% with exemplars held out). Source: `results/figures_v2/span_model.json`, aggregated by `tmp_analysis/figs_v2/span_model.py` from `src/analysis/generality/span_model.py` (`slurm/zoo_span.sh`); the per-network files are outside the repository.

network	trained rate: primary / forced	span: reported / in sample / forced	forced span, M0 / M1	z / λ	shape term / texture term (texture share)	random-direction span, absolute: median / largest
ImageNet-trained (torchvision IMAGENET1K_V1)						
VGG-11-BN	0.105 / 0.117	0.235 / 0.153 / 0.197	0.206 / 0.300	−1.11 / 0.69	0.17 / 3.24 (0.95)	0.007 / 0.018

network	trained rate: primary / forced	span: reported / in sample / forced	forced span, M0 / M1	z / λ	shape term / texture term (texture share)	random-direction span, absolute: median / largest
VGG-16-BN	0.115 / 0.117	0.226 / 0.157 / 0.177	0.174 / 0.289	−1.07 / 0.64	0.17 / 3.01 (0.95)	0.012 / 0.025
Wide-ResNet-50-2	0.230 / 0.225	0.261 / 0.274 / 0.306	0.312 / 0.348	−0.61 / 0.54	0.16 / 2.71 (0.94)	0.012 / 0.036
VGG-16	0.110 / 0.115	0.268 / 0.216 / 0.253	0.253 / 0.336	−1.04 / 0.72	0.25 / 3.95 (0.94)	0.006 / 0.020
VGG-19-BN	0.150 / 0.149	0.241 / 0.177 / 0.190	0.195 / 0.296	−0.90 / 0.56	0.19 / 2.88 (0.94)	0.009 / 0.023
Wide-ResNet-101-2	0.279 / 0.253	0.231 / 0.271 / 0.287	0.292 / 0.309	−0.50 / 0.45	0.19 / 2.42 (0.93)	0.008 / 0.022
EfficientNet-V2-S	0.184 / 0.183	0.198 / 0.189 / 0.258	0.273 / 0.393	−0.76 / 0.68	0.16 / 1.87 (0.92)	0.006 / 0.016
ResNeXt-50 32x4d	0.219 / 0.219	0.316 / 0.298 / 0.305	0.307 / 0.361	−0.66 / 0.58	0.27 / 3.19 (0.92)	0.014 / 0.026
ResNet-50	0.222 / 0.210	0.290 / 0.254 / 0.264	0.276 / 0.328	−0.74 / 0.55	0.28 / 2.72 (0.91)	0.005 / 0.019
Swin-S	0.236 / 0.238	0.287 / 0.322 / 0.362	0.375 / 0.445	−0.59 / 0.70	0.23 / 2.15 (0.90)	0.011 / 0.022
GoogLeNet	0.251 / 0.247	0.296 / 0.278 / 0.278	0.286 / 0.387	−0.58 / 0.60	0.21 / 1.95 (0.90)	0.014 / 0.033
ResNet-18	0.233 / 0.232	0.316 / 0.326 / 0.320	0.330 / 0.390	−0.67 / 0.64	0.39 / 3.46 (0.90)	0.014 / 0.031
Swin-B	0.233 / 0.240	0.229 / 0.263 / 0.333	0.349 / 0.416	−0.60 / 0.65	0.24 / 2.15 (0.90)	0.012 / 0.021
ResNet-34	0.259 / 0.242	0.353 / 0.344 / 0.351	0.349 / 0.434	−0.66 / 0.71	0.38 / 3.36 (0.90)	0.003 / 0.032
ResNet-101	0.284 / 0.258	0.276 / 0.219 / 0.203	0.215 / 0.292	−0.58 / 0.44	0.26 / 2.21 (0.89)	0.006 / 0.019
DenseNet-121	0.224 / 0.202	0.347 / 0.363 / 0.355	0.349 / 0.432	−0.70 / 0.72	0.45 / 3.78 (0.89)	0.014 / 0.038
MobileNet-V2	0.184 / 0.185	0.346 / 0.330 / 0.306	0.318 / 0.405	−0.82 / 0.73	0.41 / 3.41 (0.89)	0.009 / 0.034
Swin-T	0.225 / 0.224	0.309 / 0.354 / 0.408	0.428 / 0.480	−0.63 / 0.78	0.29 / 2.34 (0.89)	0.011 / 0.017
ConvNeXt-T	0.241 / 0.246	0.353 / 0.379 / 0.393	0.415 / 0.502	−0.65 / 0.83	0.28 / 2.26 (0.89)	0.011 / 0.021
ConvNeXt-S	0.248 / 0.234	0.287 / 0.361 / 0.413	0.424 / 0.497	−0.62 / 0.81	0.28 / 2.23 (0.89)	0.006 / 0.019
RegNet-Y-8GF	0.244 / 0.227	0.365 / 0.324 / 0.337	0.332 / 0.382	−0.64 / 0.61	0.44 / 3.21 (0.88)	0.008 / 0.036
Inception-v3	0.305 / 0.283	0.246 / 0.305 / 0.302	0.300 / 0.360	−0.41 / 0.51	0.24 / 1.68 (0.88)	0.013 / 0.039
ConvNeXt-B	0.269 / 0.261	0.334 / 0.410 / 0.425	0.437 / 0.502	−0.56 / 0.79	0.31 / 2.18 (0.88)	0.009 / 0.019
DenseNet-201	0.273 / 0.228	0.334 / 0.325 / 0.338	0.333 / 0.419	−0.60 / 0.66	0.51 / 3.50 (0.87)	0.010 / 0.025
MNASNet-1.0	0.226 / 0.210	0.395 / 0.391 / 0.349	0.352 / 0.436	−0.73 / 0.75	0.55 / 3.62 (0.87)	0.009 / 0.022

network	trained rate: primary / forced	span: reported / in sample / forced	forced span, M0 / M1	z / λ	shape term / texture term (texture share)	random-direction span, absolute: median / largest
ResNeXt-101 32x8d	0.294 / 0.265	0.283 / 0.303 / 0.322	0.326 / 0.363	−0.48 / 0.53	0.44 / 2.89 (0.87)	0.017 / 0.036
ConvNeXt-L	0.293 / 0.289	0.332 / 0.360 / 0.372	0.388 / 0.476	−0.51 / 0.72	0.35 / 2.19 (0.86)	0.009 / 0.023
MaxViT-T	0.295 / 0.297	0.239 / 0.272 / 0.338	0.353 / 0.406	−0.43 / 0.58	0.33 / 1.88 (0.85)	0.005 / 0.017
ResNet-152	0.293 / 0.255	0.246 / 0.235 / 0.219	0.226 / 0.286	−0.55 / 0.43	0.38 / 2.12 (0.85)	0.013 / 0.030
ShuffleNet-V2 x1.0	0.234 / 0.248	0.381 / 0.414 / 0.350	0.352 / 0.414	−0.62 / 0.66	0.48 / 2.71 (0.85)	0.010 / 0.069
RegNet-Y-16GF	0.261 / 0.242	0.300 / 0.291 / 0.317	0.318 / 0.352	−0.56 / 0.53	0.53 / 2.93 (0.85)	0.021 / 0.076
AlexNet	0.261 / 0.244	0.504 / 0.497 / 0.447	0.435 / 0.502	−0.68 / 0.85	0.77 / 3.50 (0.82)	0.010 / 0.020
EfficientNet-B0	0.271 / 0.278	0.327 / 0.361 / 0.361	0.373 / 0.450	−0.49 / 0.67	0.43 / 1.94 (0.82)	0.010 / 0.016
EfficientNet-B3	0.291 / 0.275	0.310 / 0.361 / 0.386	0.395 / 0.465	−0.42 / 0.68	0.46 / 1.88 (0.80)	0.010 / 0.022
MobileNet-V3-L	0.265 / 0.268	0.435 / 0.450 / 0.434	0.437 / 0.508	−0.57 / 0.81	0.92 / 3.50 (0.79)	0.009 / 0.020
ViT-L/16	0.319 / 0.310	0.337 / 0.363 / 0.408	0.421 / 0.481	−0.40 / 0.70	0.59 / 2.03 (0.77)	0.008 / 0.017
ViT-B/16	0.399 / 0.362	0.282 / 0.357 / 0.376	0.376 / 0.422	−0.24 / 0.57	0.61 / 1.48 (0.71)	0.005 / 0.015
ViT-B/32	0.617 / 0.520	0.239 / 0.356 / 0.415	0.418 / 0.453	0.14 / 0.61	1.15 / 0.93 (0.45)	0.008 / 0.023
same architecture, different training recipe						
ResNet-50 (V2 recipe)	0.185 / 0.191	0.334 / 0.234 / 0.288	0.304 / 0.389	−0.72 / 0.66	0.12 / 1.48 (0.92)	0.013 / 0.024
Stylized-ImageNet controls (Geirhos et al., 2019)						
ResNet-50 SIN+IN, ft IN	0.227 / 0.223	0.277 / 0.265 / 0.273	0.267 / 0.341	−0.66 / 0.55	0.26 / 2.75 (0.91)	0.011 / 0.029
ResNet-50 SIN+IN	0.363 / 0.327	0.350 / 0.358 / 0.370	0.373 / 0.428	−0.37 / 0.60	0.73 / 2.20 (0.75)	0.019 / 0.026
ResNet-50 SIN	0.810 / 0.703	0.166 / 0.255 / 0.318	0.328 / 0.347	0.58 / 0.53	2.43 / 0.31 (0.11)	0.015 / 0.028

References

Allen, E. J., St-Yves, G., Wu, Y., Breedlove, J. L., Prince, J. S., Dowdle, L. T., Nau, M., Caron, B., Pestilli, F., Charest, I., Hutchinson, J. B., Naselaris, T., & Kay, K. (2022). A massive 7T fMRI dataset to bridge cognitive neuroscience and artificial intelligence. *Nature Neuroscience*, 25(1), 116–126. https://doi.org/10.1038/s41593-021-00962-x

Bakhtiari, S., Mineault, P., Lillicrap, T., Pack, C., & Richards, B. (2021). The functional specialization of visual cortex emerges from training parallel pathways with self-supervised predictive learning. *Advances in Neural Information Processing Systems*, 34, 25164–25178.

Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1), 37–46. https://doi.org/10.1177/001316446002000104

Derrington, A. M., & Lennie, P. (1984). Spatial and temporal contrast sensitivities of neurones in lateral geniculate nucleus of macaque. *The Journal of Physiology*, 357(1), 219–240. https://doi.org/10.1113/jphysiol.1984.sp015498

Derrington, A. M., Krauskopf, J., & Lennie, P. (1984). Chromatic mechanisms in lateral geniculate nucleus of macaque. *The Journal of Physiology*, 357(1), 241–265. https://doi.org/10.1113/jphysiol.1984.sp015499

Dupré la Tour, T., Eickenberg, M., Nunez-Elizalde, A. O., & Gallant, J. L. (2022). Feature-space selection with banded ridge regression. *NeuroImage*, 264, 119728. https://doi.org/10.1016/j.neuroimage.2022.119728

Ferrera, V. P., Nealey, T. A., & Maunsell, J. H. R. (1994). Responses in macaque visual area V4 following inactivation of the parvocellular and magnocellular LGN pathways. *Journal of Neuroscience*, 14(4), 2080–2088. https://doi.org/10.1523/JNEUROSCI.14-04-02080.1994

Gao, P., Trautmann, E., Yu, B., Santhanam, G., Ryu, S., Shenoy, K., & Ganguli, S. (2017). A theory of multineuronal dimensionality, dynamics and measurement. *bioRxiv*. https://doi.org/10.1101/214262

Gavrikov, P., Lukasik, J., Jung, S., Geirhos, R., Lamm, B., Mirza, M. J., Keuper, M., & Keuper, J. (2024). Are vision language models texture or shape biased and can we steer them? *arXiv preprint*, arXiv:2403.09193.

Geirhos, R., Rubisch, P., Michaelis, C., Bethge, M., Wichmann, F. A., & Brendel, W. (2018a). ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. *International Conference on Learning Representations* (ICLR 2019). https://openreview.net/forum?id=Bygh9j09KX

Geirhos, R., Meding, K., & Wichmann, F. A. (2020). Beyond accuracy: Quantifying trial-by-trial behaviour of CNNs and humans by measuring error consistency. *Advances in Neural Information Processing Systems*, 33. https://doi.org/10.48550/arXiv.2006.16736

Geirhos, R., Narayanappa, K., Mitzkus, B., Thieringer, T., Bethge, M., Wichmann, F. A., & Brendel, W. (2021). Partial success in closing the gap between human and machine vision. *Advances in Neural Information Processing Systems*, 34, 23885–23899. https://doi.org/10.48550/arXiv.2106.07411

Goodale, M. A., & Milner, A. D. (1992). Separate visual pathways for perception and action. *Trends in Neurosciences*, 15(1), 20–25. https://doi.org/10.1016/0166-2236(92)90344-8

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *IEEE Conference on Computer Vision and Pattern Recognition*, 770–778. https://doi.org/10.1109/CVPR.2016.90

Hermann, K., Chen, T., & Kornblith, S. (2020). The origins and prevalence of texture bias in convolutional neural networks. *Advances in Neural Information Processing Systems*, 33. https://doi.org/10.48550/arXiv.1911.09071

Islam, M. A., Kowal, M., Esser, P., Jia, S., Ommer, B., Derpanis, K. G., & Bruce, N. (2021). Shape or texture: Understanding discriminative features in CNNs. *International Conference on Learning Representations*. https://doi.org/10.48550/arXiv.2101.11604

Jagadeesh, A. V., & Gardner, J. L. (2022). Texture-like representation of objects in human visual cortex. *Proceedings of the National Academy of Sciences*, 119, e2115302119. https://doi.org/10.1073/pnas.2115302119

Jagadeesh, A. V., & Livingstone, M. S. (2024). Texture bias in primate ventral visual cortex. *ICLR 2024 Workshop on Representational Alignment*. https://openreview.net/forum?id=D6YvyvZ7P7

Jang, H., & Tong, F. (2021). Convolutional neural networks trained with a developmental sequence of blurry to clear images reveal core differences between face and object processing. *Journal of Vision*, 21(12), 6. https://doi.org/10.1167/jov.21.12.6

Kaplan, E., & Shapley, R. M. (1982). X and Y cells in the lateral geniculate nucleus of macaque monkeys. *The Journal of Physiology*, 330(1), 125–143. https://doi.org/10.1113/jphysiol.1982.sp014333

Kingma, D. P., & Ba, J. (2017). Adam: A method for stochastic optimization. *arXiv*. https://doi.org/10.48550/arXiv.1412.6980

Komatsu, H., & Goda, N. (2018). Neural mechanisms of material perception: Quest on Shitsukan. *Neuroscience*, 392, 329–347. https://doi.org/10.1016/j.neuroscience.2018.09.001

Kriegeskorte, N., Mur, M., & Bandettini, P. A. (2008). Representational similarity analysis — connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2, 4. https://doi.org/10.3389/neuro.06.004.2008

Lakens, D. (2017). Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4), 355–362. https://doi.org/10.1177/1948550617697177

Lannelongue, L., Grealey, J., & Inouye, M. (2021). Green algorithms: Quantifying the carbon footprint of computation. *Advanced Science*, 8(12), 2100707. https://doi.org/10.1002/advs.202100707

Livingstone, M. S., & Hubel, D. H. (1987). Psychophysical evidence for separate channels for the perception of form, color, movement, and depth. *Journal of Neuroscience*, 7(11), 3416–3468. https://doi.org/10.1523/JNEUROSCI.07-11-03416.1987

Lu, Z., Thorat, S., Cichy, R. M., & Kietzmann, T. C. (2026). Adopting a human developmental visual diet yields robust and shape-based AI vision. *Nature Machine Intelligence*, 8(5), 735–748. https://doi.org/10.1038/s42256-026-01228-6

Manocchio, B. (2026). *Investigating two-stream artificial neural networks using a developmental visual diet* [Bachelor's thesis, University of Osnabrück, Institute of Cognitive Science; supervised by Z. Lu and T. C. Kietzmann].

Mishkin, M., Ungerleider, L. G., & Macko, K. A. (1983). Object vision and spatial vision: Two cortical pathways. *Trends in Neurosciences*, 6, 414–417. https://doi.org/10.1016/0166-2236(83)90190-X

Nassi, J. J., & Callaway, E. M. (2009). Parallel processing strategies of the primate visual system.

Nature Reviews Neuroscience, 10(5), 360–372. https://doi.org/10.1038/nrn2619

Okazawa, G., Tajima, S., & Komatsu, H. (2015). Image statistics underlying natural texture selectivity of neurons in macaque V4. *Proceedings of the National Academy of Sciences*, 112(4), E351–E360. https://doi.org/10.1073/pnas.1415146112

Schuirmann, D. J. (1987). A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6), 657–680. https://doi.org/10.1007/BF01068419

Subramanian, A., Sizikova, E., Majaj, N. J., & Pelli, D. G. (2023). Spatial-frequency channels, shape bias, and adversarial robustness. *Advances in Neural Information Processing Systems*, 36.

Tartaglini, A. R., Vong, W. K., & Lake, B. M. (2022). A developmentally-inspired examination of shape versus texture bias in machines. *Proceedings of the 44th Annual Conference of the Cognitive Science Society*, 1180–1186.

Thorat, S., & Kietzmann, T. C. (2024). MiniEcoset. https://doi.org/10.17605/OSF.IO/MSNA2

Wang, L., Mruczek, R. E. B., Arcaro, M. J., & Kastner, S. (2015). Probabilistic maps of visual topography in human cortex. *Cerebral Cortex*, 25(10), 3911–3931. https://doi.org/10.1093/cercor/bhu277

Supplementary Note S1 — claims deliberately not made

- That the two-stream split is what makes shape bias a readout property: networks with no split show the same dissociation — a reciprocal gain on their shape- and texture-predictive subspaces moves the rate by 0.14–0.20 in six ImageNet networks within 1.5 points of top-1 on exemplars the subspaces never saw (0.25–0.33 with exemplars shared), and in our trunks extrastriate cortex loads on the texture-leaning half relative to V1 — with the cortical part failing under the unit-level rule in all three ImageNet networks fitted (ResNet-50, ViT-B/16 and ConvNeXt-B, the last two post hoc) (Fig. 6; Results 2.4).
- That the two-stream split causes a shape-bias advantage over a single stream: at matched capacity the arms do not separate at $n = 3$, and capacity alone moves the rate by a comparable amount (Results 2.4).
- Any dorsal/ventral stream mapping. The thesis's framing "magno→dorsal, parvo→ventral" was tested rather than dropped: the pre-declared dorsal−ventral interaction is +0.0245 on the diet model and +0.0122 ($p = .195$) without the low-pass front end, so it does not survive its own control. Read directly on the Wang atlas, dorsal (V3A/B, IPS0–2) and ventral (VO1/2, PHC1/2) regions are indistinguishable under both arms (dorsal − ventral +0.015 with the diet, +0.005 without, $p > .1$; Extended Data Fig. 4c): the diet's magno stream contributes no unique variance anywhere beyond V3.
- hMT+ specificity: at $n = 8$ the two statistics disagree ($p = .195$ normalised, .039 raw) and the effect is dead in the blur control.
- A foveal/peripheral headline: the foveal band's sign flips when the blur is removed (−0.057 → +0.051), so it is a front-end artefact.
- That the chromatic stream predicts colour-biased cortex: pooled VO is +0.0045 ($p = .46$), bounded within ±0.017; hV4's parvo preference is weak (−0.012, 2/8). Substituting the koniocellular axis makes the prediction worse, and worse specifically in the colour set (hV4 −0.029, VO −0.024, each $p = .0078$, 0/8).
- Where in cortex the diet's shape-bias effect lands: a diet-specific hV4/LO step does not survive the full control set.

- That integration produces human-like shape bias, or that the diet improves neural predictivity: it does neither.
- That the diet makes the model's image-level difficulty profile more human-like: the crossed 2×2 shows train/test matching, not a weights effect.
- That human cortex uses this mechanism: a single-stream account in which texture-like ventral representations feed shape-biased behaviour directly is not excluded by these data.

Why the three regional claims above failed together, and what it implies for the survivor: without the low-pass front end essentially every ROI is magno-preferring, so the blur manufactures apparent regional specificity by handicapping the magno stream until the parvo stream can win somewhere — the same general-factor pathology that defeated the thesis's RSA, reappearing in encoding form. Any regional claim on these models must be shown to survive removal of the blur, and the ventral gradient is reported because it does.

Supplementary Note S2 — the koniocellular control

The thesis omitted the koniocellular pathway for simplicity. Because L−M fails to predict colour-biased cortex, two readings were live: that the chromatic stream does not predict colour cortex, or that L−M is the wrong chromatic axis. Swapping S−(L+M) into stream 2 of the untrained network, contrast-matched to L−M per image, with the luminance stream bit-identical, makes the ventral prediction worse and worse where colour lives (seed 0: hV4 −0.029, VO −0.024, the pre-declared colour set −0.026, each p = .0078 with 0/8 subjects positive on both statistics; colour set minus the other 17 ROIs −0.014, p = .0078), unchanged when the channel keeps its native 2.05× contrast, and replicated at two further untrained seeds, each against the L−M network of its own seed (colour set −0.027 and −0.020, colour set minus the other ROIs −0.016 and −0.014, every contrast p = .0078 with 0/8 subjects positive on both statistics; `results/konio_verdict/`). Both cone-opponent axes have now been tested against colour-biased cortex and neither predicts it. These fits are untrained, the native-contrast variant was run at seed 0 only, no third stream was added, and no blur was applied to the S channel; whether a three-band model beats a two-band one is a separate question requiring three-way variance partitioning.

Supplementary Note S3 — the cortical gradient against its nuisance axes and the instrument's bias

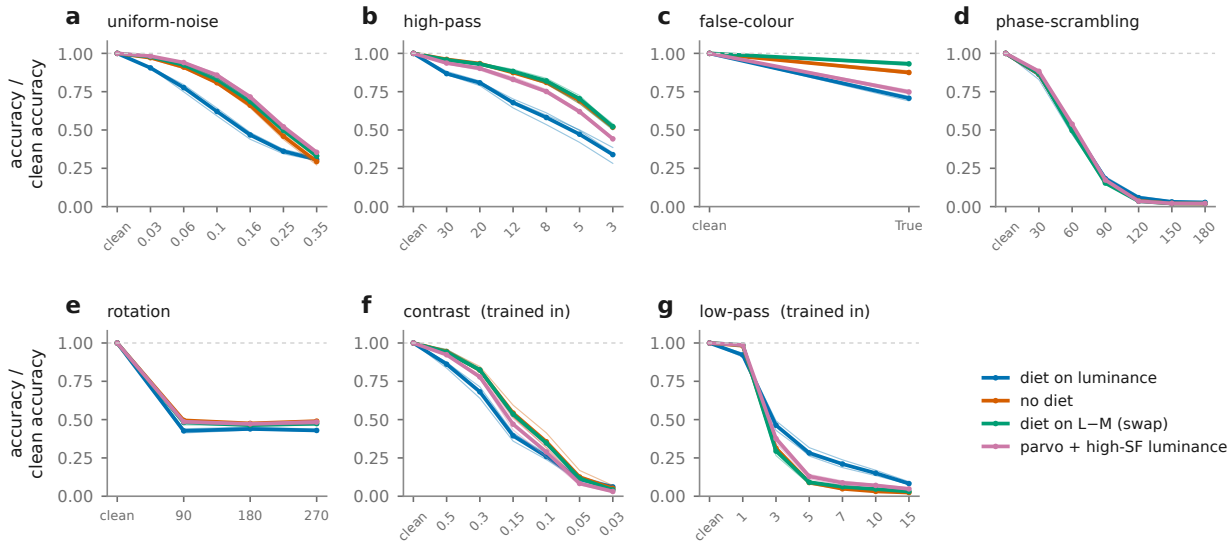
Results 2.3 states the outcome; this note carries the analyses. See Extended Data Fig. 3 and Extended Data Tables 8 and 9.

The two nuisance axes that co-vary with hierarchical rank. Receptive-field size and eccentricity coverage both rise from V1 to hV4, and the diet's magno stream is a low-pass instrument by construction, so a stream-preference gradient that merely tracked receptive-field size would be expected from the front end alone. We tested this three ways (Extended Data Fig. 3; Extended Data Table 9). First, the gradient computed within each of NSD's five eccentricity bands — V1 and hV4 voxels compared only with voxels of the same eccentricity — is negative under the diet in every band (6/7 subjects in the foveal band, 8/8 in the other four) while the no-diet arm stays between −0.10 and +0.07 in every band; the diet arm's gradient does grow with eccentricity (−0.26 in the foveal band, −0.93 beyond 4°), so eccentricity modulates it, but the contrast is present at matched eccentricity and is therefore not an eccentricity contrast in disguise. Second, stratifying the same voxels by pRF size gives the same result within size tertiles (−0.71, −0.77 and −0.90 under the diet against −0.08 in each tertile without it). Third, regressing the per-voxel index on log pRF eccentricity and log pRF size jointly within subject (NSD's pRF estimates, voxels with pRF $R^2 > 5\%$;

the two predictors are only modestly collinear, VIF 1.66) shows that under the diet, stream preference moves toward parvo with increasing pRF size at fixed eccentricity (partial slope −0.086, 8/8 subjects, p = .0078) and toward magno with increasing eccentricity at fixed size (+0.119, 8/8, p = .0078), while the no-diet arm shows neither the size effect in that direction (+0.035, opposite sign) nor any eccentricity effect (−0.004, p = 1.00); the arm interaction on the eccentricity partial is +0.134 (8/8, p = .0078). A learning-free Gabor bank on the diet front end carries no such structure (+0.035, p = .15), so it is not a property of the input filter alone; an untrained network carries it on either front end, however (diet front end +0.083, 7/8, p = .016 on the eccentricity partial and −0.038, 6/8, p = .039 on the size partial; undieted front end +0.089, 7/8, p = .023 and −0.038, 8/8, p = .0078), and training with the diet leaves it where random weights put it — trained minus untrained is +0.036 (6/8, p = .11) on the normalised statistic and −0.077 (2/8, p = .11) on the raw one, opposite in sign and neither significant — while the network trained without the diet carries none of it on either statistic (+0.028 raw, −0.004 normalised; the paired decrease from the untrained network is significant on the raw statistic only, −0.374, 0/8, p = .0078, against −0.087, 2/8, p = .15). The diet preserves a dissociation that random convolutional weights already produce; it neither installs nor amplifies it (Extended Data Table 9). Within the diet arm, then, the ventral gradient and the pRF-size dependence are the same fact seen on two axes — larger receptive fields, further along the hierarchy, are increasingly explained by the texture-biased stream — and the network trained without the diet shows neither.

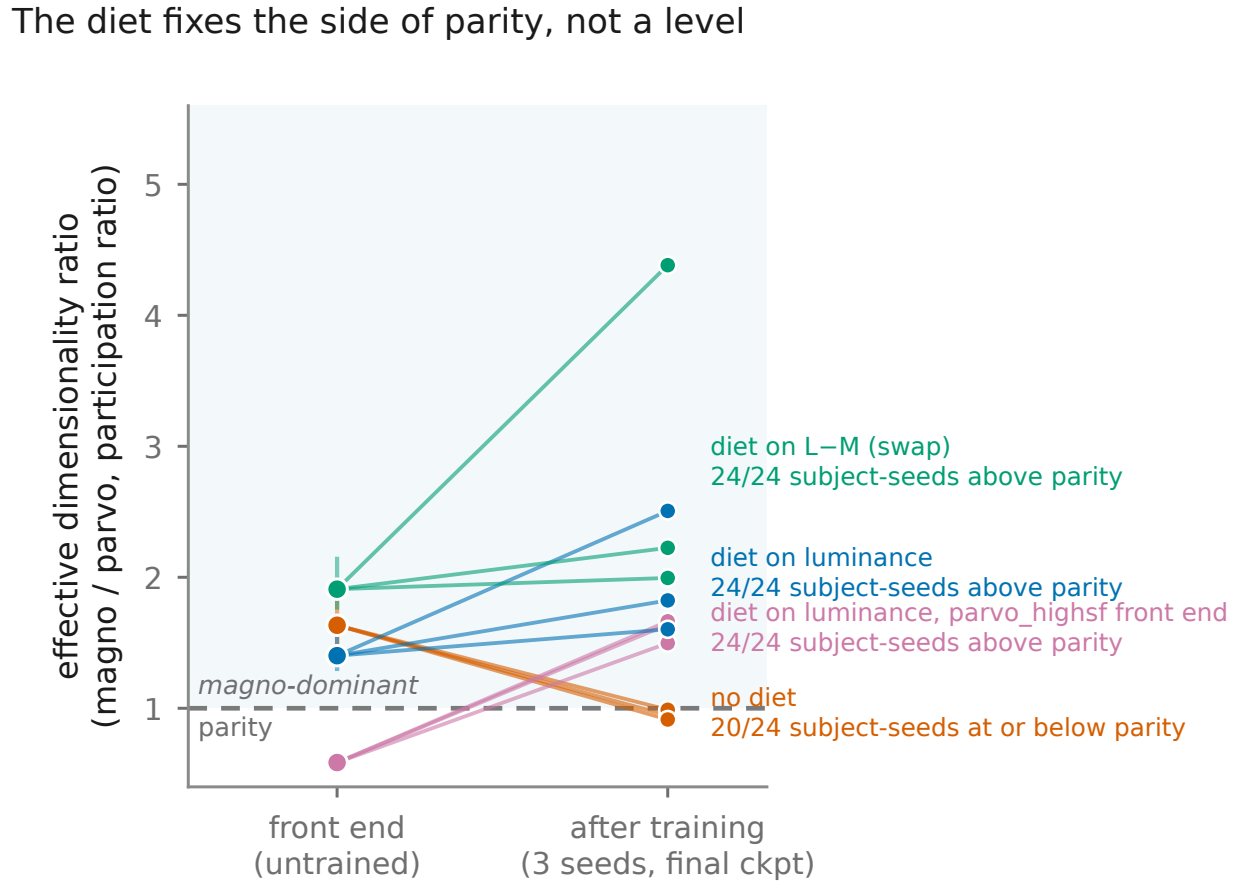
The estimator's own bias points the wrong way. The mean noise ceiling declines from V1v to hV4 ($\rho = -0.89$ against rank), a nuisance falling along the axis of the effect, so we simulated the instrument: synthetic voxels from each arm's fitted feature spaces, the measured noise ceilings imposed, the true magno − parvo difference set to zero everywhere. The recovered contrast is a small, positive floor (+0.038 with the diet's feature geometry, +0.051 without, 8/8 subjects each), flat across ROIs (+0.025 to +0.027 per ROI), and indistinguishable from zero on the raw statistic — opposite in sign to the observed effect and without its gradient. Because the floor is a signed bias, the right comparison is a paired displacement: the real contrast minus each subject's own floor is −0.871 with the diet and −0.130 without, each non-zero in every subject, the diet arm's 6.7 times larger (Extended Data Table 8). The floor subtraction relocates the recovery simulation's residual misfit rather than removing it, and we do not correct that residual, because the correction is under-specified: six variants swing the no-diet arm between 4/8 and 6/8 subjects and between p = .055 and p = 1.00, while the diet arm is non-zero in all eight subjects at p = .0078 under all six (Extended Data Table 8). The no-diet arm's non-zero gradient therefore rests on the floor correction alone, and since the simulation is not magnitude-faithful (it recovers the true contrast with a slope of 1.21 to 1.28), we read −0.834 as a sign-and-ordering result, not a calibrated effect size. The raw statistic per ROI is drawn beside the normalised one in Extended Data Fig. 3a; the diet arm's normalised index approaches −0.84 at hV4, near the index's bound, and the raw profile carries the same ordering without that bound.

Extended Data



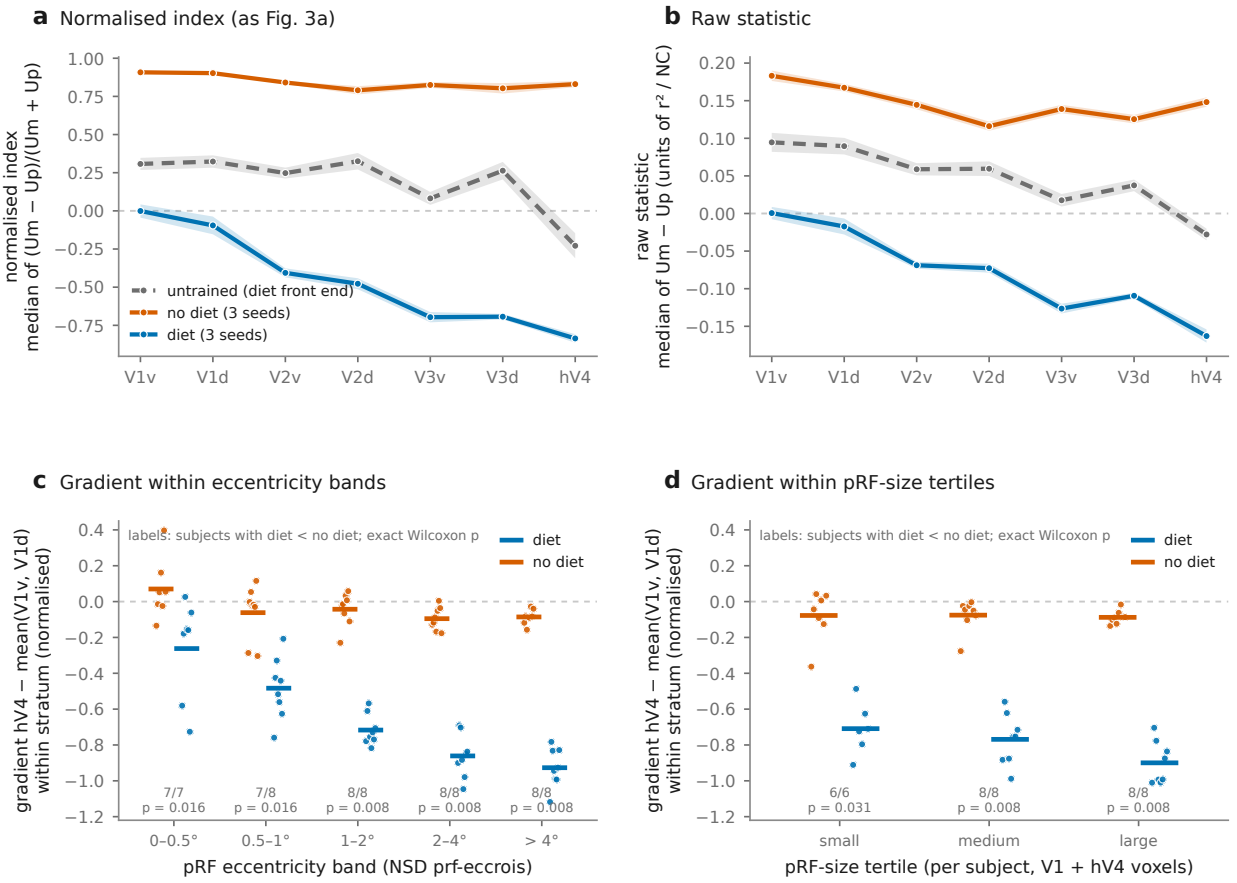
Extended Data Figure 1 | Out-of-distribution degradation curves behind Figure 5a.

Accuracy relative to each seed's own clean accuracy as a function of corruption severity, for seven corruption families (a–g) and the four front-end conditions, three seeds each (thin lines) with the seed mean (thick). Corruptions are applied to the RGB image before the front end, so each model meets the corruption through its own optics, and are scored in the models' native 112-class label space at each seed's best saved checkpoint. Five families were declared untrained-on before the run (uniform noise, high-pass, false colour, phase scrambling, rotation) and two as trained-in positive controls (contrast, low-pass), since the developmental transform is itself a low-pass and contrast-sensitivity manipulation. The diet-trained model degrades faster on uniform noise, high-pass, false colour and rotation, is indistinguishable from the control on phase scrambling, and is more robust only on low-pass. The two conditions that carry no shape bias (swap, `parvo_highsf`) track the no-diet control.



Extended Data Figure 2 | Effective dimensionality: the diet fixes which side of parity training ends on, not a level.

Ratio of the participation ratio of the magno stream's features to the parvo stream's, computed on 1,000 NSD images per subject (n = 8; Methods). Left: each arm's front end before any training, marker at the subject mean with the subject range as a whisker. Right: every training seed at the final checkpoint, each the mean over 8 subjects, fanned from the front end it started at. The four front ends hand the network very different asymmetries before learning (0.59 to 1.91, a 3.3-fold range), including one that is inverted: in `parvo_highsf` the parvo stream is the more dimensional stream in 8/8 subjects. After training every diet-trained seed ends magno-dominant (24/24 subject-seeds above parity in each of the three diet arms) while the no-diet arm collapses to parity (20/24 at or below). No level is labelled: the within-arm seed range (up to 2.39 at the final checkpoint, 0.79 at the best saved one) exceeds the gaps between arm means, so any statement of the form "training converges to X" is unsupported at three seeds. The `parvo_highsf` case also separates dimensionality from shape bias: its second stream is the more dimensional one while remaining texture-biased.

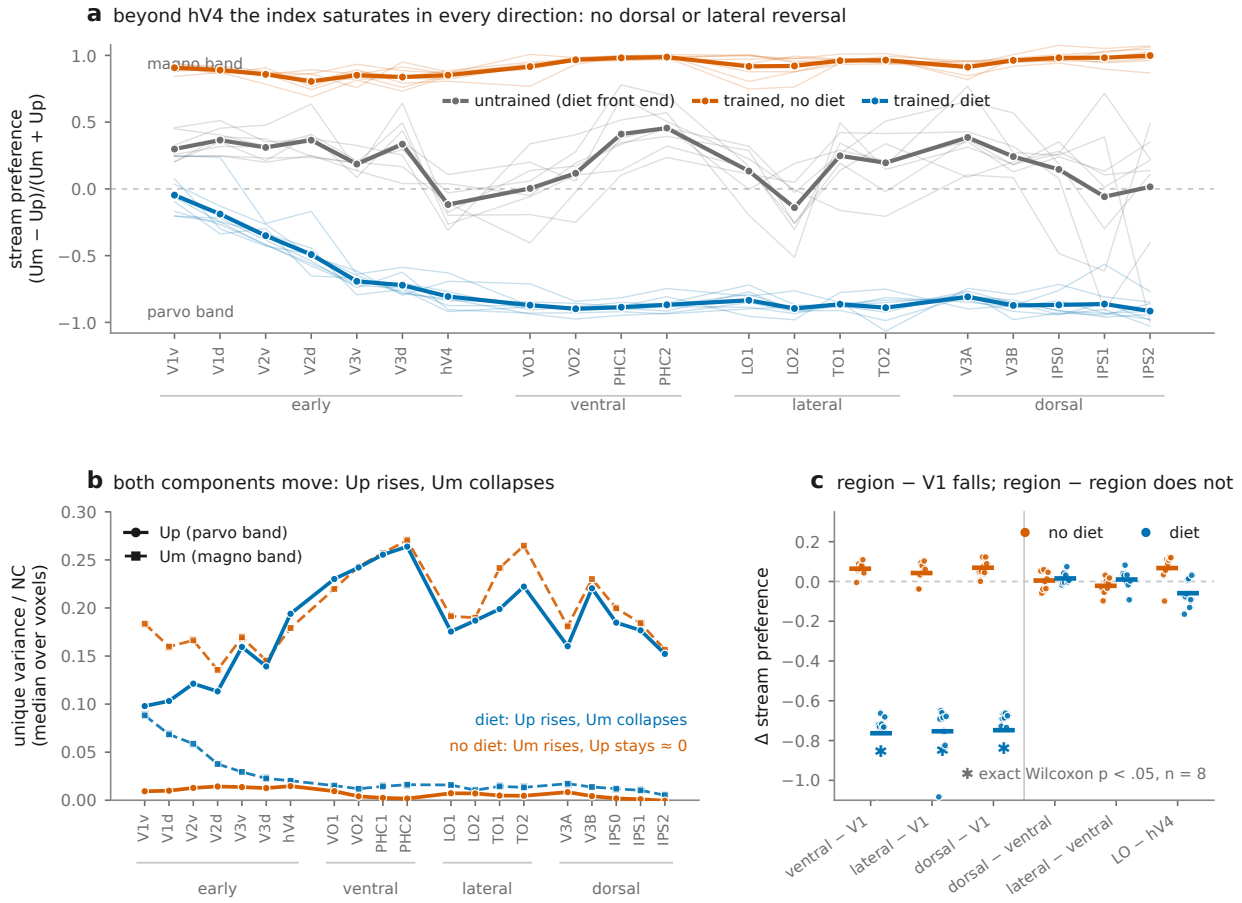


Extended Data Figure 3 | The cortical gradient against its nuisance axes: the raw statistic, eccentricity and receptive-field size.

(**a, b**) The stream-preference profile of Fig. 3a on both statistics: (a) the normalised per-voxel index (U_m − U_p)/(U_m + U_p), median over reliable voxels per ROI, and (b) the raw per-voxel difference U_m − U_p in units of noise-ceiling-normalised r². Lines are means over 8 subjects, shading ± SEM; the trained arms average three seeds per subject, the untrained network on the diet front end is a single model. The diet arm's normalised index approaches −0.84 at hV4, near the index's bound of −1; the raw statistic carries the same ordering (V1 near zero, hV4 at −0.16) without a bound, and the two-statistics rule of Methods 4.7 requires every claim to hold on both.

(**c**) The gradient hV4 − mean(V1v, V1d) computed within each of NSD's five eccentricity bands (`prf-eccrois`), so that V1 and hV4 voxels are compared only with voxels of matched eccentricity. One point per subject per band (seed-mean for the trained arms), bars = mean; a cell needs at least 20 voxels per ROI or the subject is dropped from that band. In the foveal band this drops subject 02 in every arm, whose hV4 has 19 reliable foveal voxels, and leaves subject 04's V1 value resting on V1v alone, its V1d having 5 (the voxel selection is identical across arms, so the same subject is missing from the diet and the no-diet points). Under the diet the gradient is negative in every band (6/7 subjects in the foveal band, 8/8 in the other four) and grows with eccentricity, from −0.26 in the foveal band to −0.93 beyond 4°; without the diet it stays between −0.10 and +0.07 in every band. Labels give the number of subjects in which the diet arm is more negative than the no-diet arm and the exact Wilcoxon p for that band. Eccentricity therefore modulates the diet arm's gradient but does not produce it.

(**d**) The same gradient within pRF-size tertiles, boundaries set per subject on the pooled V1 + hV4 voxels using NSD's pRF size estimates (voxels with pRF $R^2 > 5\%$). Under the diet the gradient is negative in every tertile (−0.71, −0.77, −0.90; all subjects), without the diet it is near zero in every tertile (−0.08 each), so the hierarchical contrast is not a receptive-field-size contrast in disguise. The within-subject regressions of Extended Data Table 9 give the same picture on the continuous predictors.

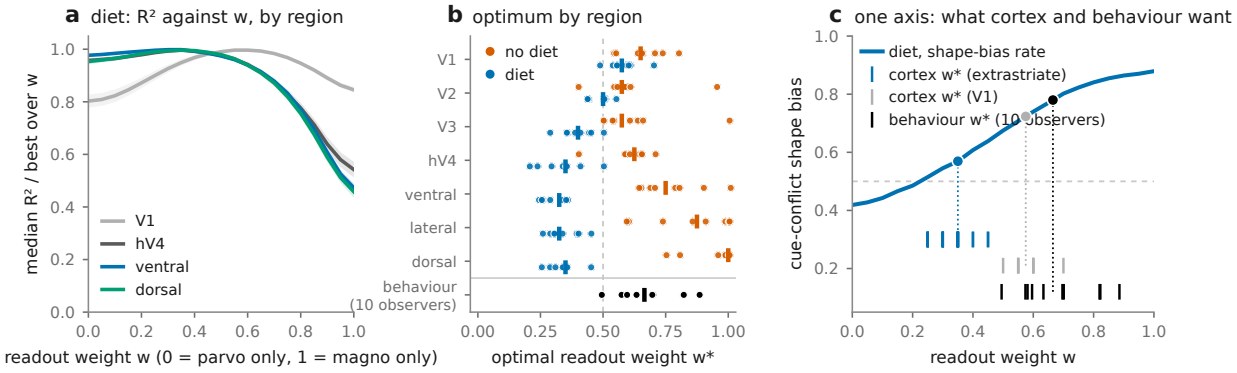


Extended Data Figure 4 | Beyond hV4 the gradient saturates in every direction: the texture-biased stream carries extrastriate cortex whole, ventral, lateral and dorsal alike.

(**a**) Stream-preference index, the median over reliable voxels of (Um − Up)/(Um + Up) (noise ceiling > 0.1, Um + Up > 10^{-3}, ≥ 20 voxels per ROI per subject), on the 22 ROIs of the Wang et al. (2015) atlas that pass the voxel rule in NSD's nsdgeneral mask, grouped as early (V1–hV4), ventral (VO1/2, PHC1/2), lateral (LO1/2, TO1/2) and dorsal (V3A/B, IPS0–2). Thick lines are the mean over the eight subjects, thin lines individual subjects; the trained arms are the mean over three training seeds per subject, the untrained network is one seed on the diet front end. Under the diet the index falls from −0.05 in V1v to −0.81 in hV4 and then sits between −0.80 and −0.94 in every extrastriate ROI. Without the diet the index is +0.80 to +1.00 everywhere. The same saturation holds on NSD's stream atlas and on the face-, place-, body- and word-selective fLoc ROIs (Extended Data Table 11).

(**b**) The two components of the index, both normalised by the noise ceiling: unique variance of the parvo band (Up, circles, solid) and of the magno band (Um, squares, dashed), median over the same voxels. Under the diet Up rises monotonically from 0.10 (V1v) to 0.26 (PHC2) while Um collapses from 0.09 to about 0.01 beyond V3; without the diet the luminance stream shows the same rise (0.18 to 0.27) and the L−M stream's unique variance stays near 0.01 throughout. The rise toward higher areas therefore belongs to whichever stream carries the full-resolution image; the diet moves it from the luminance stream to the chromatic one.

(**c**) Regional contrasts per subject (points) with the mean as a bar. Left of the divider: each region's mean index minus the V1 mean, which is large and negative under the diet in all three directions (ventral −0.76, lateral −0.75, dorsal −0.75; 8/8 subjects, exact Wilcoxon p = .0078) and small and positive without it. Right of the divider: region-to-region contrasts (dorsal − ventral, lateral − ventral, LO − hV4), none of which departs from zero under either arm (all $|\Delta| < 0.06$, $p > .1$). The model reproduces no magnocellular preference of dorsal cortex and no shape-selective reversal in LO.



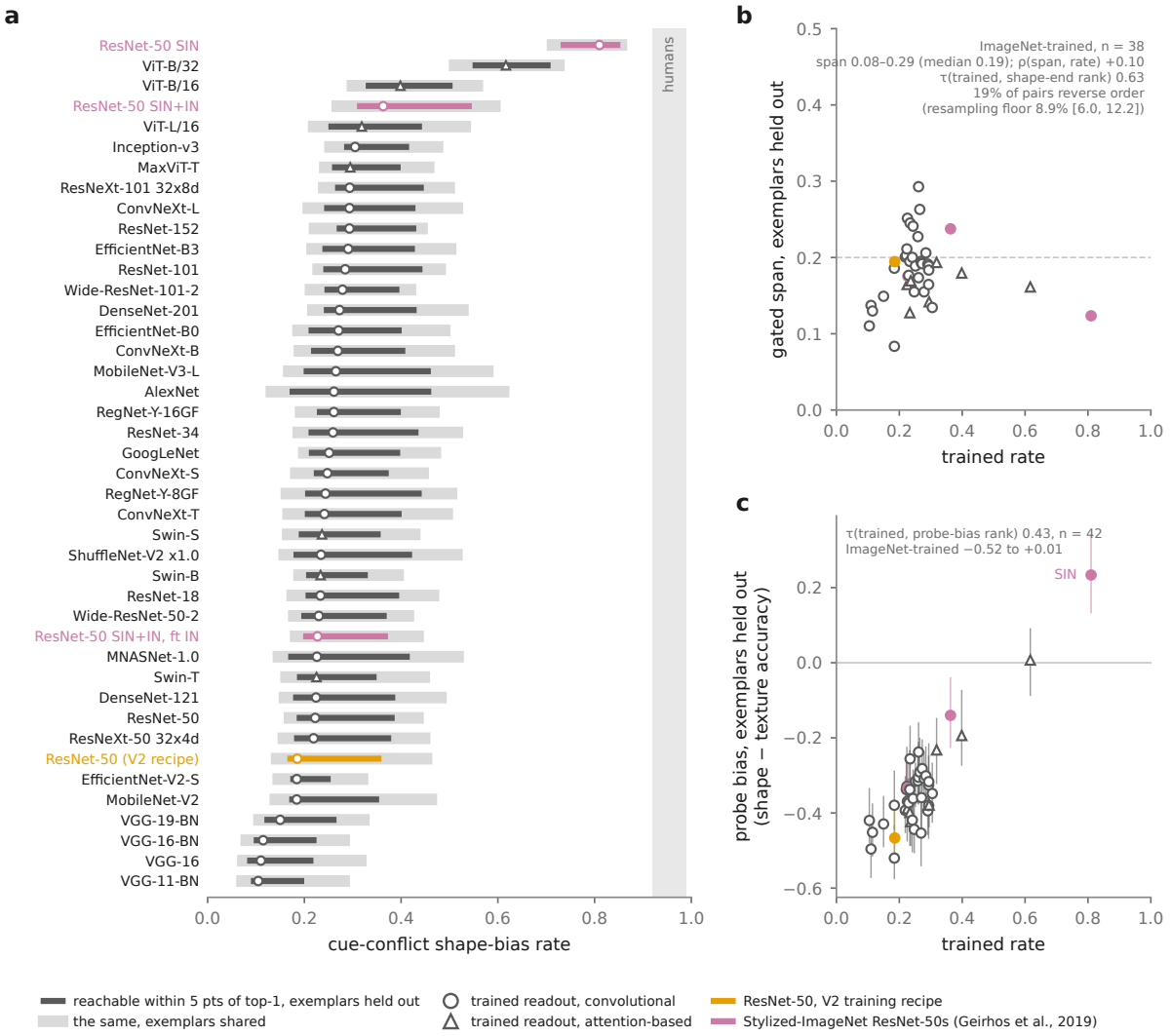
Extended Data Figure 5 | The readout weight cortex prefers is not the one behaviour prefers.

The classifier's input is $z(w) = w \cdot y_1 + (1 - w) \cdot y_2$, where y_1 and y_2 are the 512-dimensional outputs of the magno and parvo streams (Methods 4.4). Results 2.4 sweeps w and reads the cue-conflict rate; here the same $z(w)$, for w from 0 to 1 in steps of 0.05, is the design of a ridge encoding model of every voxel in NSD's general visual mask (same split, grouped folds and alpha grid as the banded model; Methods 4.5). Because $z(w)$ sums the streams dimension by dimension, the ridge cannot undo the mixture: w constrains the design rather than re-parameterising it.

(**a**) Diet arm, three seeds averaged per subject: median test r^2 over an ROI's reliable voxels (noise ceiling $>$ 0.1) as a function of w, normalised by its maximum over w, mean $\pm$ s.e.m. over eight subjects. V1 peaks near the balanced readout; hV4, ventral cortex (VO1/2, PHC1/2) and dorsal cortex (V3A/B, IPS0–2) peak on the parvo side. The curves are shallow: at the paper's $w = 0.5$ every ROI is within 3% of its best r^2, and the height differences are at the ridge's own reproducibility (Extended Data Table 12), so the result is an ordering of optima and never an r^2 difference.

(**b**) The optimum w* per subject (points) and its median (bar) for each region, both arms. Under the diet (blue) w* falls monotonically down the hierarchy, from 0.58 in V1 to 0.50, 0.40 and 0.35 in V2, V3 and hV4 and to 0.33–0.35 in every extrastriate group; the extrastriate mean is below V1 in 8/8 subjects ($\Delta = -0.25$, exact Wilcoxon $p = .0078$) and the Spearman correlation between hierarchical rank and w* over V1–hV4 is ≤ -0.80 in every subject. Without the diet (vermillion) the optimum runs the other way, from 0.65 in V1 to 0.75–1.00 beyond hV4 (7/8 subjects, $p = .023$), as the luminance stream carries the image. The bottom row is the behaviour-optimal weight: the w that best reproduces each of ten human observers' trial-by-trial cue-conflict choices (trial fit of Fig. 4; diet final checkpoints, three seeds averaged), median 0.67, range 0.49–0.89. The extrastriate optima (0.27–0.40) and the observers' optima do not overlap (Mann–Whitney $p = .0004$).

(**c**) Both optima on the axis of the behavioural sweep: the diet network's cue-conflict shape bias as a function of w (mean of three seeds, final checkpoints). Tick marks are individual subjects (cortex) or observers (behaviour); circles mark the medians. Reading the network out with the weight extrastriate cortex prefers gives a rate of 0.56, barely above parity; the weight that best predicts human choices gives 0.78; the human rate of 0.96 lies beyond the sweep's reach. The two demands are met by the same frozen streams and differ only in w.



Extended Data Figure 6 | The readout account across the field's model zoo, on exemplars the subspaces were not fitted on.

Forty-two frozen ImageNet networks under the rule of Fig. 6a–b: y is the input of each network's last linear layer, the 16-class shape and texture linear discriminants of y (PCA to 256 dimensions) give orthogonal shape- and texture-predictive subspaces, and logits(w) = W·(P_R y + 2w·P_S y + 2(1 − w)·P_T y) + b with the head untouched; the rate is scored by the Geirhos 16-class rule. Because the 1,200 cue-conflict images reuse 160 object and 48 texture exemplars, the drawn estimator splits both into three folds, fits the subspaces on the images whose object and texture exemplars both lie outside one fold and scores the rate on the images whose exemplars both lie inside it (ten repetitions, mean curve; Methods 4.8). The pale bars are the pre-registered estimator of `docs/PLAN_ZOO_2026-09-15.md`, fitted on one stratified half of the images and scored on the other, which shares 98% of its object exemplars and all of its texture exemplars with the fitting half. Thirty-eight networks are torchvision `IMAGENET1K_V1` classifiers (grey), one is the same ResNet-50 under torchvision's 2021 training recipe (orange), and three are the ResNet-50s of Geirhos et al. (2019) trained on Stylized-ImageNet (SIN), on SIN + ImageNet, and on SIN + ImageNet then fine-tuned on ImageNet (purple).

(**a**) Networks sorted by their trained rate (symbol; circles convolutional, triangles attention-based), which is scored on all 1,200 images; the held-out curves are scored on the images of the held-out folds only and pass within 0.02 of it at w = 0.5. Bars: the rates reachable within 5 points of the trained top-1 on a stratified 10-per-class ILSVRC-2012 validation subset (10,000 images), with exemplars held out (dark) and shared (pale). In every network the whole sweep lies inside the gate (top-1 changes by at most 4.4 points, median 0.7, measured on the subspaces of the exemplar-shared fit, whose w-range the held-out fits inherit). With exemplars held out the reachable span of the 38 ImageNet-trained networks is 0.08–0.29 (median 0.19; 11

of the 38 reach the pre-registered 0.20), against 0.20–0.50 (median 0.30) with exemplars shared, and the held-out mean curve rises monotonically with w in all 42 networks ($\rho \geq 0.99$). Grey band, the ten human observers (0.92–0.99). The SIN-trained network has the highest trained rate of all 42 (0.81) and a held-out span of 0.12 ($0.73 \rightarrow 0.85$), the 3rd smallest of 39; with exemplars shared its span is 0.17 ($0.70 \rightarrow 0.87$), and on the forced two-way estimator 0.32, the 16th smallest (Extended Data Tables 16 and 18).

(**b**) Held-out gated span against trained rate over the 38 ImageNet-trained networks: the span is unrelated to the rate (Spearman $\rho = +0.10$; $+0.04$ with exemplars shared) and to top-1 (-0.27), and averages 0.19 in convolutional and 0.16 in attention-based networks. Ranking by the trained rate and by the rate at the shape end of each network's gate agree at Kendall $\tau = 0.63$; 131 of the 703 pairs (19%) reverse order, against 8.9% [6.0, 12.2] for the trained ranking re-estimated on two independent resamples of the images ($\tau = 0.47$ and 27% with exemplars shared). The dashed line is the pre-registered 0.20.

(**c**) The exemplar-balanced probe bias — shape minus texture accuracy of a linear probe on the same features with three exemplars per class for both labels, one exemplar left out per fold and PCA and scaling fitted inside each fold; mean of 20 draws, bars the 5–95% range over draws — against the trained rate. Thirty-seven of the 38 ImageNet-trained networks are texture-leaning (-0.52 to $+0.01$, the exception ViT-B/32 at $+0.01$ [-0.09, $+0.09$]), and the two rankings agree only at $\tau = 0.43$. The SIN network is the only shape-leaning network of the 42 ($+0.23$ [$+0.13$, $+0.34$]): its bias is in the representation, a distinction the rate alone cannot make, and the gain moves it mainly by adding shape evidence where it moves 37 of the 38 ImageNet-trained networks mainly by removing texture evidence (Extended Data Table 18). The paper's original stratified-fold probe, whose test images share exemplars with its training folds, gives -0.30 to -0.66 for every ImageNet-trained network, -0.11 for SIN and $\tau = 0.44$ (Extended Data Table 16). Error consistency with the observers (Extended Data Table 16, not drawn) is 2–23% of the human–human ceiling at the trained readouts, rises with the rate across networks ($\rho = 0.80$) and within every gate (38 of 38), and reaches at most 37% at any gated readout of an ImageNet-trained network; the SIN network reaches 40% trained and 53% at its shape end.

Pre-specification and outcomes: `docs/PLAN_ZOO_2026-09-15.md` (P-Z1, P-Z2 and P-Z4 met; P-Z3 met in parts a and c, not in part b, the SIN span falling short of 0.20) and `docs/PLAN_EXEMPLAR_2026-09-21.md` (the held-out estimators; P-X1–P-X10, of which P-X7 and P-X8 failed). Per-network values: Extended Data Table 16 and `results/figures_v2/exemplar_zoo.json`. Preprocessing Resize 256 / CenterCrop 224 except Inception-v3 (342/299), EfficientNet-B3 (320/300) and EfficientNet-V2-S (384/384).